

## 12. Architectural Out-of-Band Management §

### 12.1 Introduction §

“Sidebands” are physical interfaces other than the high-speed Lanes connecting two Ports, typically used for out-of-band platform management functionality. Out-of-band (OOB) management refers to the ability to communicate with peripheral subsystems, such as adapters and components, without interacting with firmware/software running on the host CPU(s).

This chapter provides a “toolkit” with a variety of potentially applicable technology ingredients, including, for example [SMBus] addressing, discrete signaling consolidation and Flexible I/O signal function negotiation, and how to utilize enhanced interfaces such as [I3C-Basic] and managed USB 2.0. The intent of centralizing this content in this document is to yield major improvements in capability and consistency to the physical, electrical, and logical domains for both systems and existing or new future form factor specifications or revisions.

The following table provides relative comparisons of typical architectural out-of-band interfaces and basic use guidance. *Bandwidth* is shown as the theoretical interface speed and is not indicative of practical throughput. The term *Isolated Control Plane* means that the interface does not involve the Link. For example, PCIe Vendor-Defined Messages (VDMs) are typically used for out-of-band management that does not involve firmware/software running on the host CPU(s) but does traverse Links. Some applications choose to isolate such traffic for reasons including host bandwidth or security.

*Table 12-1 Relative Comparisons of Typical Architectural Out-of-Band Interfaces §*

| General Out-of-Band Interface Usage Guidance | Transfer Rate            | Isolated Control Plane | Example Applications                                                                                |
|----------------------------------------------|--------------------------|------------------------|-----------------------------------------------------------------------------------------------------|
| Discrete/Physical Signals                    | Signal propagation delay | Y                      | PERST#, PRSNTNT#, PWRBRK#, PWRDIS                                                                   |
| Serialized/Virtual Signals (e.g., PESTI)     | ≈250 Kb/s                | Y                      | Real-time virtual signals, OOB Controls/status                                                      |
| [SMBus] or [I2C]                             | 100, 400, 1000 Kb/s      | Y                      | EEPROM, I/O Expanders, Sensors, MCTP, Security                                                      |
| [I3C-Basic]                                  | 12.5 Mb/s SDR mode       | Y                      | Hub Control, MCTP                                                                                   |
| USB 2.0                                      | 480 Mb/s                 | Y                      | Serial, Ethernet (NC-SI and TCP/IP), Mass Storage, MCTP, Security, Bridges to SPI, [I2C], and [I3C] |
| PCIe VDM                                     | Link-specific            | N                      | MCTP                                                                                                |

### 12.2 Framework for Sidebands §

Sideband signals refer to all signals that cross a defined interface, other than those associated with the high-speed Link itself. Sideband signals impact platforms such as through added cost, additional conductors and circuits for optional signals that might be rarely or never used. This negatively impacts cross compatibility and establishes legacy constraints that might impede more efficient solutions. Although sideband signals typically travel in parallel with the Link, sideband signals often interact with components other than the two that implement the Ports at each end of the Link.

The architectural framework defined here provides:

- A partial catalog of sideband signals already in common use for defined form factors.
- Reference physical topologies representing how sideband signals are used at the platform level.
- Extensible mechanisms for sideband capability discovery and control.
- Roles and responsibilities of various platform elements in relation to sideband signals.

The goal of the framework is to simplify the definition and evolution of form factors and platforms by providing form factor agnostic consistency and structure to sideband signal handling.

## 12.3 Sideband Signaling Mechanisms §

This section describes three methods of sideband handling for form factor and other interface specifications to use as tools. These include fixed function interfaces using discrete sidebands, Flexible I/O (Flex I/O) and Peripheral Sideband Tunneling Interfaces (PESTI). Each of these mechanisms are utilized where appropriate with the following sections providing specific guidance.

### 12.3.1 Discrete Sidebands §

Discrete sideband signals are physical wires and/or pins that have diverse functions such as those related to platform power up (e.g., **PERST#**), Link control (e.g., **CLKREQ#**), or non-Link functionality (e.g., 2-wire). These physical wires can be transmitted using different methods (e.g., copper, optical). Discrete sideband signals are not limited to digital, two-state or a single functional purpose or direction. Considerations for incorporating discrete sideband signals include architectural needs (e.g., **PERST#**), low latency needs, (e.g., **PWRBRK**), or fault domain minimization control (e.g., **PWRDIS**). Form factor specifications are permitted decide to incorporate such signal types when weighing functionality against interconnect size and cost.

In general, vendor specific platform requirements drive fan-in or fan-out requirements for each signal (e.g., whether **CLKREQ#** is independent for each function).

Examples of discrete signals incorporated in multiple form factors:

- **PERST#** - PCIe Reset (and similar component reset mechanisms) is a signal that causes a component to execute a Fundamental Reset. See § Section 6.6.1 for details.
- **PRSNT#** - Presence (and similar component presence mechanisms) is a class of signal that provides an indication of the presence of the adapter, component, or cable, and optionally the insertion or removal of an adapter, component or cable. See § Section 6.7.1 for details.
- **PWRDIS** - Power Disable (and similar component power disable mechanisms) is a signal that causes an adapter or component to disable use of its own power. See § Section 6.7.3.3 for details.
- **PWRBRK#** - Emergency Power Reduction (and similar adapter or component emergency power reduction mechanisms) is a signal that causes an adapter or component to reduce its own power consumption to a safe level. See § Section 6.24 for details.
- **CLKREQ#** - Clock Request (and similar platform clock request mechanisms) is a signal that causes a platform to provide reference clock to the adapter or component. See § Section 8.6.8 for details.
- **WAKE#** - Wake Up (and similar platform wakeup mechanisms) is a signal that causes a sleeping platform to restore main power to the adapter or component. See § Section 5.3.3.2 for details.

An example of a signal with alternative solutions:

- **SMRST#** – SMBus Reset is a discrete sideband for recovering SMBus hang or stuck busy type conditions without interfering with the form factor's primary workload. This signal is a good candidate for exclusion or serialization, since it is high latency and may be rarely or ever used. It is common practice for applications that do not have a discrete **SMRST#** signal to use bus reset methods such as SMBus clock low reset or by executing SCL pulses followed by a STOP condition.

### 12.3.2 Flex I/O Sidebands §

Flexible input/output signals/interfaces (Flex I/O) utilize the concept where interface pins are permitted to be flexibly (re)purposed following discovery and determination of mutual capabilities and applying rules that guarantee safe and backward compatibility, where applicable, across the electrical and logical domains. When represented as a signal, form factor specifications should use the **FLEXIO-[N:0]** nomenclature.

Flex I/O interfaces provide the following benefits:

- Improved hardware interface utilization compared to fixed function pins.
- Opportunity to continually leverage evolving high-speed management interfaces.
- Flexibility to include, at present or future, value-added out-of-band functionality between platforms and peripheral subsystems, such as adapters or components.

#### IMPLEMENTATION NOTE: FLEX I/O PINS ARE NOT CALLED RESERVED §

Flex I/O signals are purposefully not called Reserved (RSVD) pins, which are typically specified to include no defined usage within a specific form factor specification version. RSVD pins typically call out both the platform and adapter sides to not connect any functional circuit, except high frequency filters such as on the CEM connector.

Flex I/O signal/interface engagement requires these steps, with guidelines described below:

1. Default State
2. Discovery
3. Compatibility Check
4. Control Negotiation

#### 12.3.2.1 Flex I/O Default State Guidelines §

A Flex I/O interface pin is permitted to default to:

1. **Unused:** Unconnected (electrical open) or with a specific termination as dictated by a specific form factor or connector/cable specification.
2. **Pre-Wired/Inert:** This is where a pin/interface is wired with a predetermined circuit and connectivity but does not perform the intended functionality until after negotiation is complete. Voltage bias before negotiation is allowed. An example is a form factor specific required Flex I/O signal connected to platform logic that defaults to tristate/high impedance with a platform-side pullup resistor to the platform side's +3.3 V auxiliary power rail. Then after negotiation is complete, the signal is allowed to be used for its intended purpose. Form factor

specifications must ensure that any allowed default circuit conditions are electrically safe and do not impose logical domain constraints. An example is a cross power domain electrical stress or leakage condition.

3. **Switchable Function:** This is where a pin/interface defaults to a pre-defined function (often a legacy function), but then is electrically or logically switchable to an alternate function. An example is a form factor interface that defaults to JTAG functionality, with all the associated requirements such as default terminations and connectivity. Then following a positive discovery and compatibility check (defined below), a control disengages the default functionality and then engages the intended alternate functionality.

### 12.3.2.2 Flex I/O Discovery Phase Guidelines §

When used, Flex I/O capabilities are required to be described in FRU Information (See § Section 12.6). Form factor specified usages of Flex I/O signal(s) or interface(s) must be provided for discovery purposes via form factor specified FRU Information descriptors. Examples include, but are not limited to, pin number, pin functionality, supported power domain(s), bus topology and any signal integrity attributes deemed necessary to achieve the intended functionality. The [CEM-6.0] provides an example.

### 12.3.2.3 Flex I/O Compatibility Check Guidelines §

Failure to conform to the following guidance could result in undefined behaviors or effects not limited to electrical damage.

1. The platform must compare the adapter's advertised Flex I/O capabilities with the platform side interface capabilities and policy, if applicable, before attempting to perform any of these actions:
  - a. Change from a pin or circuit default state.
  - b. Communicate new control settings.
  - c. Utilize Flex I/O pin functionality in any way, such as toggling Flex I/O connected GPIO.
2. Mismatch: If a Flex I/O pin or interface functionality mismatch is detected between a platform capability or policy and interposer, adapter, or component, then the above three actions must be avoided.
3. Single-ended Flex I/O functionality on differentially optimized signals is allowed. Implementers must be aware of potential effects between such single ended signals, such as crosstalk being an unintended signal aggressor on a victim, impedance mismatches and other unintended effects along the signaling paths.
4. Differential interfaces utilizing single-ended optimized Flex I/O signals is prohibited due to differences, such as target impedance, improper GND shielding, board routing, connector design, etc.
5. Flex I/O usage and signal engagement may be applicable to one or more specific peripheral supported power states (such as Main power state only versus Auxiliary power and Main power states). If a peripheral advertises more than one supported power state, it is allowed to have a control command (if such a command is advertised as supported) for selecting a subset of power states that the peripheral must adhere in reference to specific Flex I/O interface(s). It is the responsibility of the peripheral to provide the necessary cross power domain isolation to avoid inadvertent behaviors.
6. Flex I/O usages are permitted to be independent of an adapter's or component's Link state.
7. Single-ended signal voltage negotiation for Flex I/O pins is outside the scope of this specification.
8. Using Flex I/O pins as power supply rails is prohibited, due to possible ill effects such as an incorrect or malicious software setting causing electrical stress or damage.

### 12.3.2.4 Flex I/O Control Negotiation Guidelines §

Failure to conform to the following guidance could result in undefined behaviors or effects not limited to electrical damage. See § Section 12.7 for the control/command protocol detail. Dynamic reconfiguration of Flex I/O is left to the implementer.

There are four scenarios for how Flex I/O controls assume their intended functionality:

1. Pre-wired circuit with configurable direction, purpose, signaling or protocol: This scenario occurs when an interface is pre-wired before any negotiated functionality. The interface must remain inactive until supported uses are discovered, and the requisite compatibility and policy check are complete, and the appropriate negotiations or controls are executed. Upon completing the negotiation, communication is permitted to commence from either side, as applicable.
2. Only platform side control is necessary: Following a successful discovery and compatibility check, engaging an interface might only require platform side control. For example, [CEM] supports legacy JTAG terminations by default. Upon discovery of a Flex I/O USB capability and compatible platform support, the platform must change from the default JTAG pin functions to USB 2.0. Note that this can be done with one control since no special break-before-make is needed to be electrically safe and USB 2.0 specification compliant.
3. Only peripheral or adapter side control is required: Commanding an adapter to perform specific engage or disengage operations must use a to-be-defined MCTP-based control method.
4. Both the platform and adapter side controls are necessary: Commanding a peripheral subsystem to perform specific engage and/or disengage operations must use a to-be-defined MCTP-based control method. Commanding the system side to perform specific engage and/or disengage operations is permitted to use implementation specific controls, such as a multiplexor select. It is important to use the engage and disengage order and other guidelines below.

### 12.3.2.5 General Flex I/O Control Guidelines §

1. Engagement Order: “Engaging” a Flex I/O is defined as enabling desired interface circuitry, connectivity and/or starting communication. Caution should be taken when electrically engaging an interface. Where applicable, the following order is advised.
  - **First:** Platform side. This is the PCI Express Link downstream facing port side.
  - **Second:** Interposer, adapter, or component side. This is the PCI Express Link upstream facing port side.
2. Disengagement Order: “Disengaging” a Flex I/O is defined as disabling or reverting to default interface circuitry, severing connectivity and/or ceasing communication. Where applicable, the following order is advised.
  - **First:** Interposer, adapter, or component side. This is the PCI Express Link upstream facing port side.
  - **Second:** Platform side. This is the PCI Express Link downstream facing port side.
3. Re-configuration Order: Reconfiguration might be desired between two functions or from an active Flex I/O interface type back to default state. The “break-before-make” paradigm is recommended by using the disengagement, followed by engagement order prescribed above. The duration between the break and the make steps is form factor specific. An example is to ensure that a higher voltage function is sufficiently drained before engaging a lower voltage function that might be intolerant to the higher voltage.
4. Forced Disengagement: Flex I/O functions are required to disengage upon the power state of the platform, interposer, adapter, or component dropping below the minimum negotiated power level or outside of the specified electrical operating conditions.

5. Flex I/O Run-time Reconfiguration: Run-time reconfiguration of Flex I/O is allowed when adhering to the guidelines stated within this specification. The logical domain handling of run-time change in Flex I/O functionality is implementation specific.
6. Reset Flex I/O back to defaults: If a Flex I/O signal changes functionality when main power is enabled, it must go back to its default state upon loss of the main power. If Flex I/O functionality changes when only auxiliary power is applied, it must go back to default upon loss of auxiliary power. Adapter or component use of non-volatile information to automatically revert Flex I/O to a previous state is prohibited.

### 12.3.3 Peripheral Sideband Tunnelling Interface (PESTI) Sidebands §

#### 12.3.3.1 PESTI Introduction §

Peripheral Sideband Tunnelling Protocol (PESTI) is a single wire, half-duplex, bidirectional Universal Asynchronous Receive Transmit (UART) communication channel with a protocol that enables optional discovery and optional, real-time, virtual wire tunnelling between platform and peripheral agents. PESTI enables the minimization of the number of discrete sidebands signals, while extending functionality via scalable, form factor specific, extensible, and real-time virtual wires. PESTI is point-to-point with fanout support that utilizes a bus snooping and bus switching command method. PESTI provides message integrity detection support. PESTI protocol leverages common UART framing to create a simple, low logic, low cost out-of-band payload and signal tunnel. PESTI communication is based on a command and response method (no interrupt) where the peripheral/target is permitted to only transmit data upon request by the initiator. In some applications, PESTI protocol is permitted to occur on top of a traditionally static presence signal, thereby not incurring any extra signal cost for this interface.

This specification describes the base physical and protocol layers, from which form factor specifications are permitted to customize for specific attributes and virtual wire purposes. § Figure 12-11 shows a typical PESTI example for tunnelling virtual wired through a PCI Express Cable for managing the discrete sidebands on an interposer, adapter, or component.

PESTI is not intended to duplicate or replace what can be done over other non-real time interfaces. PESTI also does not currently support higher level protocols inclusive of message encryption.

#### NOTE: Open Compute Project Modular-PESTI §

The Open Compute Project also supports the identical “Modular-PESTI” [<https://www.opencompute.org/wiki/Server/DC-MHS>] interface and protocol.

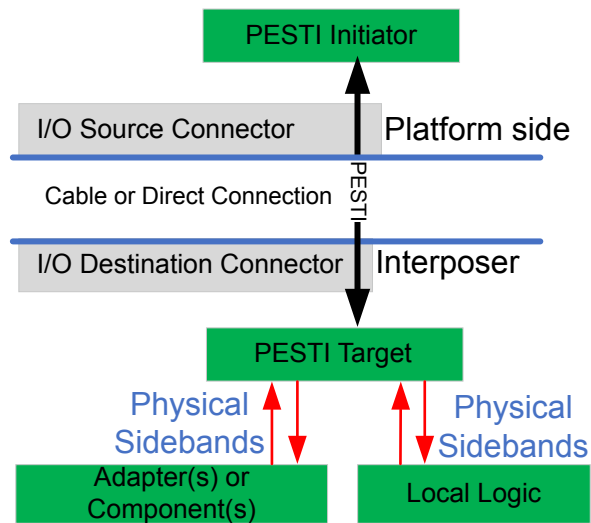


Figure 12-1 Example PESTI Application §

### 12.3.3.2 PESTI Physical Interface §

PESTI utilizes UART framing, operates directly at 3.3 V logic levels, and is a single, half-duplex signal:

- +3.3 Volt LVCMOS (open drain)
- 250,000 BAUD +/- 2%
- 8-O-1 (8 data bits, Odd Parity, 1 stop bit)
- Half-duplex

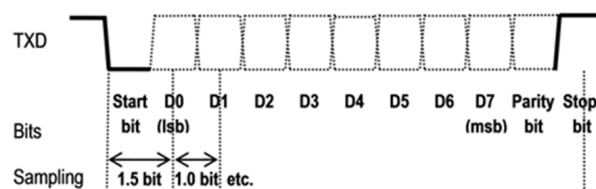


Figure 12-2 UART Data Framing §

### 12.3.3.3 PESTI Electrical Circuit §

The circuit shown in § Figure 12-3 is such that an unpowered ( $V_{3P3\_Target} = 0$  Volts) component holds the PESTI wire low indicating simple presence. Once the component has power and can respond to a command, the target General Purpose Input/Output (GPIO) sourcing the gate of the P-FET will be driven low. Once  $V_{3P3\_Target}$  is engaged by turning the P-channel FET on, a rising edge of the PESTI signal is observed at the initiator. For this mechanism to operate as intended, the default state of the target GPIO must be an input (high impedance).

When used as simple presence, the initiator side circuit must remain the same as shown in § Figure 12-3. The adapter or component side circuit can be simplified to a pulldown resistor with a maximum value of 2.27 Kohm +5% tolerance and minimum nominal value of 200 ohm -5% tolerance.

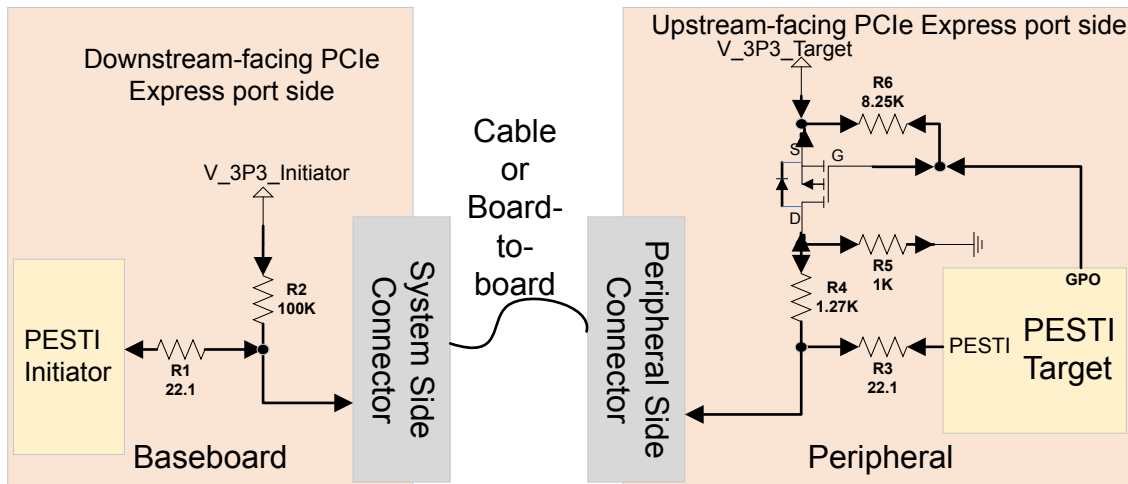


Figure 12-3 PESTI Circuit Diagram §

For non-hot-plug applications, PESTI is permitted to double as the presence signal, thereby incurring no added pin cost. Doing so for hot-plug applications is discouraged due to potential ambiguity or delay in determining a physical presence change versus an unresponsive target. PESTI is permitted to be utilized to indirectly tunnel physical presence status of hot-plug capable downstream targets.



## IMPLEMENTATION NOTE: ELECTRICAL COMPONENT SELECTION FACTORS §

Referring to § Figure 12-3 above, **R1** and **R3** are recommended series termination resistor values. The values might require adjustment to the driver and transmission line characteristics.

**R2** must be selected considering:

- Select the maximum value to minimize the current sourced into an unpowered target.
- Logic high=1 must be guaranteed to meet the baseboard initiator logic Vin High Minimum when cable/PESTI target is not attached/present.

**R4/R5** must be selected considering:

- **R4** and **R5** (pull-down) and **R2** (pull-up) must guarantee a logic low=0 at the baseboard initiator logic if detection of an unpowered target is desired.
- **R4** value provides the rise time to a logic high at the initiator or target for an open-drain interface.
- **R5** provides a path to GND for the current sourced by **R2** on the source board to drain or bleed voltage accumulation at an unpowered target.

### 12.3.3.4 PESTI DC Specifications §

Table 12-2 PESTI DC Specifications §

| Symbol | Parameter                | Min   | Max   | Units | Comments     |
|--------|--------------------------|-------|-------|-------|--------------|
| VDD    | Bus Voltage              | 3.135 | 3.465 | V     | 3.3 +/-5%    |
| VIH    | HIGH level input voltage | 2.0   |       | V     | 3.3 V LVCMOS |
| VIL    | LOW level input voltage  |       | 0.8   | V     | 3.3 V LVCMOS |
| VOL    | LOW level output voltage |       | 0.4   | V     |              |

### 12.3.3.5 PESTI Target Detection §

Presence of a PESTI target is characterized by a rising edge of PESTI. The default input state of PESTI at an initiator is HIGH=1 indicating that there is no adapter or component present. If the PESTI signal is observed as static low for more than  $t_{DBREAK}$  (See § Table 12-5), that target adapter or component is indicating simple presence. A rising edge of the PESTI signal indicates to the initiator that a target is ready to receive and process commands.

A UART BREAK event is when the wire is being driven/held low for a time greater than the entire frame length. Example: At 250 KBAUD, the PESTI frame length is nominally  $4 \mu s * 11\text{-bit positions}$  which is  $44 \mu s$ . PLD logic is permitted to detect a BREAK condition by starting a timer at the falling edge of the signal. A valid BREAK assertion does not require a falling edge to be detected. In some cases, target UART receiver logic is permitted to not implement a BREAK detect status register. Depending upon the UART implementation, a BREAK event is permitted to present itself in status registers as an abnormal frame with a data value = 00h that has both a Parity Error (PE) and a Framing Error (FE: Stop bit = 0).

A valid PESTI frame is characterized by a Start bit of LOW=0, Stop bit of HIGH=1, and an odd number of ones (including Parity).

Target Detection Rules:

1. An initiator must not attempt communication while PESTI is held low (simple presence) by the target.
2. A PESTI target must not release BREAK until it is ready to respond to the payload request command.
3. Minimum low (BREAK) assertion width required to guarantee detection at the initiator is  $t_{DBREAK}$ .
4. An adapter or component is permitted to request a re-start of the discovery process by asserting and releasing BREAK at any time prior to the transition to the active phase (see § Section 12.3.3.6.2).

The PESTI circuit shown in § Figure 12-3 guarantees that BREAK will be asserted from the time that platform input power is applied to the time that the target is powered and ready to respond to a command.

### 12.3.3.6 PESTI Protocol Commands §

#### 12.3.3.6.1 Discovery Payload Request (DPR) §

PESTI compliant targets must support the following command.

- Discovery Payload Request: Data Byte Value = 00h

Following PESTI target presence detection, the initiator will transmit the payload request command to the target. The target response is to transmit the contents of its discovery payload for capture by the initiator. See § Table 12-6 for payload requirements. The command is permitted to be sent autonomously by a PLD or under the direction of Systems Management Firmware (SMFW). See § Section 12.3.3.9 for initiator interface registers accessible to SMFW.

#### 12.3.3.6.2 PESTI Virtual Wire Exchange (VWE) §

PESTI compliant targets must support the following command.

- Virtual Wire Exchange: Data Byte Value = 01h

Once a target's attributes (number, type, and order of virtual wires) have been identified by parsing the payload, an exchange of virtual wires between initiator and target is permitted to commence. This phase of the protocol is known as the active phase. Virtual wires are optional, and a target is not required to support any virtual wire bytes in or out following the VWE command.

If a PESTI target does not support virtual wires, it must respond with an acknowledge data byte (value = 00h) after receiving the VWE command.

The acknowledge feature can be utilized as a health monitor of the target. Alternatively, SMFW is permitted to disable the active phase by clearing  $APEN=0$  (see § Section 12.3.3.9) to disable communication to a target after the discovery phase.

#### 12.3.3.6.3 PESTI Fanout MUX Control §

The MUX switch control command must be supported in PESTI fanout control components only.

- MUX/switch control: Data Byte Value = 02h

See § Section 12.3.3.14 for additional details.

### 12.3.3.7 PESTI Initiator Abort §

A PESTI transaction sequence is defined as a series of data bytes exchanged between an initiator and a target. A sequence begins with the initiator transmitting one or more data bytes to the target. The sequence ends when the initiator receives the expected number of data bytes from the target. During the active phase, a target is required to recognize that a sequence is being terminated prior to its completion. The conditions for requiring termination of a sequence are architecture and implementation specific.

During Initiator Transmission:

In place of a valid data frame, the initiator is permitted to assert a BREAK condition to cancel a sequence that is in progress. The initiator must not interrupt a data frame after the Start bit has been sent. If an event occurs that requires an abort after a Start bit has been transmitted, the initiator must not assert BREAK until  $t_{TIDLE}$  (See § Table 12-5) after the previous Stop bit has been transmitted. When this sequence is observed by a target, it must ignore previous data bytes in that sequence and prepare for the start of the next sequence.

During Target Transmission:

The initiator abort assertion is permitted to occur at any point within a target transmitted frame. It is permitted to also occur during the active phase turn around period between initiator transmit and target transmit. In this case, it would be asserted following the Stop bit of the final byte transmitted by the initiator. The assertion width ( $t_{ABREAK}$ ; See § Table 12-5) is guaranteed to be detected by a target that samples the PESTI logic state prior to transmitting any Start bit in the sequence. When an abort is detected, a target must not attempt to transmit any additional bytes and prepare itself for a new sequence start from the initiator.

### 12.3.3.8 PESTI Broadcast §

To minimize initiator logic, a single UART instance is permitted to be shared (multiplexed) among several targets. Multiplexing requires that targets be serviced in a round-robin rotation. Round-robin servicing has a limitation that responses cannot be captured from multiple targets simultaneously. However, an initiator is permitted to implement logic to transmit a command to several targets in parallel.

A broadcast sequence is characterized by the first byte of the sequence beginning with a data byte value of FFh. The second byte of the sequence is the command. An initiator must not use the broadcast command to request data or an acknowledge from any target. A target must not respond to the broadcast command with data or an acknowledge back to the initiator.

An example application is the Power Brake (e.g., PWRBRK#) sideband signal is permitted to be a virtual wire used to rapidly reduce power consumption of multiple adapters or components. PESTI initiators that implement a shared UART must support broadcast of a Power Brake event.

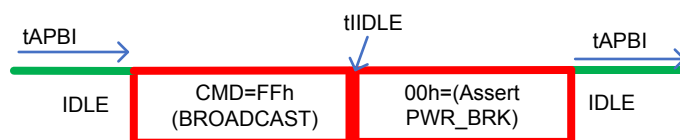


Figure 12-4 PESTI Broadcast Command §

Round-robin servicing naturally staggers PWR\_BRK de-assertion during a virtual wire exchange which is a common platform preference to avoid excessive inrush current from high power loads. Additional delay between PWRBRK# de-assertion to multiple targets is implementation specific and controlled via SMFW and initiator logic that feeds into the appropriate PESTI channel.

An initiator is permitted to issue broadcast commands back-to-back without waiting for a response.

### 12.3.3.9 PESTI Initiator Control and Status Registers §

§ Table 12-3 defines the initiator register bits enable control and status indicators for each connected PESTI wire.

Table 12-3 PESTI Initiator Control and Status Registers §

| Field Name                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Attributes |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>DISCOVERY_STATUS</b> [1:0]<br>(a.k.a. <b>DSTAT</b> ) | <p><b>00b</b> No BREAK detected; PESTI wire is static HIGH=1 (<i>Default</i>)</p> <p><b>01b</b> PESTI wire is static LOW=0 (simple presence).</p> <p><b>10b</b> Discovery payload has been received with a good checksum.</p> <p><b>11b</b> BREAK release detected, but payload has not been successfully received.</p> <p>Initiator implementation requires the following behavior:</p> <ul style="list-style-type: none"> <li>Once a BREAK release is detected, the allowable <b>DSTAT</b> values must exclude 00b until the next fundamental PLD reset or cycle of platform input power.</li> <li>If <b>APEN</b>=1b and <b>DSTAT</b>=10b, then the target component is in the active phase and a BREAK condition on the PESTI wire must not transition <b>DSTAT</b> to a value of 01b. If <b>APEN</b>=0b, then a BREAK asserted by the target will result in <b>DSTAT</b>=01b. See § Section 12.3.3.13 for more information.</li> </ul> | RO         |

| Table 12-4 PESTI Discovery Status State Transitions § |                          |                                                                                                                |
|-------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------|
| DSTAT<br>Current<br>State                             | DSTAT<br>Future<br>State | State Input                                                                                                    |
| xxb<br>(Any<br>State)                                 | 00b<br>(Absent)          | Power on Reset (POR)                                                                                           |
| 00b<br>(Absent)                                       | 01b<br>(PRSNT#)          | BREAK asserted following POR de-assertion or an unpowered adapter or component is present or a static presence |

| Field Name                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |                                                                                                                                                                                                                                                                                 | Attributes  |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                                          | DSTAT<br>Current<br>State                                                                                                                                                                                                                                                                                                                                                                                                                      | DSTAT<br>Future<br>State   | State Input                                                                                                                                                                                                                                                                     |             |
|                                                          | 01b<br>(PRCNT#)                                                                                                                                                                                                                                                                                                                                                                                                                                | 11b<br>(PESTI<br>Presence) | BREAK release by the target, simple presence or PESTI component removal prior to target BREAK release while platform power is on.                                                                                                                                               |             |
|                                                          | 11b<br>(PESTI<br>Presence)                                                                                                                                                                                                                                                                                                                                                                                                                     | 10b<br>(Payload<br>Good)   | Discovery Payload received without any errors                                                                                                                                                                                                                                   |             |
|                                                          | 10b<br>(Payload<br>Good)                                                                                                                                                                                                                                                                                                                                                                                                                       | 11b<br>(PESTI<br>Presence) | BREAK assertion and release by target if not in active phase. Payload Good is protected/locked during the active phase against target resets. This transition also occurs if DPEN is toggled from 0 to 1 while the component is still in the discovery phase ( <u>APEN</u> =0.) |             |
| <b>DISCOVERY_PAYLOAD_ENABLE</b><br>(a.k.a. <b>DPEN</b> ) | <b>0b</b> (Default) Initiator will not send payload request command.<br><b>1b</b> Initiator will send payload request command, with retries, until DSTAT=10b.<br><br>Initiator logic implementation requires that a transition of <u>DPEN</u> from LOW=0 to HIGH=1 by SMFW must clear any “Discovery Complete” flag indicated by DSTAT=10b. As a result, DSTAT would transition to 11b and the DPR command would be transmitted to the target. |                            |                                                                                                                                                                                                                                                                                 | <u>RW</u>   |
| <b>ACTIVE_PHASE_ENABLE</b><br>(a.k.a. <b>APEN</b> )      | <b>0b</b> Initiator will not enter the virtual wire exchange phase for that PESTI instance.<br><b>1b</b> (Default) Initiator will enter the active phase if DSTAT[1:0]=10b                                                                                                                                                                                                                                                                     |                            |                                                                                                                                                                                                                                                                                 | <u>RW</u>   |
| <b>ACTIVE_PHASE_ERROR</b><br>(a.k.a. <b>APERR</b> )      | <b>0b</b> (Default) Response RX error (i.e., parity, framing) or timeout has not occurred.<br><b>1b</b> Response RX timeout or data error has occurred since last cleared (sticky). SMFW must write a 1b to this bit to clear the error once it has been acknowledged.                                                                                                                                                                         |                            |                                                                                                                                                                                                                                                                                 | <u>RW1C</u> |
| <b>ACTIVE_PHASE_STAT</b><br>(a.k.a. <b>ASTAT</b> )       | <b>0b</b> (Default) Target component is in the discovery phase.<br><b>1b</b> The Initiator and Target communication is in the active phase<br>Completion of discovery is characterized by both of the following being true. <ol style="list-style-type: none"> <li>Discovery payload has been captured by the Initiator without any errors.</li> <li><u>APEN</u>=1</li> </ol>                                                                  |                            |                                                                                                                                                                                                                                                                                 | <u>RO</u>   |

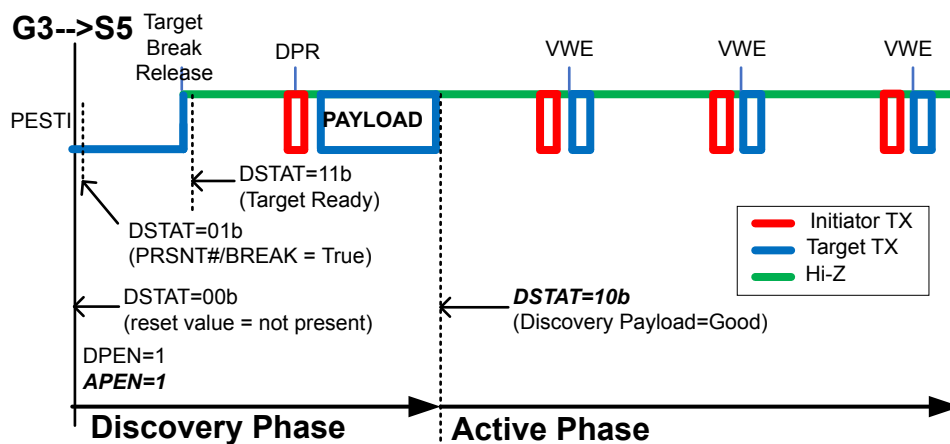


Figure 12-5 PESTI Protocol Phases §

### 12.3.3.10 PESTI AC Specifications §

Table 12-5 PESTI AC Specifications §

| Symbol       | Parameter                             | Min    | Max    | Units   | Comments                                                                                                                          |
|--------------|---------------------------------------|--------|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| $t_{BAUD}$   | BAUD rate                             | 245000 | 255000 | Hz      | 250000 +/- 2%                                                                                                                     |
| $t_{FRAME}$  | Start<br>+8b Data<br>+Parity<br>+Stop | 42.7   | 45.3   | $\mu s$ | $1/t_{BAUD} * 11$ bits                                                                                                            |
| $t_F$        | Fall Time                             | -      | 120    | ns      | Same as 1 MHz [SMBus] (VIH,MIN + 0.15 V) to (VIL,MAX - 0.15 V)                                                                    |
| $t_R$        | Rise Time                             | -      | 120    | ns      | Same as 1 MHz [SMBus] (VIL,MAX - 0.15 V) to (VIH,MIN + 0.15 V)                                                                    |
| $t_{SPIKE}$  | Noise Spike suppression time          | 0      | 50     | ns      | Same as 400 kHz [SMBus]. Noise suppression is recommended.                                                                        |
| $t_{ABREAK}$ | Initiator Abort BREAK assertion time  | 50     | 55     | $\mu s$ | Synchronous to a normal START of frame when the initiator is transmitting. Asynchronous to START when the target is transmitting. |
| $t_{DBREAK}$ | Target Discovery BREAK assertion time | 50     | -      | $\mu s$ | PESTI target BREAK can persist                                                                                                    |
| $t_{BPAUSE}$ | Broadcast Pause                       | 20     | -      | $\mu s$ | Pause to allow fanout snooper completion of channel enables                                                                       |

|                |                                              |      |      |         |                                                                                                                                                                                                                                                   |
|----------------|----------------------------------------------|------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $t_{IDLE}$     | Initiator<br>End of<br>STOP to<br>START      | 0    | -    | ns      | STOP is typically sampled at the midpoint of the bit, so there is approximately 2 $\mu$ s of time to the next START.                                                                                                                              |
| $t_{TIDLE}$    | Target End<br>of STOP to<br>START            | -    | 1000 | ns      | This is required for the target to sample for initiator abort prior to START of target TX.                                                                                                                                                        |
| $t_{MARK}$     | End of<br>BREAK to<br>START time             | 3.88 | 4.12 | $\mu$ s | MARK time is 1 BAUD period between end of initiator abort BREAK and START of new command                                                                                                                                                          |
| $t_{DPTAR}$    | Discovery<br>Phase<br>Turnaround<br>Time     | 100  | -    | ns      | MAX not specified. It is bound by payload size and $t_{DPRTO}$ .                                                                                                                                                                                  |
| $t_{APTAR}$    | Active<br>Phase<br>Turnaround<br>Time        | 0.1  | 200  | $\mu$ s | Between Target RX and Target TX of a response. MAX reduces time to sample initiator BREAK/Abort signal just before START. Target MCU should not have trouble meeting minimum time required to allow the initiator to prepare for RX following TX. |
| $t_{DPRTO}$    | Discovery<br>payload<br>receive<br>timeout   | -    | 250  | ms      | Allows for 150 ms $t_{DPTAR}$ + 2048-byte payload size.                                                                                                                                                                                           |
| $t_{APRTO}$    | Active<br>Phase<br>receive<br>timeout        | -    | 500  | $\mu$ s | Includes margin beyond $t_{APTARMAX} + 8 * t_{FRAME MAX} + 7 * t_{TIDLE MAX} = 389 \mu$ s                                                                                                                                                         |
| $t_{APBI}$     | Active<br>phase bus<br>IDLE time             | 10   | -    | $\mu$ s | Between initiator RX from target and initiator TX to same target                                                                                                                                                                                  |
| $t_{MSC\_WDT}$ | Mux Switch<br>Control<br>Watchdog<br>Timeout | 100  | -    | ms      | Time for a snoopers to detect a stuck bus following a downstream channel select operation                                                                                                                                                         |

### 12.3.3.11 PESTI Discovery Phase §

A PESTI target is required to support the DPR command when simple presence is not implemented. Payload format requirements enable the initiator implementation to be common and target agnostic. The discovery payload is typically consumed by the initiator logic (e.g., SIZE, CHECKSUM) and SMFW. The payload contents can be scaled to meet vendor requirements.

#### Discovery Phase Command and Target Response

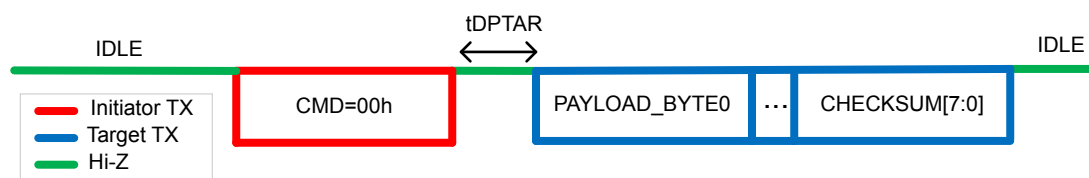


Figure 12-6 PESTI Discovery Command and Response Format §

Discovery Payload Turnaround Time ( $t_{DPTAR}$ ): Maximum time allowed between initiator completing transmission of the DPR command and the response beginning to be received at the initiator.

### Discovery Rules

Following the BREAK de-assertion by a PESTI target, the initiator must transmit the discovery request command to that target if  $DPEN=1$ .

- Turnaround time from target command received to start of response transmission has a minimum requirement of  $t_{DPTAR}$ .
- A target must complete the discovery payload response within the payload RX timeout of  $t_{DPRTO}$ .
- If a target BREAK condition is observed by the initiator with  $APEN=0$ ,  $DSTAT[1:0]$  must be transitioned by logic to 11b (Discovery not complete) until a new payload is received successfully.
- The initiator must not transmit the VWE command unless a successful discovery payload is received and  $APEN=1$ .
  - Success means within the RX timeout period ( $t_{DPRTO}$ ) with no byte parity or framing errors and a verified checksum.
- While  $DPEN=1$ , an initiator must continuously attempt to retrieve a discovery payload from a PESTI target that is present.
  - Example: Round-robin servicing by a single initiator UART to multiple targets:  
If the discovery payload is not successfully received after an initial attempt plus two retries per target, the next target in the round-robin rotation will be serviced. When servicing returns to the target, the discovery request command is sent again as a set (initial + two retries) until successful.

If a target is unresponsive, the minimum time between successive transmissions of the DPR command from initiator to any target is  $t_{DPRTO}$ . Retries described in the example above are applicable to round-robin implementations only and not a requirement of this specification and is left to the initiator implementation.

### Discovery Payload

The minimum required payload (8 bytes) is shown in § Table 12-6 for the case when simple presence is not implemented. The payload size must be a multiple of eight bytes including the checksum.

Table 12-6 PESTI Discovery Payload §

| Byte/Bit | 7                                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------|----------------------------------|---|---|---|---|---|---|---|
| 00h      | **PAYLOAD_FORMAT_VERSION [7:0]** |   |   |   |   |   |   |   |



| Byte/Bit      | 7                                                                   | 6    | 5    | 4    | 3    | 2    | 1               | 0                    |
|---------------|---------------------------------------------------------------------|------|------|------|------|------|-----------------|----------------------|
| 01h           | **DISCOVERY_PAYLOAD_SIZE [7:0] **                                   |      |      |      |      |      |                 |                      |
| 02h           | **VENDOR_ID [15:0] **                                               |      |      |      |      |      |                 |                      |
| 03h           | **VENDOR_ID [7:0] **                                                |      |      |      |      |      |                 |                      |
| 04h           | **DEVICE_CLASS [7:0] **                                             |      |      |      |      |      |                 |                      |
| 05h           | **DEVICE_ID [15:8] **                                               |      |      |      |      |      |                 |                      |
| 06h           | **DEVICE_ID [7:0] **                                                |      |      |      |      |      |                 |                      |
| 07h           | RSVD                                                                | RSVD | RSVD | RSVD | RSVD | RSVD | **PEC_SUPPORT** | **VENDOR_ID_FORMAT** |
| 08h to (N-1)h | Optional Vendor Specific Region (Varies)+ **Padding (0 – 7 Bytes)** |      |      |      |      |      |                 |                      |
| Nh            | **Checksum**                                                        |      |      |      |      |      |                 |                      |

Notes:

\*\*Required region

\*\*\*Minimum value of N=1, Maximum value of N=2040

#### **PAYLOAD\_FORMAT\_VERSION[7:0]**

Indicates the payload format version. This field must be 02h.

01h is used by the Open Compute Project.

#### **DISCOVERY\_PAYLOAD\_SIZE[7:0]**

Indicates the total static payload size. The value of this bit field represents (payload total # of bytes / 8) - 1.

#### **VENDOR\_ID[15:0]**

Utilize the PCI-SIG assigned Vendor ID, when available. If the vendor does not have a PCI-SIG Vendor ID (which applies to PESTI included subsystems such as a power converter board or a cable), then set VENDOR\_ID\_FORMAT = 1.

#### **DEVICE\_CLASS[7:0]**

Indicates the device class of the PESTI target peripheral

Note that a given form factor is permitted to define specific module classes.

#### **DEVICE\_ID[15:0]**

Vendor specific with intent to represent a unique type of device. Not intended for unit specific information such as a serial number.

#### **VENDOR\_ID\_FORMAT**

0 = PCI-SIG assigned Vendor ID of the company delivering the adapter or component with the PESTI target

1 = Non-PCI-SIG assigned Vendor ID. e.g., Open Compute Project assigned.

#### **PEC\_SUPPORT**

1 means that the target supports packet error byte checking (PEC) on command, write data, and read data returned to the PESTI initiator.

#### **RSVD**

Reserved for future use

**Vendor Specific Region**

Optional, implementation specific and is recommended to contain its own sub-payload version.

**CHECKSUM[7:0]**

CRC-8 with polynomial=0x07, Seed=0x00 is required in the discovery payload.

**12.3.3.12 PESTI Active Phase §**

Once a PESTI is transitioned to the active phase, the initiator autonomously exchanges hardware controlled virtual wires with each target component. The HW controlled virtual wire inputs and outputs are target DEVICE CLASS[15:0] specific. The number of bytes in and out and their respective usages are fixed per device class.

If the number of virtual wire bytes out is zero, the initiator will skip transmitting Virtual Wire Output (VWOUT\_0) and wait to receive any Virtual Wire Input (VWIN\_N) bytes. If the number of virtual wire bytes is zero, then an acknowledge target response of 00h is required from the target, whenever one is expected at the initiator. SMFW is permitted to clear APEN=0 so that the VWE command is not transmitted, and the bus will remain idle until the next power on reset or re-entry to the discovery phase.

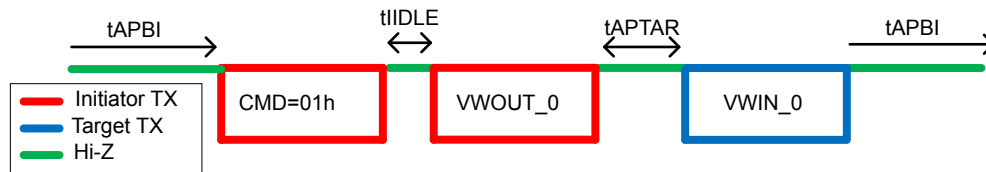
**Virtual Wire Exchange Example (1 Byte Out/In)**

Figure 12-7 Single Byte PESTI Virtual Wire Exchange §

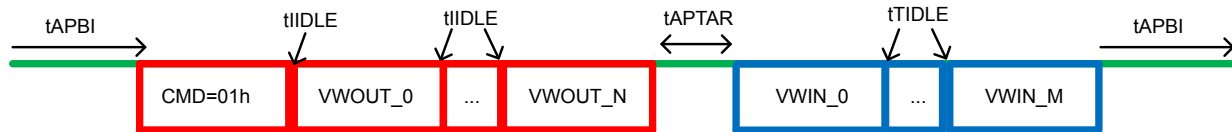
**Virtual Wire Exchange Example (N Byte Out/ M Byte In)**

Figure 12-8 Multi-byte PESTI Virtual Wire Exchange §

**Active Phase Rules**

1. Minimum PESTI idle period between initiator RX of the previous target response & TX of the next target command is  $t_{APBI}$
2. While  $ASTAT=1$ ,  $DSTAT$  must remain 10b (Discovery Complete)
3. A target assertion of BREAK while  $ASTAT=1$  must result in  $APERR=1$ .

4. A target must wait a minimum of  $t_{APTAR}$  before transmitting a VWE response.
5. A target must complete a response to the initiator within the RX timeout of  $t_{APRTO}$ .
6. The maximum period between commands sent from an initiator to that same target is not bound by this specification. The active phase of any component is under the control of SMFW and can be disabled.
7. If target supports virtual wire OUTs only, the target must respond with 00h back to the initiator as confirmation of receiving the OUTs.
8. SMFW is permitted to clear APEN to disable the active phase for a component.
9. Asymmetrical number of out and in bytes is permitted.

The method to route virtual wires to/from internal platform logic or other entities and the PESTI is outside the scope of this document. The usage of the virtual wires (internal commands/policies or connections to local physical signals) by the target is also outside the scope of this document.

### Example VWIRE Bit Mapping Definition

Below is an example of how a form factor, such as an interposer to a [CEM] slot, is permitted to map legacy wires onto virtual wires over PESTI.

*Table 12-7 Example VWIRE\_OUT\_0 (Initiator to Target) §*

| 7    | 6    | 5    | 4    | 3    | 2    | 1      | 0       |
|------|------|------|------|------|------|--------|---------|
| RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | S0_RUN | PWR_BRK |

**PWR\_BRK** Active high (1=Assert, 0=De-assert) virtual wire

**S0\_RUN** Active high (1=True, 0=False) virtual wire indicating platform power state is in ACPI S0.

**RSVD** Reserved for Future Use

*Table 12-8 Example VWIRE\_IN\_0 (Target to Initiator) §*

| 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |
|------|------|------|------|------|------|------|------|
| RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | RSVD | WAKE |

**WAKE** Active high (1=Assert, 0=De-assert) virtual wire indicating the target component request to enter the ACPI S0\_RUN state.

**RSVD** Reserved for Future Use

### Packet Error Checking

Packet Error Checking (PEC) is optional in the initiator and target. PEC is recommended in the active phase when information integrity is critical to the application (e.g., an errant control may cause more ill effects than a bit flip on an analog read that gets averaged out over time). A target's PEC support is described in a discovery payload bit. PEC utilizes CRC-8. Enabling PEC adds to active phase latency.

When supported by the target and set by the initiator, the one-byte command (CMD) bit 7 indicates to the target that the PEC byte is being delivered following with the command byte.

Example: CMD=81h, VWOUT\_0, PEC, Turnaround, VWIN\_0, PEC

If the target indicates no PEC support in the discovery payload, then the initiator must not set bit 7 in the CMD nor deliver an extra PEC byte following the command. This is because following the turnaround, the target may drive contention on against the incoming PEC byte.

If the CMD to target has bit 7 set indicating PEC:

1. The initiator must supply the PEC byte.
2. If the PEC check at the target fails, then the target must drop the command and, if applicable, receive payload. A method to indicate from target to initiator that a PEC failure occurred is out of scope of this specification.
3. The target must append a PEC byte to the end of the active phase payload response (just as it does for the discovery payload).
4. If the initiator calculates a bad PEC on the target response message, the initiator must drop the response. Further handling, such as retries, is implementation specific.

For commands such as CMD=00h for get discovery payload, PEC is optional, since the 1-byte command has parity.

#### **Example Supported Commands with and without PEC**

01h = Put 1 Virtual Wire Byte, Get 1 Virtual Wire Byte

81h = Put 1 Virtual Wire Byte with PEC, Get 1 Virtual Wire Byte with PEC

### **12.3.3.13 PESTI Target Reset and Fault Handling** §

If the target resets during the discovery phase prior to the entry to the active phase (e.g., APEN=0), the DSTAT value would revert to 11b (Payload not received.) The initiator would autonomously send the payload request command if DPEN=1. The target component would not be transitioned to the active phase until DSTAT=10b. If discovery had not previously completed or DPEN=0, the component would not be discovered and would not transition to the active phase.

If the target resets during the active phase, it must be observed as temporary unresponsiveness at the initiator. This would be reflected in the ACTIVE\_PHASE\_ERROR (APERR) register bit if any RX timeout or parity error occurred. Following target reset, a BREAK assertion and release would be observed by the initiator. During the active phase (DSTAT[1:0]=10b & APEN=1), DSTAT must be locked. Locking the discovery status and payload data is required so that FW can safely consume the contents at any warm or cold reset. Since discovery has already occurred, the initiator would resume sending the virtual wire exchange command and the target would not observe a discovery request command. Once a component has transitioned to the active phase, the only method to unlock the discovery payload and status is by SMFW clearing, then setting DPEN.

### **12.3.3.14 PESTI Fan-Out** §

Applications where the ability to fan-out a PESTI bus to multiple targets exist. Since the number N is not a priori known by the motherboard, and pre-plumbing for a maximum quantity requires additional interconnects, PESTI fan-out support helps scale the ability to support PESTI type features.

This specification version includes the scope of fanout from between 2 to 8 downstream buses. Although out of scope, nesting of multiple tiers of fan-out within a single hierarchy is possible with additional command codes and circuitry. In all fanout cases, it is left to the designer to understand the latency effects of fanout width (and depth). Two methods are shown where the MUX method is for typical fanout needs and the Switch method is targeted for applications that require the ability to broadcast commands simultaneously with all targets.

Two fanout methods of operation include the MUX and Switch methods (see circuits below):

- MUX method: Typical 1-to-many fanout. Broadcast commands are not supported.
- Switch method: For applications requiring broadcast commands to all attached targets.

Two modes of operation for a fanout controller include Target Mode and Snoop Mode:

- Target Mode (Default):
  - Controller acts as a target supporting discovery and active phases.
  - Only the fanout controller is permitted to be attached to the initiator bus.
    - MUX Method: No channels are selected.
    - Switch Method: Fanout controller on Ch0 is always enabled; all others default disabled.
  - Must support =1 status command for the initiator reading information from the fanout controlling PESTI target such as:
    - The current MUX/switch settings
    - If an issue was observed such as a closed switch bus hang watchdog timeout
    - Live status of downstream subsegments (when the MUX or switch is open for a particular subsegment)
- Snoop Mode:
  - Fanout controller must enter snoop mode any time any other target(s) are attached to a bus.
  - Is permitted to process broadcast commands if application relevant.
  - Listen/Process only special fanout control commands (MUX Select or Switch channel enable), thereby ignoring discovery and target mode active phase commands intended for other targets.

Example MUX Select commands are permitted to include:

- Go to Target Mode and de-select all subsegments.
- Select specific subsegment(s) as connected, which can be comprised of 1, all or select groups.

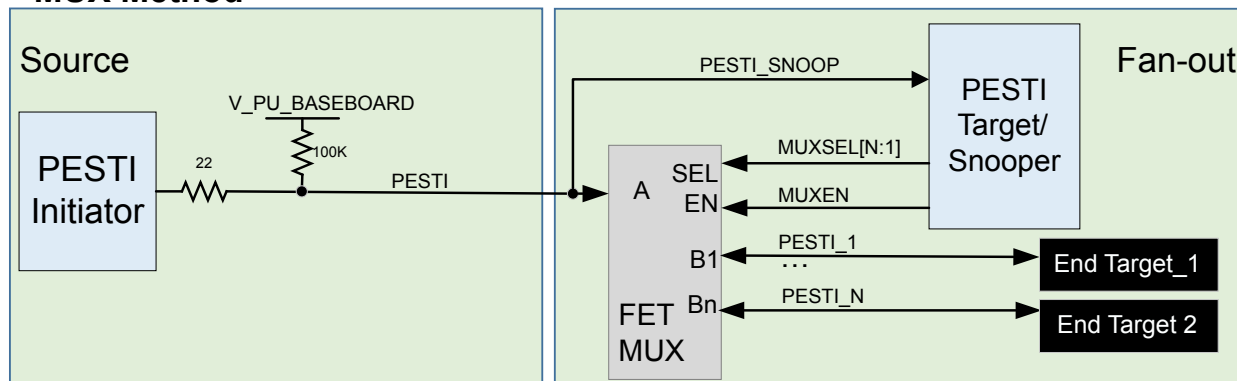
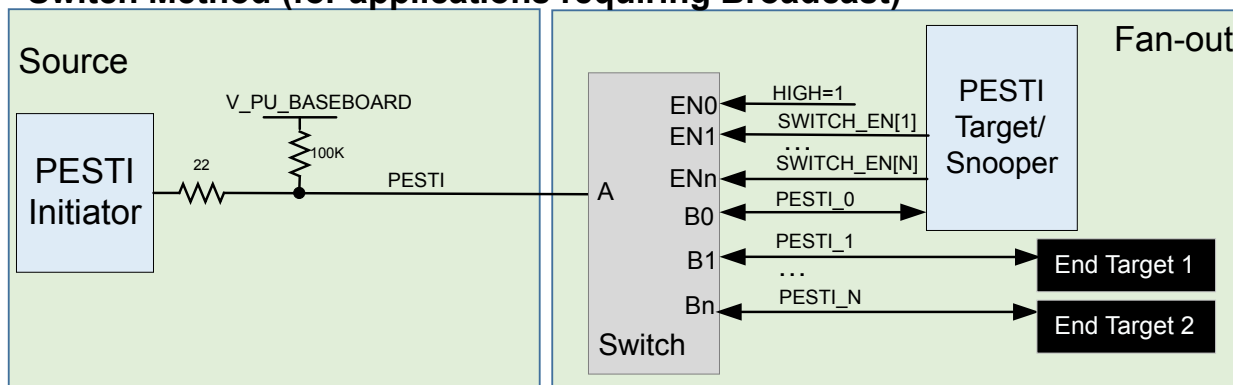
**MUX Method****Switch Method (for applications requiring Broadcast)**

Figure 12-9 PESTI Fan-out Methods §

Buffer(s) might be necessary if the Target/Snooper and the FET MUX or Switch have significant layout stubs. The designer is permitted to choose a switch with active buffers versus a FET switch to avoid reflection effects.

Since any downstream interface is permitted to have a strap (such as a strong presence pulldown resistor) or be in a stuck state driving high or low, it is imperative to not create a deadlock condition where the Fanout controller cannot observe new commands to change the MUX or switch settings. Therefore, it is required to have the fanout controller implement a watchdog timer of at least  $t_{MSC\_WDT}$  after closing one or more channels to oversee any signaling activity. If no such activity (rising or falling edges), the PESTI snooper must open the selected channel(s) after the timeout.

An alternative option is for the fanout controller to always be able to observe and report upstream the state of any downstream sub-channel so that the PESTI initiator can choose to not close a suspect channel.

**MUX Switch Control (MSC) Command Structure**

Table 12-9 MSC\_CTRL\_VAL (Initiator to PESTI Snooper Target) Data Byte Value = 02h §

|        |      |                                   |               |                |   |        |   |
|--------|------|-----------------------------------|---------------|----------------|---|--------|---|
| 7      | 6    | 5                                 | 4             | 3              | 2 | 1      | 0 |
| RSVD   | RSVD | RSVD                              | DS_ALL_CH_SEL | DS_CH_SEL[2:0] |   | ENABLE |   |
| ENABLE |      | Active high (1=Enable, 0=Disable) |               |                |   |        |   |

| 7                                                                                                                                                                  | 6 | 5                                                                                                                                                                                                                      | 4 | 3 | 2 | 1 | 0 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| When the mux or switch is disabled, only the snoopers target can receive communication from the initiator. All downstream MUX or switch channels must be disabled. |   |                                                                                                                                                                                                                        |   |   |   |   |   |
| <b>DS_CH_SEL[2:0]</b>                                                                                                                                              |   | Hex representation of downstream channel select (0...7 decimal). This bitfield is only used if ENABLE = 1b and DS_ALL_CH_SEL = 0b.                                                                                     |   |   |   |   |   |
| <b>DS_ALL_CH_SEL</b>                                                                                                                                               |   | When set to 1b, then all downstream channels are connected. Used for broadcast commands only. This bitfield is only valid if ENABLE = 1b. When this bitfield is set to 1b then DS_CH_SEL bitfield must be set to 000b. |   |   |   |   |   |
| <b>RSVD</b>                                                                                                                                                        |   | Reserved for Future Use.                                                                                                                                                                                               |   |   |   |   |   |

Table 12-10 MSC\_STAT\_VAL (PESTI Snooper Target to Initiator) §

| 7              | 6 | 5                        | 4 | 3 | 2 | 1 | 0       |
|----------------|---|--------------------------|---|---|---|---|---------|
| RSVD           |   |                          |   |   |   |   | NOT_ACK |
| <b>NOT_ACK</b> |   | Not Acknowledge          |   |   |   |   |         |
|                |   | <b>0b</b> Success        |   |   |   |   |         |
|                |   | <b>1b</b> Error          |   |   |   |   |         |
| <b>RSVD</b>    |   | Reserved for Future Use. |   |   |   |   |         |

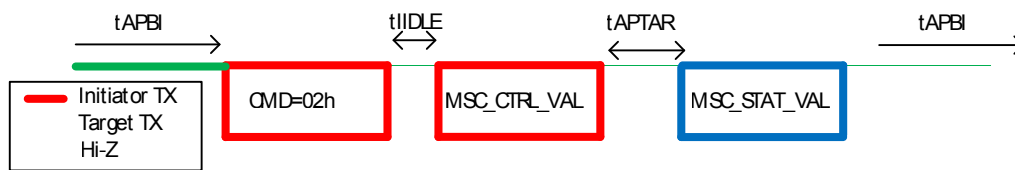


Figure 12-10 PESTI Mux Switch Control Format §

**Fanout Rules:**

1. PESTI snoopers must not be nested. Only a single fanout level is supported.
2. A PESTI snooper fanout target must default to all downstream PESTI channels disabled during the discovery phase.
3. A PESTI snooper must not respond to a discovery payload request command = 00h when any downstream channel is selected.
4. A PESTI snooper must not respond to any virtual wire exchange using command 01h when any downstream channel is selected.
5. Except for when broadcast commands are supported and transmitted, only a single downstream channel of a Switch can be selected at any given time.
6. Non-snooper targets must ignore argument(s) following the MSCMSC command = 02h.
7. On any PESTI that a fanout switch is present, the initiator must begin any broadcast with the following sequence:

- Broadcast start value = FFh
  - Wait for  $t_{\text{BPAUSE}}$  (Broadcast Pause)
  - Resume with typical broadcast sequence of FFh (repeated) followed by the broadcasted command value(s).
8. A fanout switch snooper must enable all downstream channels at any start of a broadcast indicated by 0xFF as the first, or repeated, data byte of a sequence.
  9. Once the broadcast sequence is complete, the initiator must utilize the MSC command to select a single downstream channel of any switch or MUX prior to resuming typical communication.

### 12.3.3.15 PESTI Security Considerations §

Although PESTI is a basic, low-level data communication channel the following threats are identified with possible mitigations, although the specific mitigations are implementation specific.

#### Threats

1. Physical implant/signal re-routing: Assurance, where applicable, that the PESTI target is on the same HW (or same target) as other management interfaces. Security Protocols and Data Model (SPDM) types of security capabilities are the likely current method used to address this threat.
2. Physical implant, man-in-the-middle snooping or alteration of payloads or virtual wires in flight. Potentially mitigated with encrypted payloads most likely using SPDM defined methods.

## 12.4 Managed USB 2.0 §

Numerous use cases exist for USB 2.0 as a plug-and-play, out of band management interface, such as:

- Adapter or component firmware update, telemetry, debug, and security operations such as attestation, recovery, and measurement.
- MCTP direct to an adapter or component.
- TCP/IP or Network Controller-Sideband Interface (NC-SI) traffic.
- Bridge to common low-level interfaces such as GPIOs, UARTs, SPI, I3C and JTAG.

#### USB Rules:

1. The USB Host must be on the platform side of the form factor interface.
2. USB 2.0 high speed and full speed modes are supported.
3. USB 2.0 low speed and USB 1.1 mode devices are prohibited when directly connected to the form factor interface but are allowed when attached to a downstream port of a device side USB hub.
4. USB 2.0 low speed is prohibited due to USB 2.0's support of larger packet sizes as well as common handling of Flex I/O default terminations and D+ pullup-based attach detection (e.g., CEM JTAG vs USB Flex I/O alternate function rules).
5. USB 2.0 voltage through form factor interfaces must be 3.3 V to support standard discrete components such as bridges and hubs and to provide maximum signaling margin for long buses. Voltage translation when connecting USB 2.0 to components that do not support 3.3 V is implementation specific.



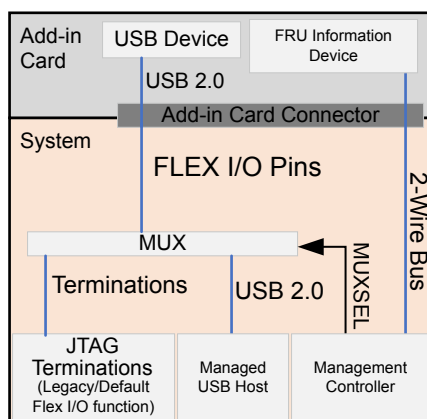


Figure 12-11 [CEM] form factor example circuit for repurposing legacy JTAG to USB 2.0 mode

## 12.5 2-Wire Interface §

[I2C] and [SMBus] are long standing and useful bus interfaces that are expected to be fully supported for the foreseeable future. Form factor specifications dictate the electrical particulars, such as the reference voltage, power domain, peripheral capacitive loading and define such an interface as being required, optional or unsupported. Although the SMBUS specification is stable in terms of revisions, especially at the electrical level, it is always encouraged that form factor specifications attempt to reference the latest versions of the [SMBus] specification when possible. [SMBus-3.2] version 3.2 is the preferred minimum version at the time of this writing due to support for the Default Target Address feature, while using inclusive terminology. The remainder of this specification refers to [SMBus] and not [I2C]. However, [I2C] compliant components that are not [SMBus] compliant, such as an EEPROM or MUX, are permitted at the designer's discretion.

MIPI [I3C-Basic] is an additional protocol option that shares the same physical 2-Wire interface as the [SMBus] interface. This specification defines a backward-compatible method of discovering and enabling the optional [I3C-Basic] mode on the existing 2-Wire interface. The [I3C-Basic] architecture includes an optional intermediary component known as 2-Wire Hub (see § Section 12.5.3.3) to addresses specific [SMBus] challenges related to asynchronous communications. [I3C-Basic] defines the protocol, including its electrical characteristics. The recommended PCI Architecture to allow interoperability with legacy adapters is for the platform to use the 2-Wire Hub which provides a method for a transition between [I2C] / [SMBus] and I3C. Legacy [SMBus] and [I3C-Basic] have different electrical requirements which needs to be considered in the platform design. The [I3C-Basic] provides a common electrical requirement definition across all PCIe form-factor connectors (interfaces).

## Common Interface and Signal Names

Historically, the [SMBus] interface and signal names were used in various form factor specifications. With the addition of optional I3C interface on the same pins, the following generic terms should be utilized, when referring to either [SMBus] or I3C:

1. “2-wire interface”
2. Signal names “SCL” for the serial clock wire and “SDA” for the serial data wire

### 12.5.1 2-Wire Interface Use Cases §

Throughout this section, the generic term 2-Wire interface references either [SMBus], [I2C], or [I3C-Basic].

*Table 12-11 2-Wire Interface Example Usages §*

| Functionality                            | Typical Applications                                                                                                       |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Field Replaceable Unit (FRU) Information | Discovery of Hardware inventory, capabilities such as bus topologies, features supported, etc.                             |
| Security                                 | Attestation of HW/FW, Root of Trust for Measurement, Root of Trust for Update, Root of Trust for Recovery, Data Encryption |
| Configuration / Controls                 | Flex I/O, Link subdivision                                                                                                 |
| Update                                   | Peripheral Subsystem Firmware                                                                                              |
| Health                                   | Error Causes, Crash Dump Logs, Temperature, External Link States                                                           |
| Telemetry                                | Power, Throughput                                                                                                          |

### 12.5.2 2-Wire Addressing §

To ease topology and component discovery operations, § Table 12-12 lists common SMBus default target addresses recommended to be used by all PCI Express-based form factors, when such component exist. Components supporting both SMBus and I3C modes must stop responding to their original SMBus address(es) upon transitioning to I3C mode, as described in § Section 12.5.4.2 .

§ Table 12-12 identifies the recommended addresses for specific SMBus usages. Note that a card carrier is defined as one with a connector edge based on a form factor that includes connector slots based on one or more instances of the same or different form factor, sometimes referred to as passenger slots. An example is a [CEM] card carrier containing one or more M.2 form factor slots.

#### IMPLEMENTATION NOTE: RECOMMENDED 2-WIRE INTERFACE ADDRESSES §

All implementations are recommended to use the addresses in § Table 12-12 for the listed usages to ease discovery. For all usages, any address other than the ones listed in § Table 12-12 are permitted.

Platform makers should avoid these listed addresses on the same subsegments.

For maximum backward compatibility, platform makers should not include any component addresses on the same (sub)-segment that connects to the form factor.

The addresses shown have synergy with other industry standards such as [NVM-Express].

Table 12-12 Baseline SMBus Recommended Default Target Addresses §

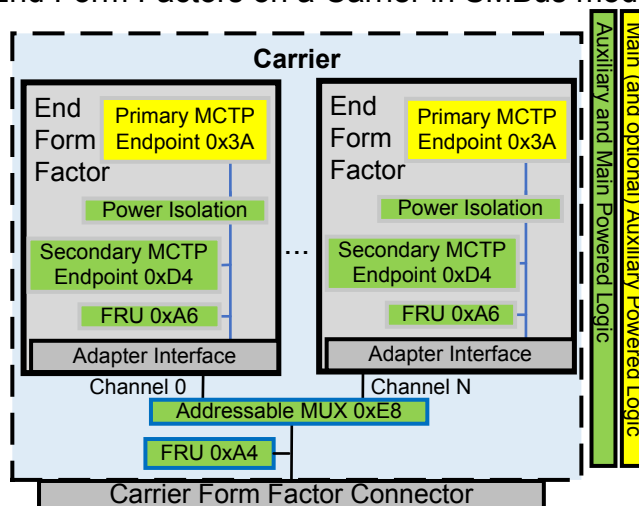
| Default Target Address: Binary with x indicating R/W<br>(Hex 8-bit/Hex 7-bit) | Recommended Usages                                                                                |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1010 010xb (A4h/52h)                                                          | Recommended for FRU Information Device on a card carrier <sup>1</sup>                             |
| 1010 011xb (A6h/53h)                                                          | Recommended for FRU Information Device on a PCI Express-based adapter or component <sup>1</sup> . |
| 0011 101xb (3Ah/1Dh)                                                          | Recommended for Primary MCTP-compliant Management Target <sup>2</sup>                             |
| 1101 010xb (D4h/6Ah)                                                          | Recommended for Secondary MCTP-compliant Management Target <sup>3</sup>                           |
| 1110 100xb (E8h/74h)                                                          | Recommended for Addressable SMBus MUX, if applicable                                              |
| 1110 000xb (E0h/70h)                                                          | Recommended for 2-Wire Hub <sup>4</sup>                                                           |

## Notes:

1. The FRU Information Device is a typically an optional element. Form factor specified features that need out-of-band discovery descriptors must make the FRU Information Device required and available in the appropriate earliest discoverable power state (auxiliary and/or main power).
2. Primary MCTP target: This target must operate when main power is applied. This interface is permitted to operate when only auxiliary power is applied.
3. Secondary MCTP target: This target is permitted to operate when auxiliary and/or main power is applied.
4. 2-wire Hubs must default to the static address listed and are then permitted to be changed via register programming or via [\[I3C-Basic\]](#) Dynamic Addressing.

Interposer discovery details are left to the implementer, with designer care needed to avoid address conflicts if the 2-Wire interface extends into standard form factors.

End Form Factors on a Carrier in SMBus mode

Figure 12-12 Example of 2-wire, 8-bit addressing for a card carrier with  $N$  end form factors in SMBus mode §

### 12.5.3 2-Wire Bus Sharing §

The subsections describe 2-Wire interface sharing options for fanout on an adapter or component or between form factor instances.

#### 12.5.3.1 2-Wire Multi-Drop Topology §

A multi-drop topology with a 2-Wire interface is permitted. In platform topologies utilizing a multi-drop 2-Wire interface, unique addresses must be assigned:

- Either statically (e.g., using a unique slot identifier control method) or
- Dynamically using SMBus Address Resolution Protocol (ARP) or [I3C-Basic]'s Dynamic Addressing.

Caution should be taken as the use of multi-drop 2-Wire topology exposes the potential for malicious adapters or components to initiate traffic to other adapters or components on the same subsegment.

#### 12.5.3.2 SMBus MUX Use §

An addressable SMBus MUX is defined as a bidirectional fan-out multiplexer with one upstream channel and one or more downstream channels configured by a 2-Wire interface command (not discrete mux select signals) from a Management Controller to select zero or exactly one downstream channel to the upstream channel.

The use of addressable switches, which allow for connecting multiple downstream subsegments to the upstream segment at one time is not recommended, for reasons that include signal integrity, security and addressing complexities.

## IMPLEMENTATION NOTE:

### SECURITY CONCERNS WITH SMBUS MUXES §

Platform implementations utilizing 2-Wire switches/MUXes are recommended to mitigate against security risks of errant or malicious adapters or components that could initiate peer-to-peer traffic to other adapters, components, or other platform adapters or components on the same 2-Wire segment.

Platform designers are recommended to consider that switches/MUXes interrupt the continuity of the shared bus breaking the arbitration and acknowledgment mechanisms of SMBus. This creates major challenges for implementations utilizing multi-initiator protocols, such as MCTP, especially when:

- MCTP messages initiated by downstream components are spread over long periods of time (e.g., as per [PLDM-Firmware-Update]), or
- asynchronous communication happens (e.g., asynchronous event or alert reporting as [PLDM-Platform-Monitoring-Control]), or
- response times may be difficult to predict, or response messages delayed (e.g., due to complex computations, such as cryptographic ones required for the SPDm attestation flows defined in [SPDM]).

Additional challenges in the above scenarios exist when switches/MUXes interrupt the continuity of the shared bus and:

1. the platform uses MCTP bridging (management controller is not the source of the MCTP communication and thus has more difficulty to predict the MCTP request/response patterns) or
2. downstream port enabling/disabling is not synchronized with SMBus transactions (e.g., truncating the MCTP packets).

Platforms utilizing MCTP protocol in such scenarios are recommended to use a 2-Wire Hub component instead (see the next section). Switching to [I3C-Basic] mode may be another alternative to address such challenges but only possible if all downstream components support [I3C-Basic] (I3C is a single-initiator protocol even when running MCTP and it also allows explicit control of I3C in-band interrupts).

#### 12.5.3.3 2-Wire Hub Use §

It may be beneficial to use a 2-Wire Hub component for reasons including:

- a. Bus capacitance isolation.
- b. Fanout of I3C and SMBus components with I3C in-band interrupt aggregation across all downstream channels.
- c. Voltage translation and power domain isolation.
- d. Using offload capabilities such as SMBus agents to send and receive asynchronous SMBus messages, such as those utilized by MCTP.

An example use of such a 2-Wire Hub is presented in § Figure 12-13. This architecture is recommended to address the interoperability challenges existing with simple MUX components, as described in § Section 12.5.3.2. SMBus endpoints can initiate asynchronous messages even if they do not support I3C mode of operation. I3C mode, which is typically available with such 2-Wire Hubs, allows protocol and voltage transition to enable coexistence of SMBus and I3C components.

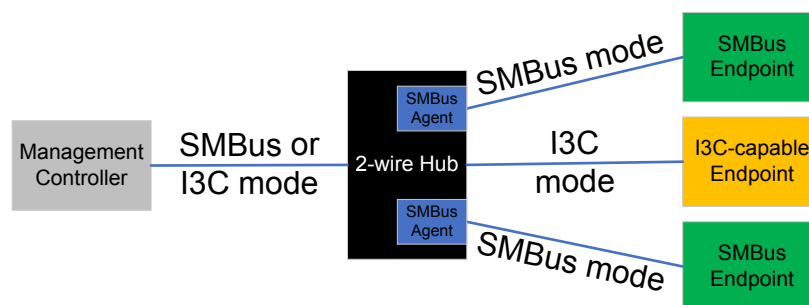


Figure 12-13 Example of 2-Wire Hub Use §

2-Wire Hub components should be discovered using the I3C DCR value of 0xC2. See the allocation at [I3C-DCR].

It should be recognized by implementers and form factor specifications that the high signaling rates drive for subsegment capacitance that is significantly lower than the SMBUS specification. This can limit the achievable bus speed or require utilizing intermediary logic such as I3C hubs when considering entire channel characteristics.

## 12.5.4 [I3C-Basic] Support on Existing SMBus Signals §

Most PCI Express-based platforms have supported [SMBus] as an optional 2-Wire interface (i.e., Serial Clock and Serial Data) for adapter or component management purposes. This section defines [I3C-Basic] as an additional option where the [I3C-Basic] interface shares the same 2 wires as the SMBus interface.

While [SMBus] was defined with different electrical requirements for different form-factors, [I3C-Basic] is defined with one interoperable set of behaviors for PCI Express. This section defines the behaviors that:

- Determine which interface is active
- Define the electrical requirements for PCI Express [I3C-Basic]

### 12.5.4.1 I3C Basic Overview §

Key attributes of [I3C-Basic] support on existing [SMBus] signals are the following:

- Supports the *MIPI™ Alliance Specification for I3C Basic, Version 1.0* available at [I3C-Basic] (See Implementation Note Optional I3C Features).
- Sideband 2-Wire bus reliant use cases should be defined using either [SMBus] only or both [SMBus] and [I3C-Basic].
- If optional I3C Basic is supported, then I3C Data shares the same signal as SMBus Data, and I3C Clock share the same signal as SMBus Clock.
- If optional I3C Basic is supported, SMBus voltage levels must also be tolerated for backwards compatibility.
- I3C operating voltage must utilize 1.8 V signaling as defined in [I3C-Basic]. The rationale is for maximum bus length and allowed capacitance which eases implementations in large platforms.

## IMPLEMENTATION NOTE: OPTIONAL I3C FEATURES §

The following I3C features from [I3C-Basic-1.1.1] should be supported if I3C is supported:

- The Target Reset pattern (RSTACT) with Common Command Code 0x2A (Broadcast) and 0x9A (Direct).
  - This allows the host to perform a I3C peripheral reset without having to go back to SMBus mode.
- Asynchronous Timing Control (Mode 0) (support determined if GETXTIME CCC is acknowledged)
  - This aids in scenarios needing real time communication by providing a timestamp.
- Grouped addressing (support determined through GETCAPS CCC)
  - This allows broadcast request to multiple I3C Basic components using a single message.
- SETBUSCON CCC
  - [I3C-Basic] recommends using SETBUSCON to inform I3C Targets of the specific use of the Bus, and that vendor specific CCCs will be used. Can also inform I3C Targets of the version of the specification that the Controller supports.

## IMPLEMENTATION NOTE: POTENTIAL ELECTRICAL CONTENTION §

Due to the nature of [I3C-Basic] utilizing push-pull active drivers, implementers should be cautious about possible corner case scenarios where electrical contention could occur (e.g., an I3C component is actively driving data high while the initiator goes through a reset and drives data open drain low as part of a new command). Actions such as the initiator beginning a recovery sequence with an SMBus clock low timeout to reset the target is advised.

### 12.5.4.2 I3C Basic Discovery and Mode Changing §

The goal of the I3C Basic adapter or component discovery flow as shown in § Figure 12-14 is to enable an I3C Basic capable host and I3C Basic capable components to establish I3C Basic communication while allowing backward compatibility with legacy [SMBus] components. If one side of communication supports both [SMBus] and I3C Basic and the other side supports SMBus only, then SMBus protocol and voltage is used. If there is a mix of I3C Basic and SMBus components that are active on the same bus, then only [SMBus] protocol and voltage is used. Mixed Fast Mode is prohibited. This flow should be initiated after any power rail state change.

The discovery flow uses reserved address 1111 110xb (FCh 8-bit/7Eh 7-bit) to determine if there are components that support I3C Basic on the bus. Address 7Eh (7-bit) is reserved in the [SMBus] specification and cannot be assigned to any SMBus component. Address 7Eh (7-bit) is defined for I3C Basic and every I3C Basic component will respond as per the I3C Basic Specification. Any time a component that supports both SMBus and I3C Basic receives a cycle for address 7Eh (7-bit), it should disable SMBus (not drive or stretch the clock, nor ACK SMBus addresses) until reset to help the host determine if there are SMBus only components on the bus. If only I3C Basic components are detected and the host chooses to use I3C Basic, then the component requires a transition time from 3.3 V to the I3C Basic voltage within the time  $T_{\text{smb2i3c}}$  (see § Table 12-14). A reset by driving a system management hardware mechanism (e.g., a hardware reset or power removal) if supported by a given formfactor or driving the clock low for a specified period reverts the interface

to SMBus at 3.3 V signaling within the time  $T_{i3c2smb}$  (see § Table 12-14). An I3C Basic Target Reset issued by the host resets the I3C Basic interface but does not impact the signaling voltage. See § Figure 12-14 for more details of this flow.



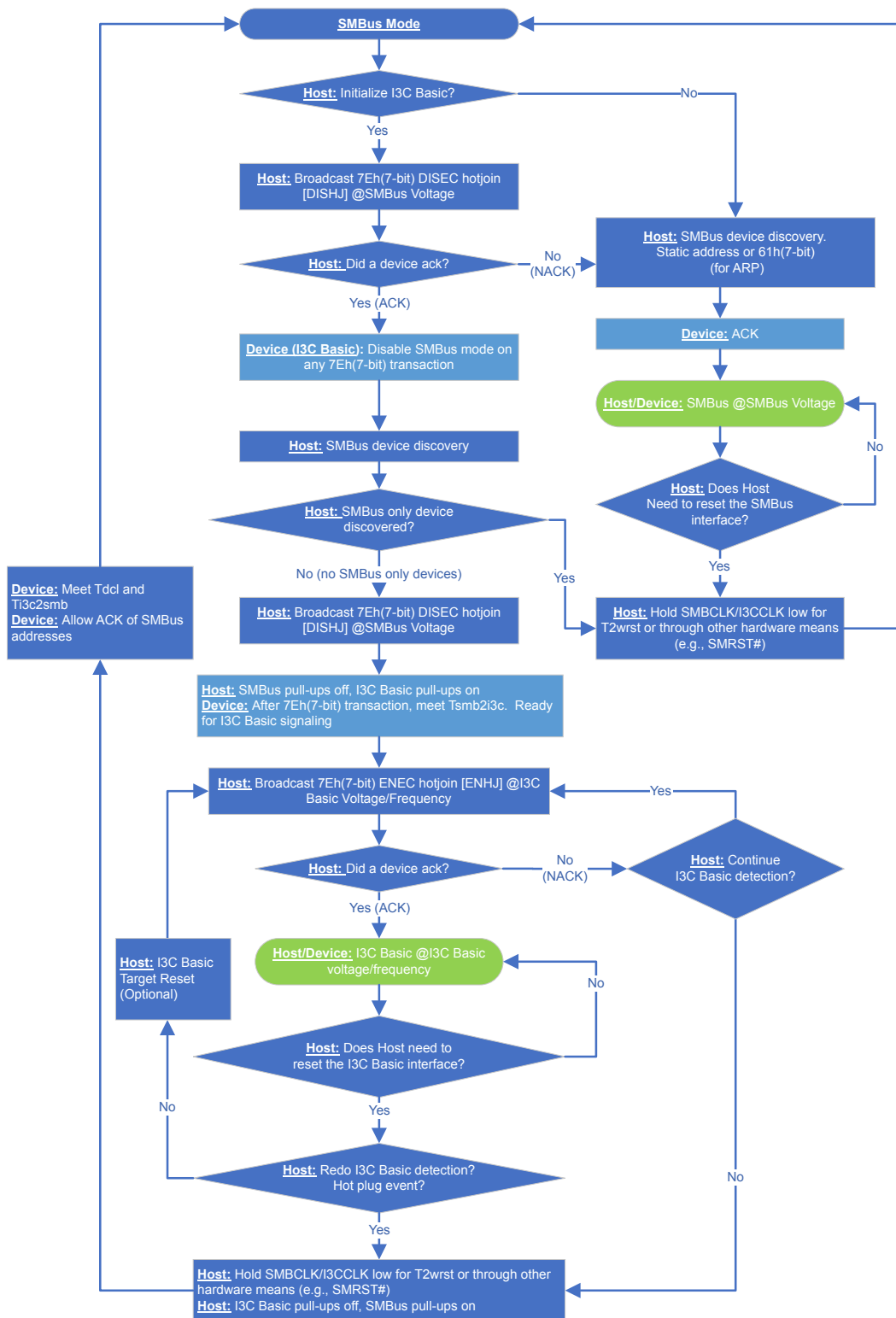


Figure 12-14 SMBus to I3C Transition Flow 5

### 12.5.4.3 I3C Basic DC and AC Signal Requirements §

The I3C Basic operating voltage and capacitance for SCL and SDA is defined in § Table 12-13.

For more information on logic levels or bus timings, refer to [I3C-Basic].

*Table 12-13 I3C Basic Logic Signaling DC Specification §*

| Symbol      | Parameter                                               | Min  | Nominal | Max  | Unit | Notes |
|-------------|---------------------------------------------------------|------|---------|------|------|-------|
| $V_{ddi3c}$ | I3C Basic Operating Voltage                             | 1.65 | 1.80    | 1.95 | V    |       |
| $C_{i3c}$   | Total adapter and component capacitance for I3C support |      |         | 20   | pF   | 1     |

Notes:

1. Total capacitance the platform board will see from the adapter including all connector, board routing, component package wires, and component on die parasitic effects when the target component(s) is/are in I3C Basic mode. The capacitance of SMBus components on the I3C Basic interface even if not operational is also included.

§ Table 12-14 defines the component maximum transition times between [SMBus] and I3C. During  $T_{smb2i3c}$ , the host will transition the pull-ups from  $V_{ddsmb}$  to  $V_{ddi3c}$  and the component will transition to  $V_{ddi3c}$  based signaling. During  $T_{i3c2smb}$ , the host will transition the pull-ups from  $V_{ddi3c}$  to  $V_{ddsmb}$  and the component will transition to  $V_{ddsmb}$  based signaling.

*Table 12-14 I3C Timing Requirements §*

| Symbol        | Parameter                                         | Min | Max | Unit | Notes |
|---------------|---------------------------------------------------|-----|-----|------|-------|
| $T_{smb2i3c}$ | Component transition time from SMBus to I3C Basic |     | 20  | ms   | 1, 2  |
| $T_{dcl}$     | Component Clock low reset time                    | 25  | 35  | ms   | 2, 3  |
| $T_{i3c2smb}$ | Component transition time from I3C Basic to SMBus |     | 20  | ms   | 2, 3  |
| $T_{2wrst}$   | Host clock low timeout                            | 50  |     | ms   |       |

Notes:

1. The host must wait a minimum of  $T_{smb2i3c}$  before sending I3C commands.
2. No SMBus or I3C transmissions must occur during  $T_{smb2i3c}$ ,  $T_{dcl}$ , and  $T_{i3c2smb}$ .
3. The host must wait a minimum of  $T_{dcl} + T_{i3c2smb}$  before sending SMBus commands.

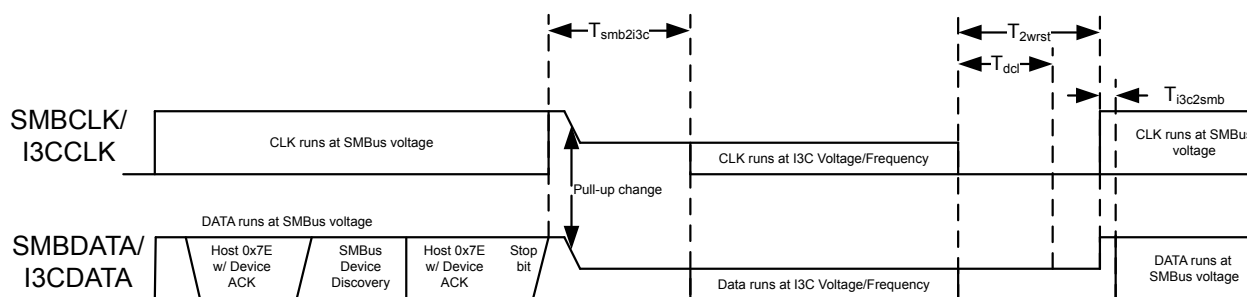


Figure 12-15 Component Timing Diagram for Transition to I3C Signaling Voltage §

## IMPLEMENTATION NOTE: RATIONALE FOR T<sub>2WRST</sub> §

MIPI [I3C] reset mechanisms provide useful granularity. That specification's sections titled "Controller Stuck SDA Handling" and "Controller Recovery After a Crash or Unexpected Reset" describe various typically infrequent situational, diagnosis, and recovery methods. These methods provide numerous stimuli with various responses, inclusive of reducing drive strength to limit effects of possible electrical contention via a target actively driving SDA bus high or low.

Since:

1. The recovery methods are cumbersome.
2. A crash or unexpected reset of a controller may result in a lack of awareness of any target device's current mode of operation.
3. Any chance of causing electrical contention is not preferred.

Then, holding SCL low for the  $T_{2wrst}$  minimum duration (which may start at any point in a transaction) is the additional required recovery method, which makes the solution:

1. Easy to implement in software/firmware.
2. Easy to observe and debug on a logical bus trace.
3. Removes all concerns over the necessary electrical tuning and/or selection of controller and/or devices with specific drive strength granularity.

## IMPLEMENTATION NOTE: DEVICE HOT PLUG WITH MULTI-DROP 2-WIRE BUS CONSIDERATIONS §

This specification explicitly prohibits running in mixed mode, which means at least one device runs in SMBus mode while other(s) run in I3C mode. If a system implements a multi-drop 2-Wire bus between multiple hot pluggable devices, then:

- a. Hot removal of a target is no issue, where discovery is outside the scope of this specification.
- b. An end form factor target that is inserted, internally reset or power cycled is required to start in SMBus mode with the respective form factor native voltage. If a multi-drop bus exists and is in I3C mode at the time of new device insertion, reset or power cycle, then the I3C mode 1.8 V voltage does not cause stress to the new target. Unexpected bus behavior may occur if the newly added, reset or power cycled device(s) apply internal SMBus pullup resistors or do not support the requisite glitch filter to ignore I3C traffic. How the system detects new device insertion and knows to send  $T_{2wrst}$  to force all targets back to SMBus mode is outside the scope of this specification.

## 12.6 Field Replacement Unit (FRU) Information §

FRU Information is data that describes an adapter or component. FRU Information includes data about the adapter or component such as:

- Inventory information (e.g., serial number, model number, etc.).
- Capability information (e.g., power consumption, performance, etc.).
- Information required by the host to configure the adapter or component (e.g., supported connector subdivision combinations, clocking modes supported, etc.).

FRU Information may be made available to a Baseboard Management Controller or service processor using the following mechanisms:

- Over the 2-Wire interface with use of the protocol defined in § Section 12.6.1 .
- Over 2-Wire interface, USB, or PCIe VDMs with use of the MCTP protocol.

In-band access of FRU Information via a method such as Management Message Passthrough (MMPT) or *Memory-Mapped BMC Interface (MMBI)* [MMBI] is outside the scope of this chapter.

### 12.6.1 FRU Information Device Requirements §

FRU Information is contained in a FRU Information Device. The following list contains requirements and recommendations for FRU Information Devices regardless of the operating mode being [SMBus]/[I2C] or [I3C]:

1. A 2-Wire discoverable Field Replaceable Unit (FRU) Information Device must be implemented as a required element when dictated by form-factor specifications such as to support specific features.

2. The FRU Information Device is permitted to be any implementation that meets the requirements in this specification (e.g., a physical integrated circuit or virtually emulated by firmware).
3. The FRU Information Device is recommended to be protected from write operations that may corrupt or replace FRU Information. The exception is when the FRU Information is accessed via MCTP or while in a manufacturing mode. An adapter or component is permitted to update the FRU Information in conjunction with a firmware update.
4. The FRU Information Device must support at least 8 full writes.
5. The FRU Information Device must be accessible in any of the following power states:
  - a. Main power only.
  - b. Main power and auxiliary power if auxiliary power is supported.
  - c. Auxiliary power only if auxiliary power is supported.
6. If auxiliary power is supplied, then the availability and transitions of main power must not disrupt FRU Information Device access.
7. The FRU Information Device must be accessible on the 2-Wire interface within 1 second after power becomes valid and must remain accessible while power is valid unless otherwise specified.
8. Clock-low timeout-based reset recovery must be supported per the [SMBus] Specification or [I3C-Basic] section of this specification. The FRU Information Device must be accessible on the 2-Wire bus within 1 second after the de-assertion of clock-low timeout-based reset. It is recommended to reset the current address pointer due to clock-low reset.

#### **12.6.1.1 FRU Information Device Requirements Specific to SMBus/I2C Mode §**

This section describes the FRU Information Device requirements and recommendations when using SMBus/I2C read and write transactions.

1. The FRU Information Device must use a 7-bit target address.
2. FRU Information Device capacity must be an even power of 2 in the range of 256 Bytes up to 64 KB.
3. The FRU Information Device must support at least 400 kHz operation in SMBus/I2C mode. The adapter or component should limit bus capacitance. The appropriately tuned pullup resistors are the platform's responsibility. The adapter or component should have no or weak pullup resistors only.
4. FRU Information Device accesses must use 2-byte addresses for read or write offsets. One-byte offsets supplied by the platform during random reads should be zero extended to two-byte offsets. See § Figure 12-17.
5. The FRU Information Device must maintain a current address pointer. The current address pointer must auto increment to the next byte after each byte that is read. Refer to the “Current Address Read” and “Current Sequential Read” in § Figure 12-17. A read of the last byte of the last memory page of the FRU Information Device must cause the current read pointer to roll over to the first byte of the first memory page of the FRU Information Device. For example, for a 256-byte FRU Information Device, a read of the last byte of the FRU Information Device at offset FFh causes the current address pointer to roll over to 00h. The current address pointer must be cleared to 0h upon a power cycle of the FRU Information Device.
6. Any controller on the adapter or component must not initiate traffic to the FRU Information Device on the same adapter or component unless the FRU Information Device is hidden from host view (e.g., by detaching the FRU Information Device from the system-facing 2-Wire interface). For example, the FRU Information Device must be hidden from host view during a firmware update that includes new FRU Information. While the FRU Information Device is disconnected from the form-factor connector, accesses to the FRU Information Device via the 2-Wire bus must be Not Acknowledged.

7. If the FRU Information over the 2-Wire interface is initially provided using a discrete SMBus/I2C FRU Information Device (e.g., a physical EEPROM), the component must be detached from the 2-Wire interface when entering I3C mode. If the FRU Information Device is being emulated by a programmable component, such as a microcontroller, then when switching to I3C mode, the virtual FRU Information Device SMBus address must be logically disabled. As mixed fast mode is not supported, designs must not assume that the I3C bus frequency and short SCL high time allows for SMBus/I2C devices with 50 ns glitch filters to coexist.
8. Placing the FRU Information Device behind a 2-Wire MUX or hub is prohibited due to possible variability in discovery method.

### 12.6.1.2 [SMBus]/[I2C] Access Protocol §

The following figures describe the proper logical sequence of supported FRU Information Device read and write commands when in SMBus/I2C mode. Note that the address example is specific to 4 KB component.

#### Byte Write



#### Page Write

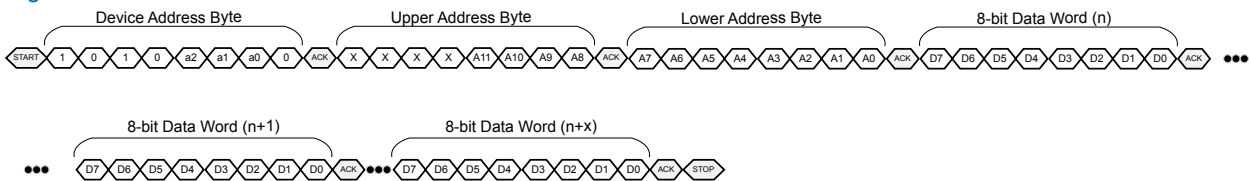
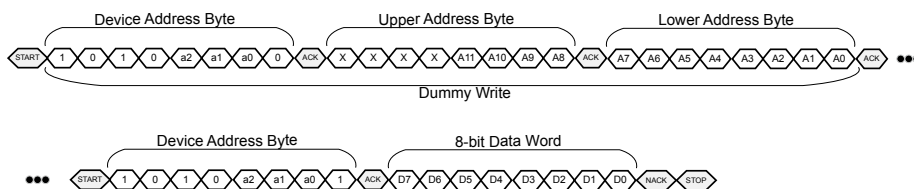


Figure 12-16 SMBus/I2C-based FRU Information Device Writes with Two-Byte Addressing §

#### Random Read



#### Current Address Read



#### Current Sequential Read



Figure 12-17 FRU Information Device Reads with Two-Byte Addressing §

## 12.6.2 FRU Information Format §

This section defines the format of the FRU Information stored in the FRU Information Device. The FRU Information format must comply with the requirements defined in the IPMI Platform Management FRU Information Storage Definition ([IPMI-FRU]).

In addition to CRC-based data integrity, assurance of data authenticity is permitted to support an optional cryptographic signature of the FRU Information Device. It is recommended that the FRU Information be covered by a digital signature of the adapter's or component's firmware and validated when the firmware is updated. If the content is validated during adapter or component firmware initialization, care must be taken to ensure the component and information are available to system management firmware as needed, such as when only auxiliary power is supplied.

The IPMI Platform Management FRU Information Storage Definition ([IPMI-FRU]) allows more than 1 MultiRecord to be in a single FRU Information Device. The follow table describes the specific MultiRecord format that applies to all PCI Express adapters or components.

*Table 12-15 PCI-SIG MultiRecord §*

| Bytes                      | Factory Default         | Description                                                                                                                                                                                                        |                                                                                               |
|----------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| PCI-SIG MultiRecord Header |                         |                                                                                                                                                                                                                    |                                                                                               |
| 0h                         | 0C0h                    | PCI-SIG Record Type ID: This field indicates the IPMI Record Type ID of the PCI-SIG MultiRecord. This field must be set to a value of C0h.                                                                         |                                                                                               |
| 1h                         | 2h or 82h               | Record Format: This field indicates the format of this MultiRecord.                                                                                                                                                |                                                                                               |
|                            |                         | Bits                                                                                                                                                                                                               | Definition                                                                                    |
|                            |                         | 7                                                                                                                                                                                                                  | When Set, this bit indicates that this is the last MultiRecord in the list.                   |
|                            |                         | 6:0                                                                                                                                                                                                                | This field indicates the MultiRecord format version. This field must be set to a value of 2h. |
| 2h                         | Implementation Specific | Record Length (RLEN): This field indicates the length of the PCI-SIG MultiRecord Body in bytes (i.e., the bytes of the MultiRecord excluding the 5-byte MultiRecord header that is common to all MultiRecords).    |                                                                                               |
| 3h                         | Implementation Specific | Record Checksum: This field is used to give the PCI-SIG MultiRecord Body a zero checksum (i.e., the modulo 256 sum of bytes from byte offset 5h to the end of this MultiRecord plus this checksum byte equals 0h). |                                                                                               |
| 4h                         | Implementation Specific | Header Checksum: This field is used to give the PCI-SIG MultiRecord Header a zero checksum (i.e., the modulo 256 sum of offset 0h of the PCI-SIG MultiRecord Header through this checksum byte equals 0h).         |                                                                                               |
| PCI-SIG MultiRecord Body   |                         |                                                                                                                                                                                                                    |                                                                                               |
| 7h:5h                      | E8FFh                   | [PCI-SIG IANA Private Enterprise Number]: This field indicates the PCI-SIG's IANA Private Enterprise Number. This field must be set to a value of E8FFh (i.e., 59,647).                                            |                                                                                               |
| 8h                         | 0h                      | Version Number: This field indicates the version number of this MultiRecord. This field must be 0h in this version of the specification.                                                                           |                                                                                               |
| 9h                         | Implementation Specific | Number of Descriptors: This field indicates the number of descriptors in this MultiRecord. The value of 0h is reserved.                                                                                            |                                                                                               |

| Bytes                   | Factory Default         | Description                                                                                                                                                            |
|-------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation Specific | Implementation Specific | <b>Descriptor 0:</b> This field contains the first descriptor in this MultiRecord.                                                                                     |
| Implementation Specific | Implementation Specific | <b>Descriptor 1:</b> This field contains the second descriptor in this MultiRecord if Number of Descriptors is greater than one; otherwise, this field is not present. |
| ...                     | ...                     | ...                                                                                                                                                                    |

Table 12-17 PCI-SIG MultiRecord Descriptor §

| Bytes          | Factory Default                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
|----------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|-------------|----------|-----------|--------------------------------|-----------|-------------------------------|-----------|-------------------------------|-----------|-----------------------------|-----------|--------------|-----------|---------------------|-----------|------------------------------|-----------|--------------------------------------|-----------|--------------------------------------|---------------|----------|
| 1h:0h          | Implementation Specific              | <p><b>Descriptor Type:</b> This field indicates the type of this descriptor.</p> <p>Bits:</p> <table><tr><td><b>15:13</b></td><td>Reserved</td></tr><tr><td><b>11:8</b></td><td>Group ID</td></tr><tr><td><b>0h</b></td><td>PCI Express Base Specification</td></tr><tr><td><b>1h</b></td><td>PCI Express CEM Specification</td></tr><tr><td><b>2h</b></td><td>PCI Express M.2 Specification</td></tr><tr><td><b>3h</b></td><td>PCI Express SFF-8639 Module</td></tr><tr><td><b>4h</b></td><td>PCI Firmware</td></tr><tr><td><b>5h</b></td><td>PCI Express OCuLink</td></tr><tr><td><b>6h</b></td><td>PCI Express External Cabling</td></tr><tr><td><b>7h</b></td><td>PCI Express CopprLink Internal Cable</td></tr><tr><td><b>8h</b></td><td>PCI Express CopprLink External Cable</td></tr><tr><td><b>Others</b></td><td>Reserved</td></tr></table> <p><b>7:0</b> Sub-Type (Defined by Group ID owner)</p> | <b>15:13</b> | Reserved | <b>11:8</b> | Group ID | <b>0h</b> | PCI Express Base Specification | <b>1h</b> | PCI Express CEM Specification | <b>2h</b> | PCI Express M.2 Specification | <b>3h</b> | PCI Express SFF-8639 Module | <b>4h</b> | PCI Firmware | <b>5h</b> | PCI Express OCuLink | <b>6h</b> | PCI Express External Cabling | <b>7h</b> | PCI Express CopprLink Internal Cable | <b>8h</b> | PCI Express CopprLink External Cable | <b>Others</b> | Reserved |
| <b>15:13</b>   | Reserved                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>11:8</b>    | Group ID                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>0h</b>      | PCI Express Base Specification       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>1h</b>      | PCI Express CEM Specification        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>2h</b>      | PCI Express M.2 Specification        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>3h</b>      | PCI Express SFF-8639 Module          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>4h</b>      | PCI Firmware                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>5h</b>      | PCI Express OCuLink                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>6h</b>      | PCI Express External Cabling         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>7h</b>      | PCI Express CopprLink Internal Cable |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>8h</b>      | PCI Express CopprLink External Cable |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| <b>Others</b>  | Reserved                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| 2h             | Implementation Specific              | <p><b>Descriptor Version Number:</b> This field indicates the version number of this descriptor. This field must be 0h.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| 3h             | Implementation Specific              | <p><b>Descriptor Length:</b> This field indicates the number of bytes in this descriptor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |
| Length N-1: 4h | Implementation Specific              | <p><b>Descriptor Data:</b> This field contains the type-specific data associated with this descriptor where N is the total length in bytes of this descriptor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |          |             |          |           |                                |           |                               |           |                               |           |                             |           |              |           |                     |           |                              |           |                                      |           |                                      |               |          |

Each Group ID owner can define a list of sub-types with associated Descriptor Data. The Sub-Type field in the PCI-SIG MultiRecord Descriptor is 8 bits allowing for 256 sub-types.

Descriptor Data can be anything defined by the Group ID owner but should follow IPMI FRU conventions and should be kept as small as possible.

Types of FRU Information that are common across many form factors are defined in PCI-SIG MultiRecord Descriptors in this specification as part of Group ID 0h. Types of FRU Information that are not common across many form factors are defined in PCI-SIG MultiRecord Descriptors in other specifications as part of Group IDs other than 0h.



### 12.6.3 Common PCI-SIG MultiRecord Descriptors §

This section defines the PCI-SIG MultiRecord Descriptor sub-types for the PCI Express Base Specification group (i.e., Group ID 0h). These descriptors are common across many PCI Express-based adapters and components.

*Table 12-18 Descriptor Sub-Types for Group ID 0h §*

| Descriptor Sub-Type | Description           |
|---------------------|-----------------------|
| 00h                 | Connector Subdivision |
| All other encodings | Reserved              |

#### 12.6.3.1 Connector Subdivision (Group ID 0h, Sub-Type 0h) §

In some use cases, the host requires the ability to discover connector subdivision combinations of an adapter's or component's upstream-facing connector so that the host can configure the subdivision of the Downstream Port connected to the adapter or component. A couple of examples include:

- Two x8 Ports on an adapter with a x16 upstream-facing connector.
- Four M.2 x4 connectors on a CEM card carrier with a x16 upstream-facing connector.

This section defines the FRU descriptors that describe the adapter's or component's supported connector subdivision combination(s). If only one connector subdivision combination is supported, then no Connector Subdivision Combinations Descriptor (refer to § Table 12-19) is required. If more than one connector subdivision combination is supported and the implementation supports the PCI-SIG MultiRecord, then a Connector Subdivision Combinations Descriptor with an instance of the Connector Subdivision Descriptor (refer to § Table 12-20) for each supported connector subdivision combination must be present.

The Connector Subdivision Descriptor at the lowest offset in the Connector Subdivision Combinations Descriptor should include the most common expected usage and must be the default Connector Subdivision Descriptor. This is unless a form factor specific method is applied (e.g., the **DUALPORTEN#** sideband signal defined by the [SFF-8639]) or the host commands the form factor to select an alternate Connector Subdivision Descriptor using a mechanism outside the scope of this specification. The default Connector Subdivision Descriptor is defined as the adapter configuration if no control function is sent to select an alternate mode of operation. The adapter or component must revert to the default Connector Subdivision Descriptor only on a Cold Reset.

Each Lane on the upstream-facing connector within each connector subdivision combination that is connected to a Port or another connector on the adapter or component must be described within a single Connector Subdivision Descriptor. Lanes on the upstream-facing connector that are not connected to a Port or another connector on the adapter or component must not be included in the Connector Subdivision Combinations Descriptor. The Lanes associated with a Connector Subdivision Descriptor may be organized as a single subdivision or multiple subdivisions. Each subdivision within each Connector Subdivision Descriptor must be listed in ascending order starting from the adapter's or component's lowest-numbered Lane.

The lowest-numbered Lane must always be with respect to the adapter or component's designated lane 0 without lane reversal. The starting lane number of any subdivision within a Connector Subdivision Descriptor must be less than or equal (if the subdivision width is x1) to the ending lane number of the respective subdivision.

The details of how connector subdivision information is passed from an out-of-band agent (e.g., a Baseboard Management Controller) responsible for reading this information to the platform entity (e.g., the platform BIOS) responsible for configuring the appropriate Downstream Port(s) is implementation specific.

It is recommended that adapters include the Connector Subdivision Combinations Descriptor, even if only a single width configuration is supported, because it enables system firmware to determine degraded link width conditions in situations where an adapter advertises a value in the Maximum Link Width field in the Link Capabilities Register that is less than or greater than the physical adapter connector edge connectivity (e.g., a x8 capable component on an adapter that only connects x4 to the adapter card edge).

*Table 12-19 Connector Subdivision Combinations Descriptor* §

| Byte Offset(s)          | Factory Default         | Description                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1h:0h                   | 0001h                   | <b>Descriptor Type:</b> This field indicates the type of this descriptor.<br>Bits:<br><b>15:12</b> Reserved<br><b>11:8</b> Group ID = 0h (PCI Express Base Specification)<br><b>7:0</b> Sub-Type = 00h (Connector Subdivision)                                                                                                                        |
| 2h                      | 01h                     | <b>Descriptor Version Number:</b> This field indicates the version number of this descriptor. This field must be 0h.                                                                                                                                                                                                                                  |
| 3h                      | 04h                     | <b>Header Length:</b> This field indicates the length of the header portion of this descriptor (meaning from offset 0h until but not including the Connector Subdivision 0 field). This serves the purpose of being able to insert future bytes after the Descriptor Length and before the Connector Subdivision list if new useful attributes arise. |
| 4h                      | Implementation Specific | <b>Descriptor Length:</b> This field indicates the length of this descriptor in bytes.                                                                                                                                                                                                                                                                |
| Implementation Specific | Implementation Specific | <b>Connector Subdivision 0:</b> This field contains the first connector subdivision.                                                                                                                                                                                                                                                                  |
| Implementation Specific | Implementation Specific | <b>Connector Subdivision 1:</b> This field contains the second connector subdivision, if applicable. If there is only one connector subdivision, then this field is not present.                                                                                                                                                                      |
| ...                     | ...                     | ...                                                                                                                                                                                                                                                                                                                                                   |

The Connector Subdivision Descriptor is shown in § Table 12-20. An upstream PCIe connector is made up of one or more subdivisions. Each subdivision is routed to a Port or another connector on the adapter or component. The Connector Subdivision Descriptor indicates the number of subdivisions and the starting and ending Lane numbers for each subdivision that make up each supported connector subdivision combination.

*Table 12-20 Connector Subdivision Descriptor* §

| Byte Offset(s) | Factory Default         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0h             | Implementation Specific | <b>Descriptor Length:</b> This field indicates the length of this descriptor in bytes.                                                                                                                                                                                                                                                                                                                                                        |
| 1h             | Implementation Specific | <b>Number of Subdivisions:</b> This field indicates the number of subdivisions in this descriptor. The permitted range of values is from 1 (e.g., 1 subdivision of maximum supported width) to the number of individual lanes used at the maximum supported width. All other values are Reserved. As an example, a x16 capable adapter can have values from 1 (e.g., one subdivision of x16 width) to 16 (e.g., 16 subdivisions of x1 width). |
| 2h             | Implementation Specific | <b>Subdivision 0:</b> This field indicates the starting and ending lanes of the first subdivision of this descriptor.<br>Bits:                                                                                                                                                                                                                                                                                                                |

| Byte Offset(s) | Factory Default         | Description                                                                                                                                                                                                                                                                                                                                |
|----------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |                         | <b>7:4</b> Starting Lane is the lowest-numbered Lane of the first subdivision of this descriptor.<br><b>3:0</b> Ending Lane is the highest numbered Lane of the first subdivision of this descriptor.                                                                                                                                      |
| 3h             | Implementation Specific | <b>Subdivision 1:</b> This field indicates the starting and ending lanes of the second subdivision of this descriptor.<br>Bits:<br><b>7:4</b> Starting Lane is the lowest-numbered Lane of the second subdivision of this descriptor.<br><b>3:0</b> Ending Lane is the highest numbered Lane of the second subdivision of this descriptor. |
| ...            | ...                     | ...                                                                                                                                                                                                                                                                                                                                        |

#### **Example Connector Subdivision Combinations Descriptor for an Adapter Supporting 3 Connector Subdivision Descriptors:**

1 Port of x16 Lanes, 2 Ports of x8 Lanes each, 4 Ports of x4 Lanes each

Connector Subdivision Descriptor: 0 (the default)

Number of Subdivisions: 1

Starting Lane of the first subdivision: 0; Ending Lane of the first subdivision: 15

Connector Subdivision Descriptor: 1

Number of Subdivisions: 2

Starting Lane of the first subdivision: 0; Ending Lane of the first subdivision: 7

Starting Lane of second subdivision: 8; Ending Lane of second subdivision: 15

Connector Subdivision Descriptor: 2

Number of Subdivisions: 4

Starting Lane of the first subdivision: 0; Ending Lane of the first subdivision: 3

Starting Lane of second subdivision: 4; Ending Lane of second subdivision: 7

Starting Lane of third subdivision: 8; Ending Lane of third subdivision: 11

Starting Lane of forth subdivision: 12; Ending Lane of forth subdivision: 15

## **12.7 Out-of-Band Management Control Mechanism §**

This section defines the optional PCI Express Management Interface (PCIe-MI). PCIe-MI is an architecture and command set for message-based management of a PCIe-MI Component (e.g., an adapter or component). The following management functionality is defined.

- Host CPU-agnostic and operating system-agnostic out-of-band management.
- A common management interface used across all in-band management transports and out-of-band management transports.
- Commands to perform management operations such as the following.

- Discover the PCIe-MI Components that are connected to the host system.
- Discover the properties and capabilities of each PCIe-MI Component.
- Monitor runtime changes to the PCIe-MI Component such as changes to the speed and width of any Link on the PCIe-MI Component.
- PCIe-specific configuration of PCIe-MI Components such as the following.
  - Get and set Configuration Space bits and fields using out-of-band management transports.
  - Get and set a Function's Segment Number, Bus Number, and/or Device Number.
  - Get and set connector subdivisions.

The management messages are referred to as PCIe-MI Messages. The two types of PCIe-MI Messages are PCIe-MI Commands and PCIe-MI Completion.

- A PCIe-MI Command is transmitted from a PCIe-MI Commander, such as a Baseboard Management Controller (BMC) or in-band management system software, to a PCIe-MI Completer (e.g., an MCTP endpoint or a Function that supports MMPT) that specifies a management operation to be performed.
- In order to provide the results of a PCIe-MI Command, a PCIe-MI Completion must be transmitted from the PCIe-MI Completer that received the PCIe-MI Command to the PCIe-MI Commander that transmitted the PCIe-MI Command.

PCIe-MI is supported using out-of-band management transports (i.e., independent of system software running on a host CPU) from a management controller (see the [MCTP Base Specification - <https://www.dmtf.org/dsp/DSP0236>]) such as a BMC and is supported using in-band management transports (i.e., from system software running on a host CPU). The supported out-of-band management transports for PCIe-MI include any MCTP-based intercommunications between a management controller (e.g., a BMC) and a managed device (i.e., a PCIe-MI Component) as described in the [Platform Management Communications Infrastructure (PMCI) Architecture Whitepaper - <https://www.dmtf.org/dsp/DSP2015>]. The supported in-band management transports for PCIe-MI include the following.

- PCIe-MI over MMPT (see § Section 6.35.1.4 and the PCI Code and ID Assignment Specification).
- Any MCTP-based intercommunications between host platform software (i.e., system software running on a host CPU) and a management controller (i.e., a PCIe-MI Component) as described in the [Platform Management Communications Infrastructure (PMCI) Architecture Whitepaper - <https://www.dmtf.org/dsp/DSP2015>].

PCIe Components that implement PCIe-MI over MCTP must comply to the [MCTP Base Specification - <https://www.dmtf.org/dsp/DSP0236>], the [PCIe-MI over MCTP Binding Specification - <https://www.dmtf.org/dsp/DSP0291>], and any applicable MCTP transport or interface specifications such as the following:

- The [MCTP SMBus/I2C Binding Specification - <https://www.dmtf.org/dsp/DSP0237>].
- The [MCTP PCIe VDM Binding Specification - <https://www.dmtf.org/dsp/DSP0238>].
- The [MCTP I3C Binding Specification - <https://www.dmtf.org/dsp/DSP0233>].
- The [MCTP USB Binding Specification - <https://www.dmtf.org/dsp/DSP0283>].
- The [MCTP MMBI Binding Specification - <https://www.dmtf.org/dsp/DSP0284>].
- The [MCTP Host Interface Specification - <https://www.dmtf.org/dsp/DSP0256>].

For PCIe-MI over MCTP, PCIe-MI Messages are considered PCIe-MI Request messages per the [PCIe-MI over MCTP Binding Specification - <https://www.dmtf.org/dsp/DSP0291>]. Therefore, the Tag Owner bit defined by the [PCIe-MI over MCTP Binding Specification - <https://www.dmtf.org/dsp/DSP0291>] is Set in PCIe-MI Messages (i.e., PCIe-MI Commands and PCIe-MI Completions), and the value of the Message Tag field defined by the [PCIe-MI over MCTP Binding Specification - <https://www.dmtf.org/dsp/DSP0291>] always originates from the PCIe-MI Commander or PCIe-MI Completer that is transmitting the PCIe-MI Command or PCIe-MI Completion. Additionally, since there are no PCIe-MI Response messages,

there are no timing requirements related to PCIe-MI Response messages. For example, there is no request-to-response timing requirement and instead, the PCIe-MI Command to PCIe-MI Completion time must adhere to the time indicated by the Maximum Command-To-Completion Time field in § Table 12-39.

§ Figure 12-18 shows the PCIe-MI protocol layers on a PCIe-MI Component for the transports described in this section. PCIe-MI may also be transported over transport layers that are not shown or transported over MCTP over physical layers with MCTP bindings that are not shown. The details related to any transport or physical layer with an MCTP binding not defined by this specification are implementation specific. Note that the MCTP Host Interface as defined by the [MCTP Host Interface Specification - <https://www.dmtf.org/dsp/DSP0256>] may also be utilized for use cases such as MCTP over MMBI (not shown in § Figure 12-18).

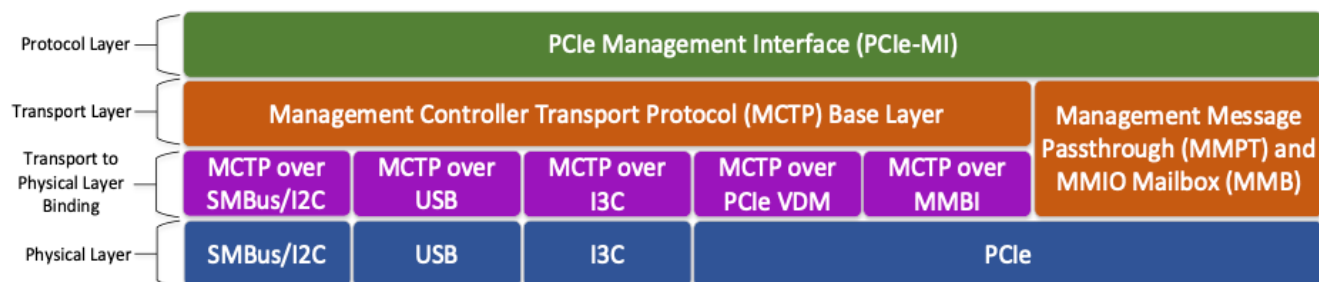


Figure 12-18 PCIe-MI Component Protocol Layers §

### 12.7.1 PCIe-MI Architectural Model §

A PCIe-MI Component is a collection of one or more Functions, such as a component (e.g., a Root Complex, Switch, or embedded Endpoint) or adapter (e.g., a network adapter or a GPU on a CEM add-in card, or an NVMe drive in a U.2 or EDSFF form factor) and has the following characteristics.

- The PCIe-MI Component must contain one or more non IOV Functions and/or Physical Functions.
- The PCIe-MI Component must support PCIe-MI on one or more PCIe-MI Completers. Examples of PCIe-MI Completers include the following.
  - A Function supporting PCIe-MI over MMPT.
  - An MCTP endpoint supporting PCIe-MI over MCTP over one or more of the following.
    - PCIe VDM.
    - SMBus/I2C.
    - I3C.
    - USB.
    - MMBI.
- A PCIe-MI Port must be one of the following.
  - A PCIe Upstream Port.
  - A PCIe Downstream Port.
  - A 2-Wire interface port.
  - A USB port.
- A PCIe-MI Component must contain one or more PCIe ports (i.e., Upstream Ports or Downstream Ports).
- A PCIe-MI Component contains zero or more non-PCIe PCIe-MI Ports.

- Each PCIe-MI Port on the PCIe-MI Component must have a PCIe-MI Port Identifier that is unique within the PCIe-MI Component (see § Section 12.7.3.1.5).
  - PCIe-MI Port Identifiers must be in the range 0 to N minus 1, where N is the number of PCIe-MI Ports on the PCIe-MI Component.
  - PCIe-MI Port Identifiers must not change value unless main power to the PCIe-MI Component is removed.
- Each PCIe-MI Port is exposed or hidden from the host system as indicated by the PCIe-MI Port Exposed bit in § Table 12-24.
  - Examples of exposed/hidden ports include the following.
    - PCIe ports:
      - A PCIe-MI Component supports a connector subdivision with a x8 PCIe port on lanes 0 to 7 and a connector subdivision with two x4 PCIe ports where one x4 PCIe port is on lanes 0 to 3 and the other x4 PCIe port is on lanes 4 to 7.
      - The x8 connector subdivision is selected.
      - In this example, the PCIe-MI Component has three PCIe ports where the x8 PCIe port is exposed, and each x4 PCIe port is hidden.
    - USB port:
      - A PCIe-MI Component supports a USB port on the JTAG pins of a CEM add-in card connector.
      - If the JTAG pins are configured for USB, then the USB port is exposed.
      - If the JTAG pins are configured for JTAG, then the USB port is hidden.
- Each PCIe-MI Port that is a PCIe Upstream Port or PCIe Downstream Port contains zero or more Functions supporting PCIe-MI over MMPT.
  - Each Function supporting PCIe-MI over MMPT contains an MMPT PCIe-MI Completer.
- Each PCIe-MI Port contains zero or more MCTP physical addresses supporting PCIe-MI.
  - Each MCTP physical address supporting PCIe-MI must contain one or more PCIe-MI Completers (i.e., MCTP endpoints, see the [MCTP Base Specification - <https://www.dmtf.org/dsp/DSP0236>]):
    - Supporting more than one MCTP endpoint per MCTP physical address requires the PCIe-MI Component to support MCTP bridging (see the [MCTP Base Specification - <https://www.dmtf.org/dsp/DSP0236>]).
- The PCIe-MI Component may support a FRU Information Device accessible in SMBus/I2C mode on the 2-Wire interface port (see § Section 12.6.1 ).

### 12.7.1.1 Operational Times §

A PCIe-MI Completer that is able to receive PCIe-MI Commands, process PCIe-MI Commands successfully, and transmit PCIe-MI Completions is operational.

For PCIe-MI over MMPT, a PCIe-MI Completer must be operational whenever MMPT is operational.

For PCIe-MI over MCTP, a PCIe-MI Completer is operational under the following situations.

- If a reset that causes a PCIe-MI Completer Reset (see § Section 12.7.1.2 ) defines an assertion and a deassertion condition (e.g., SMBRST# as defined in the [SNIA SFF-TA-1009 Enterprise and Datacenter Standard Form Factor Pin and Signal Specification - <https://www.snia.org/sff>] or PERST#), then the PCIe-MI Completer must become operational within 1 s from the de-assertion of that reset unless the PCIe-MI Completer is undergoing a subsequent PCIe-MI Completer Reset.

- If a reset that causes a PCIe-MI Completer Reset (see § Section 12.7.1.2 ) defines an assertion condition but no deassertion condition (e.g., application of power or a Target Reset Pattern as defined by the [MIPI I3C Basic Specification - <https://www.mipi.org>] with RSTACT Defining Byte value 2), then the PCIe-MI Completer must become operational within 1 s from the assertion of that reset unless the PCIe-MI Completer is undergoing a subsequent PCIe-MI Completer Reset.
- A PCIe-MI Completer must become operational 1 s after main power is applied unless the PCIe-MI Completer is undergoing a PCIe-MI Completer Reset due to a reason other than application of main power (see § Section 12.7.1.2 ).
- A PCIe-MI Completer may become operational with only auxiliary power supplied.
  - If a PCIe-MI Completer supports being operational with only auxiliary power supplied, then the PCIe-MI Completer must become operational 1 s after only auxiliary power is applied unless the PCIe-MI Completer is undergoing a PCIe-MI Completer Reset due a reason other than application of only auxiliary power.
- If main power is applied, then application or removal of auxiliary power must have no impact on the operational state of the PCIe-MI Completer.
- If the PCIe-MI Completer supports operation with only auxiliary power applied and auxiliary power is applied, then application or removal of main power must have no impact on the operational state of the PCIe-MI Completer.
- Once a PCIe-MI Completer becomes operational, the PCIe-MI Completer must remain operational unless the PCIe-MI Completer undergoes a PCIe-MI Completer Reset.
- PCIe-MI over MCTP must be operational during a Conventional Reset or a Function Level Reset unless the PCIe-MI Completer is one of the following.
  - A PCIe VDM MCTP endpoint or MMBI MCTP endpoint on a Function that is undergoing a Conventional Reset or Function Level Reset.
  - A PCIe-MI Completer that is undergoing a PCIe-MI Completer Reset due to a reason other than a Conventional Reset or a Function Level Reset.

### 12.7.1.2 PCIe-MI Completer Reset §

If a PCIe-MI Completer undergoes a PCIe-MI Completer Reset, then the PCIe-MI Completer must abort the processing of any PCIe-MI Commands, must not transmit a PCIe-MI Completion for PCIe-MI Commands whose processing has been aborted, and should abort the transmission of any PCIe-MI Messages in progress. If the transmission of a PCIe-MI Message for PCIe-MI over MCTP is aborted due to a PCIe-MI Completer Reset, then the transmission should be aborted on an MCTP packet boundary. The following rules apply to PCIe-MI Completer Resets. The impact of any reset not listed in this section is implementation specific.

- When power is applied to the PCIe-MI Completer, then a PCIe-MI Completer Reset must occur.
- If the PCIe-MI Completer is a PCIe VDM MCTP endpoint, MMPT-capable Function, or MMBI-capable Function, then a Conventional Reset or a Function Level Reset must cause a PCIe-MI Completer Reset.
- If the PCIe-MI Completer is an MCTP endpoint, then a Conventional Reset or a Function Level Reset must not cause a PCIe-MI Completer Reset unless the PCIe-MI Completer is a PCIe VDM MCTP endpoint or MMBI MCTP endpoint on a Function that is undergoing the Conventional Reset or Function Level Reset.
- If the PCIe-MI Completer is an MCTP endpoint on a 2-Wire interface port and the 2-Wire clock is held low for longer than  $T_{dcl}$  (see § Table 12-14), then a PCIe-MI Completer Reset must occur.
- If the PCIe-MI Completer is an MCTP endpoint on a 2-Wire interface port and a form factor-specific 2-Wire reset occurs (e.g., assertion of SMBRST# as defined in the [SNIA SFF-TA-1009 Enterprise and Datacenter Standard



Form Factor Pin and Signal Specification - <https://www.snia.org/sff>), then a PCIe-MI Completer Reset must occur.

- If the PCIe-MI Completer is an MCTP endpoint on a 2-Wire interface port, then a PCIe-MI Completer Reset must cause the PCIe-MI Completer to do the following in the order listed.
  1. Stop transmitting any PCIe-MI Message in flight within 5 ms from the assertion of the PCIe-MI Completer Reset as follows.
    - If in SMBus mode, generate a STOP condition to signal the end of the transmission (see the [SMBus Specification - <https://www.smbus.org>]).
    - If in I3C mode, set the T bit to 1b to signal the end of the transmission (see the [MIPI I3C Basic Specification - <https://www.mipi.org>]).
  2. Transition the transmitter to the bus idle condition (refer to the [SMBus Specification - <https://www.smbus.org>] and the [MIPI I3C Basic Specification - <https://www.mipi.org>]).
    - Keep the transmitter in the bus idle condition and ignore any incoming traffic until the PCIe-MI Completer is operational again (see § Section 12.7.1.1 ).
  3. Reset the 2-Wire interface port physical layer.
    - If the 2-Wire interface port is in I3C Mode, then the 2-Wire interface port must switch to SMBus mode as part of resetting the 2-Wire interface port physical layer due to the PCIe-MI Completer Reset, unless otherwise specified (e.g., the PCIe-MI Completer Reset is due to an I3C Target Reset Pattern with RSTACT Defining Byte value 2).
- If the PCIe-MI Completer is an MCTP endpoint on a 2-Wire interface port in I3C mode, then the Target Reset Pattern as defined by the [MIPI I3C Basic Specification - <https://www.mipi.org>] behaves as follows.
  - All Target Reset Patterns must not change the 2-Wire interface port's mode to SMBus mode (i.e., the 2-Wire interface port must remain in I3C mode).
  - The default cases for the Target Reset Pattern must reset the I3C physical layer, must not cause a PCIe-MI Completer Reset, and must not reset the I3C Dynamically Assigned Address.
  - For non-default cases for the Target Reset Pattern, the RSTACT CCC as defined by the [MIPI I3C Basic Specification - <https://www.mipi.org>] may be supported and if supported, behaves as follows.
    - Defining Byte value 0 must not cause a PCIe-MI Completer Reset and must have no effect.
    - Defining Byte value 1 must behave the same as the default cases for the Target Reset Pattern.
    - Defining Byte value 2 must cause a PCIe-MI Completer Reset and must reset the I3C Dynamically Assigned Address.
- If the PCIe-MI Completer is an MCTP endpoint on a USB 2.0 port and the port is reset using USB Reset signaling as defined in the [Universal Serial Bus Specification, Revision 2.0 – <https://www.usb.org>], then a PCIe-MI Completer Reset must occur.
- A PCIe-MI Completer Reset of a given PCIe-MI Completer must not affect the operation of any other entity, including other PCIe-MI Completers, outside of the PCIe-MI Completer undergoing the PCIe-MI Completer Reset.



### **12.7.1.3 Example PCIe-MI Component** §

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## IMPLEMENTATION NOTE:

### EXAMPLE PCIE-MI COMPONENT §

§ Figure 12-19 shows an example PCIe-MI Component with the following properties.

- Two PCIe ports with PCIe-MI Port Identifiers of 0 and 1.
  - PCIe-MI Port Identifier 0.
    - One PCIe Function with a Function Number of 0.
    - Function 0 supports PCIe-MI over MMPT.
    - Function 0 supports PCIe-MI over MCTP over PCIe VDM.
  - PCIe-MI Port Identifier 1.
    - Two PCIe Functions with Function Numbers of 0 and 1.
    - Function 0 and 1 support PCIe-MI over MMPT.
    - Function 0 supports PCIe-MI over MCTP over PCIe VDM.
  - Each Function that supports PCIe-MI over MMPT may support PCIe-MI over MCTP over MMBI instead of or in addition to PCIe-MI over MMPT or may support neither PCIe-MI over MMPT nor PCIe-MI over MCTP over MMBI.
  - Each PCIe port is an Upstream Port (e.g., a Switch Upstream Port or an Upstream Port on an Endpoint) or a Downstream Port (e.g., a Root Port on a Root Complex or a Switch Downstream Port).
- One 2-Wire interface port with a PCIe-MI Port Identifier of 2 with SMBus mode addresses shown in 8-bit/7-bit format.
  - One FRU Information Device at physical address A6h/53h in SMBus/I2C mode only (see § Section 12.6.1 ).
  - A primary MCTP physical address at address 3Ah/1Dh in SMBus mode.
    - Two MCTP endpoints (e.g., to support redundant BMCs).
    - Each MCTP endpoint is operational only while main power is applied.
  - A secondary MCTP physical address at address D4h/6Ah in SMBus mode.
    - One MCTP endpoint.
    - The MCTP endpoint is operational while auxiliary power only, main power only, or both is applied.
      - The ability to become operational with only auxiliary power applied allows a BMC to perform tasks before applying main power such as configuring connector subdivisions so that the Links train to their configured widths once main power is applied.
  - Optionally supports the ability to transition to I3C mode.

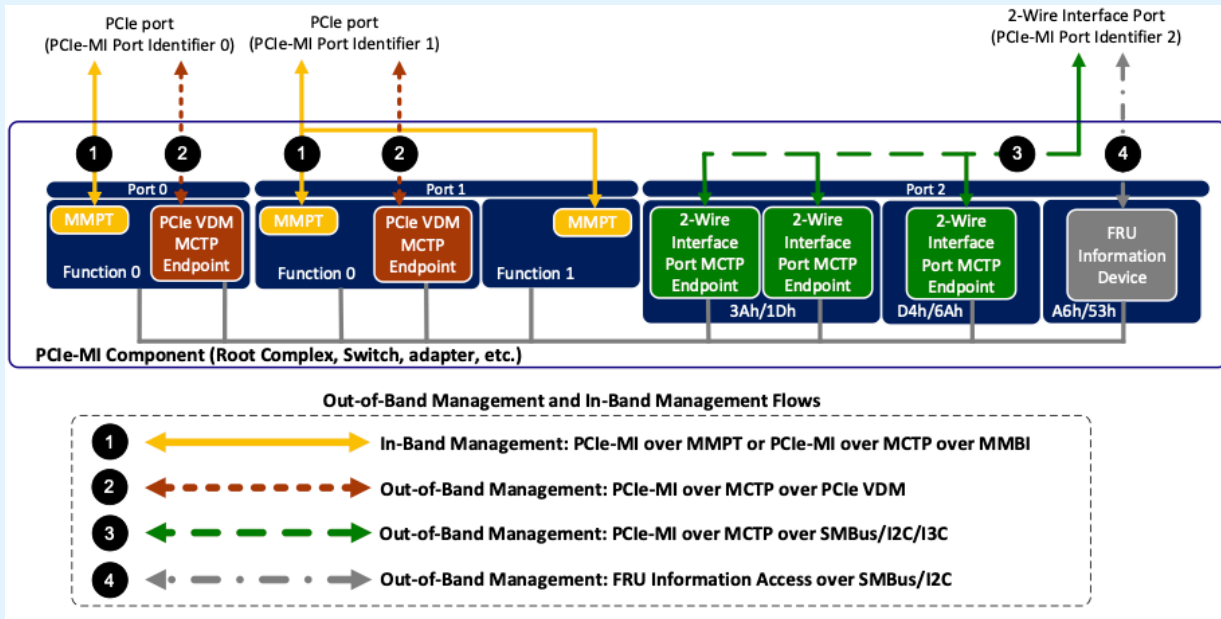


Figure 12-19 Example PCIe-MI Component §

#### **12.7.1.4 Example PCIe-MI Management Traffic Flows** §

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## IMPLEMENTATION NOTE:

### EXAMPLE PCIE-MI MANAGEMENT TRAFFIC FLOWS §

§ Figure 12-20 shows example PCIe-MI Message flows.

- In-band management system software (e.g., the operating system or UEFI firmware running on a host CPU) communicates with the PCIe-MI Component via PCIe-MI over MMPT or PCIe-MI over MCTP over MMBI as shown in management flow number 1 (see § Section 6.35.1.4 and the PCI Code and ID Assignment Specification).
- A BMC communicates with the PCIe-MI Component via PCIe-MI over MCTP (see the [PCIe-MI over MCTP Binding Specification - <https://www.dmtf.org/dsp/DSP0291>]) over PCIe VDM (see the [MCTP PCIe VDM Binding Specification - <https://www.dmtf.org/dsp/DSP0238>]) as shown in management flow number 2.
  - PCIe-MI Commands are transmitted peer-to-peer using Route by ID from the BMC to the PCIe-MI Component.
  - PCIe-MI Completions are transmitted peer-to-peer using Route by ID from the PCIe-MI Component to the BMC.
  - The Root Complex also transmits MCTP control messages to the PCIe-MI Component using Broadcast from Root Complex as defined by the [MCTP PCIe VDM Binding Specification - <https://www.dmtf.org/dsp/DSP0238>] and the PCIe-MI Component responds using Route to Root Complex (not shown in the figure).
  - The Root Complex may contain an MCTP bridge used to route MCTP messages based on MCTP endpoint ID as defined by the [MCTP Base Specification - <https://www.dmtf.org/dsp/DSP0236>] and the [MCTP PCIe VDM Binding Specification - <https://www.dmtf.org/dsp/DSP0238>] (not shown in the figure).
- A BMC communicates with each of the PCIe-MI Components via PCIe-MI over MCTP over a sideband bus that supports MCTP such as SMBus/I2C, I3C, USB, etc. (see the applicable MCTP physical layer binding specification) as shown in management flow number 3.

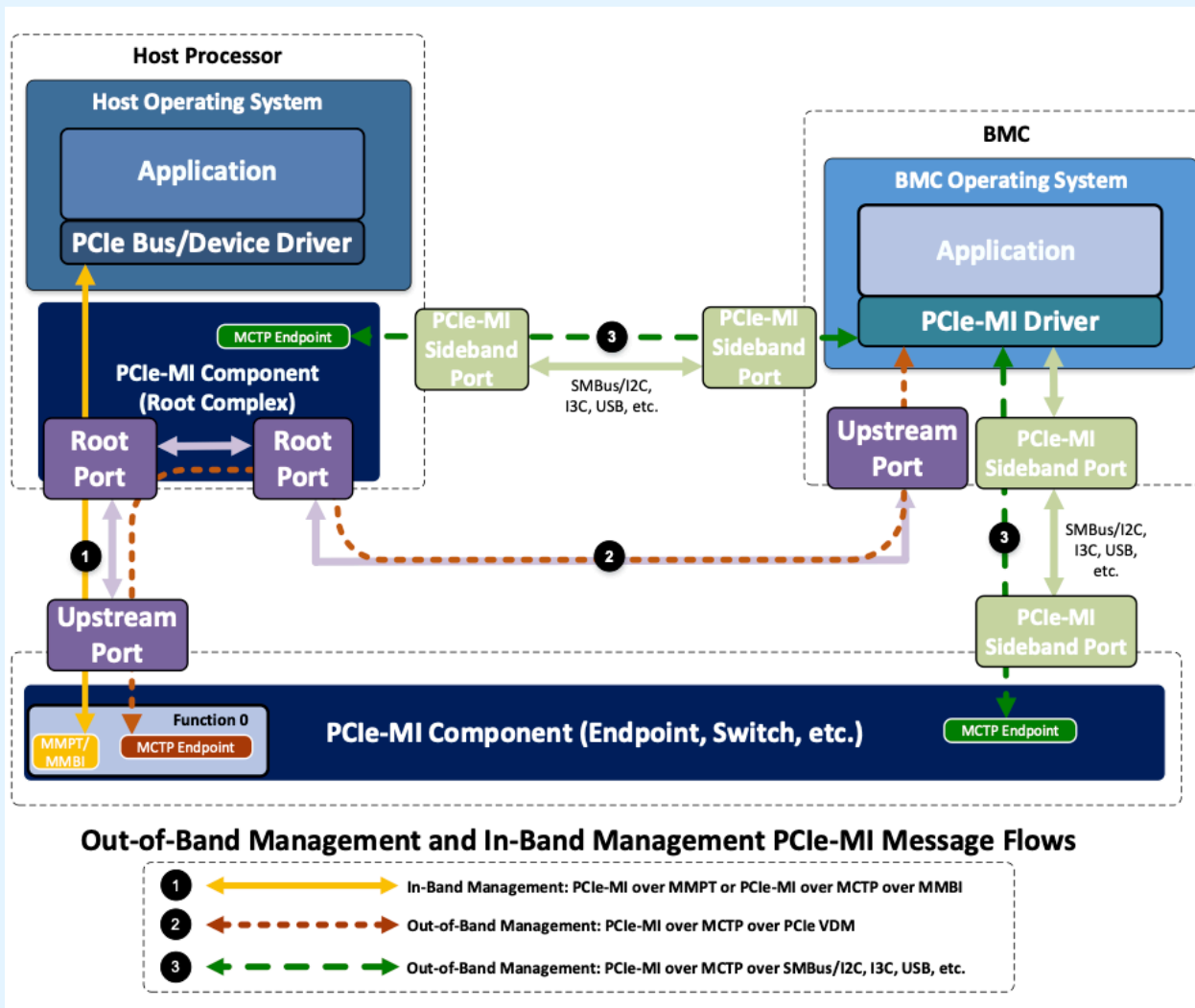


Figure 12-20 Example PCIe-MI Message Flows §

### 12.7.2 PCIe-MI Message Format §

The format of a PCIe-MI Message is shown in § Figure 12-21. The PCIe-MI Message is a PCIe-MI Command or a PCIe-MI Completion as defined by the PCIe-MI Message Type (PMT) field. The endianness of all PCIe-MI Message fields is little endian. The PCIe-MI Message format is common between PCIe-MI over MCTP and PCIe-MI over MMPT.

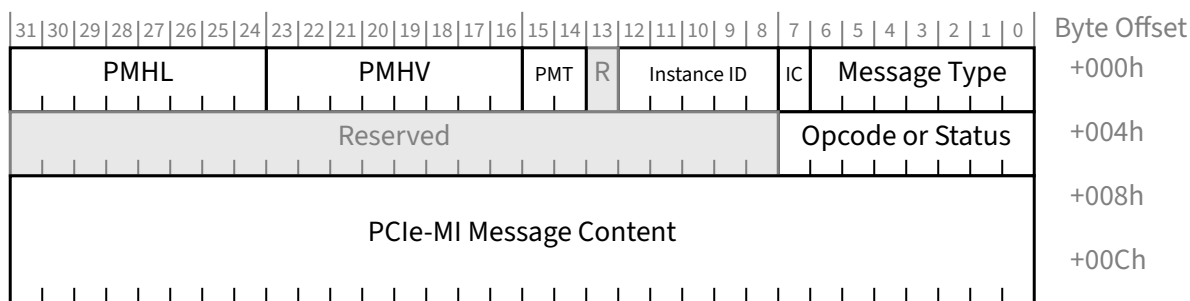


Figure 12-21 PCle-MI Message Format §

The format of a PCle-MI Command is shown in § Table 12-21.

Table 12-21 PCle-MI Command §

| Byte Location                 | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------|--------|--------------------|-----|-----------------|--------|----------|----|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>PCle-MI Message Header</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 1:0                           | <b>Message Type Information:</b> This field specifies information about the message type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
|                               | <table border="1"> <tr> <th>Bits</th><th>Description</th></tr> <tr> <td>15:14</td><td> <p><b>PCle-MI Message Type (PMT):</b> This field should be set to 01b to specify the PCle-MI Message is a PCle-MI Command. If a PCle-MI Completer receives a PCle-MI Message with this field set to 01b, then the PCle-MI Completer must handle the PCle-MI Message as a PCle-MI Command; otherwise, an Invalid Input Error Completion with the Error Location field indicating this field must be returned (see § Section 12.7.2.2.1).</p> <table border="1"> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00b</td><td>PCle-MI Completion</td></tr> <tr> <td>01b</td><td>PCle-MI Command</td></tr> <tr> <td>Others</td><td>Reserved</td></tr> </table> </td></tr> <tr> <td>13</td><td>Reserved</td></tr> <tr> <td>12:8</td><td><b>Instance Identifier (Instance ID):</b> This field specifies the instance of the PCle-MI Command. See § Section 12.7.3 for additional details.</td></tr> <tr> <td>7</td><td><b>Integrity Check (IC):</b> For PCle-MI over MCTP, this bit is Clear per the [PCle-MI over MCTP Binding Specification - <a href="https://www.dmtf.org/dsp/DSP0291">https://www.dmtf.org/dsp/DSP0291</a>]. For PCle-MI over MMPT, this bit is not applicable and should be Clear. If a PCle-MI Completer receives a PCle-MI Message with this bit Set, then an Invalid Input Error Completion with the Error Location field indicating this bit must be returned (see § Section 12.7.2.2.1).</td></tr> <tr> <td>6:0</td><td> <p><b>Message Type:</b> For PCle-MI over MCTP, this field set to 9 specifies a PCle-MI Message per the [PCle-MI over MCTP Binding Specification - <a href="https://www.dmtf.org/dsp/DSP0291">https://www.dmtf.org/dsp/DSP0291</a>].</p> <p>For PCle-MI over MMPT, this field set to 9 specifies a PCle-MI Message (see the PCI Code and ID Assignment Specification). If a PCle-MI Completer receives a PCle-MI Message over MMPT with this field set to a value</p> </td></tr> </table> | Bits  | Description | 15:14 | <p><b>PCle-MI Message Type (PMT):</b> This field should be set to 01b to specify the PCle-MI Message is a PCle-MI Command. If a PCle-MI Completer receives a PCle-MI Message with this field set to 01b, then the PCle-MI Completer must handle the PCle-MI Message as a PCle-MI Command; otherwise, an Invalid Input Error Completion with the Error Location field indicating this field must be returned (see § Section 12.7.2.2.1).</p> <table border="1"> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00b</td><td>PCle-MI Completion</td></tr> <tr> <td>01b</td><td>PCle-MI Command</td></tr> <tr> <td>Others</td><td>Reserved</td></tr> </table> | Value | Description     | 00b    | PCle-MI Completion | 01b | PCle-MI Command | Others | Reserved | 13 | Reserved | 12:8 | <b>Instance Identifier (Instance ID):</b> This field specifies the instance of the PCle-MI Command. See § Section 12.7.3 for additional details. | 7 | <b>Integrity Check (IC):</b> For PCle-MI over MCTP, this bit is Clear per the [PCle-MI over MCTP Binding Specification - <a href="https://www.dmtf.org/dsp/DSP0291">https://www.dmtf.org/dsp/DSP0291</a> ]. For PCle-MI over MMPT, this bit is not applicable and should be Clear. If a PCle-MI Completer receives a PCle-MI Message with this bit Set, then an Invalid Input Error Completion with the Error Location field indicating this bit must be returned (see § Section 12.7.2.2.1). | 6:0 |
| Bits                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 15:14                         | <p><b>PCle-MI Message Type (PMT):</b> This field should be set to 01b to specify the PCle-MI Message is a PCle-MI Command. If a PCle-MI Completer receives a PCle-MI Message with this field set to 01b, then the PCle-MI Completer must handle the PCle-MI Message as a PCle-MI Command; otherwise, an Invalid Input Error Completion with the Error Location field indicating this field must be returned (see § Section 12.7.2.2.1).</p> <table border="1"> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00b</td><td>PCle-MI Completion</td></tr> <tr> <td>01b</td><td>PCle-MI Command</td></tr> <tr> <td>Others</td><td>Reserved</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value | Description | 00b   | PCle-MI Completion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 01b   | PCle-MI Command | Others | Reserved           |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| Value                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 00b                           | PCle-MI Completion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 01b                           | PCle-MI Command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| Others                        | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 13                            | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 12:8                          | <b>Instance Identifier (Instance ID):</b> This field specifies the instance of the PCle-MI Command. See § Section 12.7.3 for additional details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 7                             | <b>Integrity Check (IC):</b> For PCle-MI over MCTP, this bit is Clear per the [PCle-MI over MCTP Binding Specification - <a href="https://www.dmtf.org/dsp/DSP0291">https://www.dmtf.org/dsp/DSP0291</a> ]. For PCle-MI over MMPT, this bit is not applicable and should be Clear. If a PCle-MI Completer receives a PCle-MI Message with this bit Set, then an Invalid Input Error Completion with the Error Location field indicating this bit must be returned (see § Section 12.7.2.2.1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| 6:0                           | <p><b>Message Type:</b> For PCle-MI over MCTP, this field set to 9 specifies a PCle-MI Message per the [PCle-MI over MCTP Binding Specification - <a href="https://www.dmtf.org/dsp/DSP0291">https://www.dmtf.org/dsp/DSP0291</a>].</p> <p>For PCle-MI over MMPT, this field set to 9 specifies a PCle-MI Message (see the PCI Code and ID Assignment Specification). If a PCle-MI Completer receives a PCle-MI Message over MMPT with this field set to a value</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                 |        |                    |     |                 |        |          |    |          |      |                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |

| Byte Location | Field Description                                                                                                                                                                                                                                                                                                |                                                                                                                                                       |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Bits                                                                                                                                                                                                                                                                                                             | Description                                                                                                                                           |
|               |                                                                                                                                                                                                                                                                                                                  | other than 9, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned (see § Section 12.7.2.2.1). |
| 2             | <b>PCIe-MI Message Header Version (PMHV):</b> This field specifies the version of the PCIe-MI Message Header. This field should be cleared to 0 for this version of the specification. See § Section 12.7.3 for additional details.                                                                              |                                                                                                                                                       |
| 3             | <b>PCIe-MI Message Header Length (PMHL):</b> This field specifies the length of the PCIe-MI Message Header in bytes. This field should be set to 8 for this version of the specification. See § Section 12.7.3 for additional details.                                                                           |                                                                                                                                                       |
| 4             | <b>Opcode:</b> This field specifies the Opcode of the PCIe-MI Command (see § Table 12-27). If a PCIe-MI Completer receives a PCIe-MI Command with an unsupported Opcode, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned (see § Section 12.7.2.2.1). |                                                                                                                                                       |
| 7:5           | Reserved                                                                                                                                                                                                                                                                                                         |                                                                                                                                                       |

**PCIe-MI Command Content**

|       |                                                                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N+7:8 | <b>PCIe-MI Command Content:</b> This optional field specifies PCIe-MI Command-specific content as defined in the following sections, where N is the size of this field in bytes. If N is 0, then this field is not present. |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The format of a PCIe-MI Completion is shown in § Table 12-24.

*Table 12-24 PCIe-MI Completion* §

| Byte Location          | Field Description                                                                                                                                                                                                         |                                                                                                                        |                    |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------|
| PCIe-MI Message Header |                                                                                                                                                                                                                           |                                                                                                                        |                    |
| 1:0                    | Message Type Information: This field indicates information about the message type.                                                                                                                                        |                                                                                                                        |                    |
|                        | Bits                                                                                                                                                                                                                      | Description                                                                                                            |                    |
|                        | 15:14                                                                                                                                                                                                                     | PCIe-MI Message Type (PMT): This field must be cleared to 00b to indicate the PCIe-MI Message is a PCIe-MI Completion. |                    |
|                        |                                                                                                                                                                                                                           | Value                                                                                                                  | Description        |
|                        |                                                                                                                                                                                                                           | 00b                                                                                                                    | PCIe-MI Completion |
|                        |                                                                                                                                                                                                                           | 01b                                                                                                                    | PCIe-MI Command    |
|                        |                                                                                                                                                                                                                           | Others                                                                                                                 | Reserved           |
| 13                     | Reserved                                                                                                                                                                                                                  |                                                                                                                        |                    |
| 12:8                   | Instance Identifier (Instance ID): This field must indicate the value in the Instance Identifier field of the corresponding PCIe-MI Command.                                                                              |                                                                                                                        |                    |
| 7                      | Integrity Check (IC): For PCIe-MI over MCTP, this bit is Clear per the [PCIe-MI over MCTP Binding Specification – https://www.dmtf.org/dsp/DSP0291]. For PCIe-MI over MMPT, this bit is not applicable and must be Clear. |                                                                                                                        |                    |



| Byte Location                     | Field Description                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | Bits                                                                                                                                                                                                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                 |
|                                   | 6:0                                                                                                                                                                                                                                                      | <b>Message Type:</b> For PCIe-MI over MCTP, this field is set to 9 to indicate a PCIe-MI Message per the [PCIe-MI over MCTP Binding Specification - <a href="https://www.dmtf.org/dsp/DSP0291">https://www.dmtf.org/dsp/DSP0291</a> ]. For PCIe-MI over MMPT, this field must be set to 9 to indicate a PCIe-MI Message (see the PCI Code and ID Assignment Specification). |
| 2                                 | <b>PCIe-MI Message Header Version (PMHV):</b> This field indicates the version of the PCIe-MI Message Header. This field must be cleared to 0 for this version of the specification.<br><br>See § <a href="#">Section 12.7.3</a> for additional details. |                                                                                                                                                                                                                                                                                                                                                                             |
| 3                                 | <b>PCIe-MI Message Header Length (PMHL):</b> This field indicates the length of the PCIe-MI Message Header in bytes. This field must be set to 8 for this version of the specification.                                                                  |                                                                                                                                                                                                                                                                                                                                                                             |
| 4                                 | <b>Status:</b> This field must indicate the status (see § <a href="#">Table 12-28</a> ) of the corresponding <a href="#">PCIe-MI Command</a> .                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                             |
| 7:5                               | Reserved                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                             |
| <b>PCIe-MI Completion Content</b> |                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                             |
| N+7:8                             | This optional field indicates PCIe-MI Completion-specific content as defined in the following sections, where N is the size of this field in bytes. If N is 0, then this field is not present.                                                           |                                                                                                                                                                                                                                                                                                                                                                             |

### 12.7.2.1 PCIe-MI Opcodes §

PCIe-MI Commands are identified by an opcode specified in the Opcode field of the PCIe-MI Command. The opcodes defined for PCIe-MI are listed in § Table 12-27.

Table 12-27 PCIe-MI Opcodes §

| Opcode     | Description     | Support <sup>1</sup> | Section            |
|------------|-----------------|----------------------|--------------------|
| 00h        | Get Information | M                    | § Section 12.7.3.1 |
| 01h        | Get Parameters  | O <sup>2</sup>       | § Section 12.7.3.2 |
| 02h        | Set Parameters  | O <sup>2</sup>       | § Section 12.7.3.2 |
| 03h to BFh | Reserved        |                      |                    |
| C0h to FFh | Vendor Specific |                      |                    |

**Notes:**

1. O/M definition: O = optional; M = mandatory over MCTP for each physical layer binding over which PCIe-MI over MCTP is supported and mandatory over MMPT if PCIe-MI over MMPT is supported.
2. If the PCIe-MI Completer does not support any Parameter Identifiers, then the Get Parameters command and Set Parameters command must not be implemented on that PCIe-MI Completer.

### 12.7.2.2 PCIe-MI Status §

The status of a PCIe-MI Command must be returned in the Status field of the corresponding PCIe-MI Completion. The status codes defined for PCIe-MI are listed in § Table 12-28. If the PCIe-MI Completer encounters an error corresponding to a status code in the range 10h to BFh while processing the PCIe-MI Command, then the PCIe-MI Command must be aborted prior to transmitting the PCIe-MI Completion.

## IMPLEMENTATION NOTE: PCIe-MI COMMAND ERROR HANDLING §

If a PCIe-MI Commander receives a PCIe-MI Completion with the Status field set to a value in the range 10h to BFh which indicates the PCIe-MI Command was aborted, then the PCIe-MI Commander may choose to reissue the PCIe-MI Command in an attempt to recover from transient errors. If retrying the PCIe-MI Command does not resolve the issue and a PCIe-MI Completer Reset is able to be performed without disturbing the operation of Functions on the PCIe-MI Component that are in use by the operating system running on the host CPU(s), then a PCIe-MI Commander may choose to reset the PCIe-MI Completer at runtime using that PCIe-MI Completer Reset and then reissue the PCIe-MI Command again.

For example, a PCIe-MI Completer Reset of an MCTP endpoint on a 2-Wire interface port does not impact Functions on the PCIe-MI Component in use by the operating system running on the host CPU(s), whereas a PCIe-MI Completer Reset of a PCIe VDM MCTP endpoint may impact Functions on the PCIe-MI Component in use by the operating system running on the host CPU(s). If a PCIe-MI Completer Reset is not able to be performed without disturbing the operation of Functions on the PCIe-MI Component in use by the operating system running on the host CPU(s), then a reset or power cycle of the system the PCIe-MI Completer is installed in may be performed to attempt recovery.

If a PCIe-MI Commander receives a PCIe-MI Completion with the Status field set to 11h (i.e., External Error), then the error may be caused by an entity external to the PCIe-MI Component whose PCIe-MI Completer is processing the PCIe-MI Command. In such cases, recovery of the error condition may require performing recovery steps on the external entity rather than the PCIe-MI Completer processing the PCIe-MI Command.

Unless otherwise specified, if multiple status codes apply, then the PCIe-MI Completer selects one of the applicable status codes to return in the Status field of the PCIe-MI Completion.

Table 12-28 PCIe-MI Status Codes §

| Status Code                                                | Description                                                                                                                                                                                                                                      |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Status Codes Indicating PCIe-MI Command Not Aborted</b> |                                                                                                                                                                                                                                                  |
| 00h                                                        | <b>Success:</b> The PCIe-MI Command was processed successfully.                                                                                                                                                                                  |
| 01h to 0Fh                                                 | Reserved                                                                                                                                                                                                                                         |
| <b>Status Codes Indicating PCIe-MI Command Aborted</b>     |                                                                                                                                                                                                                                                  |
| 10h                                                        | <b>Internal Error:</b> The PCIe-MI Command could not be processed due to a vendor-specific error internal to the PCIe-MI Component.                                                                                                              |
| 11h                                                        | <b>External Error:</b> The PCIe-MI Command could not be processed due to an error external to the PCIe-MI Component (e.g., a Configuration Write Request to a Function external to the PCIe-MI Component that is generated by the Set Parameters |

| Status Code                                 | Description                                                                                                                                                                                                                                                   |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | command specifying the Routing Information Parameter Identifier timed out or was completed with a UR or CA Completion Status).                                                                                                                                |
| 12h                                         | <b>Invalid Input:</b> The PCIe-MI Command contained a reserved, unimplemented, or conflicting value in a defined field (see § Section 12.7.2.2.1 ). This status code must only be used if there are no other status codes defined for the specific condition. |
| 13h                                         | <b>PCIe-MI Message Content Length Error:</b> The PCIe-MI Message Content contained greater than or less than the number of bytes required. See § Section 12.7.3 for additional details.                                                                       |
| 14h to BFh                                  | Reserved                                                                                                                                                                                                                                                      |
| <b>Vendor-Specific PCIe-MI Status Codes</b> |                                                                                                                                                                                                                                                               |
| C0h to FFh                                  | Vendor Specific                                                                                                                                                                                                                                               |

### 12.7.2.2.1 Invalid Input Error Completion §

An invalid input error occurs when a PCIe-MI Completer detects a reserved, unimplemented, or conflicting value in a defined bit or field in a PCIe-MI Command. An Invalid Input Error Completion is a PCIe-MI Completion with the Status field set to 12h (i.e., Invalid Input) and an Error Location field that indicates the bit or field that is invalid. The format of an Invalid Input Error Completion is shown in § Table 12-29.

Unless otherwise specified, if there are errors in multiple bits and/or fields in a PCIe-MI Command, then the PCIe-MI Completer selects which bit or field to indicate in the Error Location field.

Table 12-29 Invalid Input Error Completion §

| Byte Location                                 | Field Description                                                                                                                  |                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                           | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24 with the Status field set to 12h (i.e., Invalid Input). |                                                                                                                                                                                                                                                                                            |
| <b>Invalid Input Error Completion Content</b> |                                                                                                                                    |                                                                                                                                                                                                                                                                                            |
| 9:8                                           | <b>Error Location:</b> This field indicates the bit or field within the PCIe-MI Command that is invalid.                           |                                                                                                                                                                                                                                                                                            |
|                                               | Bits                                                                                                                               | Description                                                                                                                                                                                                                                                                                |
|                                               | 15:3                                                                                                                               | <b>Byte Location:</b> This field must indicate the byte offset of the least-significant byte in the PCIe-MI Command of the bit or field that is invalid. If the byte offset of the invalid bit or field is beyond byte offset 8,191, then the value 8,191 must be indicated by this field. |
|                                               | 2:0                                                                                                                                | <b>Bit Location:</b> This field must indicate the bit offset of the least significant bit in the least significant byte in the PCIe-MI Command of the bit or field that is invalid.                                                                                                        |
|                                               |                                                                                                                                    |                                                                                                                                                                                                                                                                                            |

### 12.7.3 PCIe-MI Commands §

This section defines the PCIe-MI Commands, which are identified by an Opcode (see § Table 12-27), and their corresponding PCIe-MI Completions. Each PCIe-MI Completer in a PCIe-MI Component may support a different set of PCIe-MI Commands.

The following PCIe-MI Command and PCIe-MI Completion processing rules apply to PCIe-MI Completers.

- Reserved bits and fields in PCIe-MI Completions must be cleared to 0.
- Reserved bits and fields in PCIe-MI Commands must be ignored.
- A PCIe-MI Component must support the concurrent processing of PCIe-MI Commands on each PCIe-MI Completer on the PCIe-MI Component. The maximum number of supported concurrent PCIe-MI Commands on each PCIe-MI Completer is indicated by the Maximum Concurrent Commands fields (see § Table 12-39).
  - As an example, if the PCIe-MI Component supports three PCIe-MI Completers (e.g., MCTP over PCIe VDM, MCTP over SMBus/I2C, and MMPT) and the Maximum Concurrent Commands field for each PCIe-MI Completer indicates a value of 2, then the PCIe-MI Component must support the concurrent processing of two PCIe-MI Commands on each of the three PCIe-MI Completers (i.e., concurrent processing of six PCIe-MI Commands in total).
- If a PCIe-MI Completer receives a PCIe-MI Command (i.e., a new PCIe-MI Command) with an Instance Identifier equal to the Instance Identifier of a PCIe-MI Command that is currently being processed, then the PCIe-MI Command currently being processed may be aborted.
  - If the PCIe-MI Command currently being processed is aborted, then a PCIe-MI Completion for the aborted PCIe-MI Command must not be returned.
  - If the PCIe-MI Command currently being processed is not aborted, then an Invalid Input Error Completion with the Error Location field indicating the Instance ID field must be returned for the new PCIe-MI Command (see § Section 12.7.2.2.1 ) and the PCIe-MI Completer must continue processing the PCIe-MI Command with that Instance ID that is currently being processed.
- If a PCIe-MI Completer receives a PCIe-MI Command (i.e., a new PCIe-MI Command) with an Instance Identifier that is not equal to the Instance Identifier of a PCIe-MI Command that is currently being processed while the PCIe-MI Completer is currently processing the maximum number of concurrent PCIe-MI Commands supported (see the Maximum Concurrent Commands field in § Table 12-39), then an Invalid Input Error Completion with the Error Location field indicating the Instance ID field must be returned for the new PCIe-MI Command and the PCIe-MI Completer must continue processing the PCIe-MI Command(s) currently being processed.

A PCIe-MI Completer designed to support version N of any PCIe-MI Command structure with a version field (e.g., the PCIe-MI Message Header Version field) behaves as follows.

- Must return a value of N in the version field in the PCIe-MI Completion, if the version field is returned in the PCIe-MI Completion, regardless of the value of the version field in the corresponding PCIe-MI Command.
- Must not generate an error for any value in the version field of the PCIe-MI Command.
- Must process the structure as defined by version N of the structure regardless of the value in the version field of the structure, unless otherwise specified.
  - If the value of the version field is less than the value the PCIe-MI Completer is compliant to, then the following rules apply.
    - If the value of any associated length field (e.g., the PCIe-MI Message Header Length field) is less than the length defined by the initial version of the structure, then an Invalid Input Error Completion with the Error Location field indicating the length field must be returned.

- If the value of any associated length field is greater than the length defined by the initial version of the structure but not equal to the length defined for the specified version of the structure, then an Invalid Input Error Completion with the Error Location field indicating the length field should be returned.
- If the value of any associated length field (e.g., the PCIe-MI Message Header Length field) is greater than the length defined by the version of the structure the PCIe-MI Completer is compliant to, then an Invalid Input Error Completion with the Error Location field indicating the length field must be returned.
- If the value of any associated length field is greater than or equal to the length defined by the initial version of the PCIe-MI Message Header but less than the value defined by the revision of this specification that the PCIe-MI Completer is compliant to, then the structure must be processed without utilizing the information in structure bytes defined in a revision of this specification after the revision of this specification that the structure is compliant to.
- If the value of the version field is greater than the value the PCIe-MI Completer is compliant to, then the following rules apply.
  - If the value of any associated length field is less than the length defined by the revision of this specification that the PCIe-MI Completer is compliant to, then an Invalid Input Error Completion with the Error Location field indicating the length field must be returned.
  - If the value of any associated length field is greater than the length defined by the revision of this specification that the PCIe-MI Completer is compliant to, then the extra bytes that are not defined in the revision of this specification that the PCIe-MI Completer is compliant to must be ignored.
- Must not perform any functionality related to the structure that is not defined by version N of that structure.

A PCIe-MI Completer designed to support revision N of this specification handles the PCIe-MI Message Content as follows.

- Must not generate an error due to PCIe-MI Message Content being compliant to a revision of this specification other than the revision of this specification that the PCIe-MI Completer is compliant to.
- Must process the PCIe-MI Message Content as defined by revision N of this specification regardless of the revision of this specification that the PCIe-MI Command is compliant to, if no errors with any length field in the PCIe-MI Command are detected, unless otherwise specified.
  - If the length of the PCIe-MI Message Content is less than the length defined by the initial version of the PCIe-MI Message Content, then a PCIe-MI Completion with the Status field set to a value of 13h (i.e., PCIe-MI Message Content Length Error) must be returned.
  - If the length of the PCIe-MI Message Content is greater than the length defined by the initial version of the PCIe-MI Message Content, less than the length defined by the revision of this specification the PCIe-MI Completer is compliant to, and not equal to the length defined by any version of the PCIe-MI Message Content that is less than the version the PCIe-MI Completer is compliant to, then a PCIe-MI Completion with the Status field set to a value of 13h (i.e., PCIe-MI Message Content Length Error) should be returned.
  - If the length of the PCIe-MI Message Content is greater than or equal to the length defined by the initial version of the PCIe-MI Message Content but less than the length defined by the revision of this specification that the PCIe-MI Completer is compliant to, then the PCIe-MI Message Content must be processed without utilizing the information in the PCIe-MI Message Content bytes defined in a revision of this specification after the revision of this specification that the PCIe-MI Command is compliant to.

- If the length of the PCIe-MI Message Content is greater than the length defined by the revision of this specification that the PCIe-MI Completer is compliant to, then the extra bytes that are not defined in the revision of this specification that the PCIe-MI Completer is compliant to must be ignored.
- Must not perform any functionality related to the PCIe-MI Message Content that is not defined the revision of this specification that the PCIe-MI Completer is compliant to.

To ensure compatibility with PCIe-MI Completers that comply to any revision of this specification, a PCIe-MI Commander should process PCIe-MI Completions, including accounting for PCIe-MI Completions of varying sizes and versions, using techniques similar to the ones described in the section for PCIe-MI Completers processing PCIe-MI Commands.

### 12.7.3.1 Get Information (Opcode 00h) §

This PCIe-MI Command returns information about a PCIe-MI Component. The type of information returned in the PCIe-MI Completion is specified by an Information Identifier in the Get Information command. The Information Identifiers supported by the Get Information command are listed in § Table 12-31. Each PCIe-MI Completer in a PCIe-MI Component may support a different set of Information Identifiers.

Table 12-31 Information Identifiers §

| Information Identifier | Description                       | Support <sup>1</sup> | Section              |
|------------------------|-----------------------------------|----------------------|----------------------|
| 00h                    | Supported Information Identifiers | M                    | § Section 12.7.3.1.1 |
| 01h                    | Supported Opcodes                 | M                    | § Section 12.7.3.1.2 |
| 02h                    | Supported Parameters              | O/M <sup>2</sup>     | § Section 12.7.3.1.3 |
| 03h                    | Component Information             | O                    | § Section 12.7.3.1.4 |
| 04h                    | Port Information                  | O                    | § Section 12.7.3.1.5 |
| 05h                    | Function List                     | O                    | § Section 12.7.3.1.6 |
| 06h                    | Function Information              | O                    | § Section 12.7.3.1.7 |
| 07h to BFh             | Reserved                          |                      |                      |
| C0h to FFh             | Vendor Specific                   |                      |                      |

**Notes:**

1. O/M definition: O = optional; M = mandatory over MCTP for each physical layer binding over which PCIe-MI over MCTP is supported and mandatory over MMPT if PCIe-MI over MMPT is supported.
2. Mandatory if any optional Parameter Identifiers are supported by the PCIe-MI Completer; otherwise, the Supported Parameters Information Identifier must not be implemented by the PCIe-MI Completer.

The PCIe-MI Command format for the Get Information command is shown in § Table 12-32.

Table 12-32 Get Information Command §

| Byte Location | Field Description                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0           | <b>PCIe-MI Message Header:</b> This field specifies values as per § Table 12-21 with the Opcode field cleared to 00h (i.e., Get Information, see § Table 12-27). |

**Get Information Command Content**

| Byte Location                                          | Field Description                                                                                                                                                                                                                                                               |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8                                                      | <b>Information Identifier:</b> This field specifies the Information Identifier (see § Table 12-31). If this field specifies an unsupported Information Identifier, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |
| <b>Information Identifier-Specific Command Content</b> |                                                                                                                                                                                                                                                                                 |
| N+8:9                                                  | <b>Information Identifier-Specific Command Content:</b> This optional field specifies Information Identifier-specific command content as defined in the following subsections, where N is the size of this field in bytes. If N is 0, then this field is not present.           |

The PCIe-MI Completion format for the Get Information command for each Information Identifier is defined in the following subsections.

#### 12.7.3.1.1 Supported Information Identifiers (Information Identifier 00h) §

This command must return the list of optional and mandatory Information Identifiers (see § Table 12-31) in ascending order of Information Identifier supported by the PCIe-MI Completer on which the Get Information command is received. This command uses the PCIe-MI Command format shown in § Table 12-32 with the Information Identifier field cleared to 00h (i.e., Supported Information Identifiers, see § Table 12-31) and with no optional Information Identifier-specific command content.

The PCIe-MI Completion format for this command is shown in § Table 12-33.

*Table 12-33 Get Information – Supported Information Identifiers Completion §*

| Byte Location                                               | Field Description                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                         | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                                                                                                                                                                                |
| <b>Supported Information Identifiers Completion Content</b> |                                                                                                                                                                                                                                                            |
| 8                                                           | <b>Number of Information Identifiers:</b> This field must indicate the number of Information Identifiers returned (see § Table 12-31).                                                                                                                     |
| 9                                                           | <b>Information Identifier 0:</b> This field must indicate the supported Information Identifier with the lowest value.                                                                                                                                      |
| 10                                                          | <b>Information Identifier 1:</b> This field must indicate the supported Information Identifier with the second lowest value, if there is more than one supported Information Identifier.                                                                   |
| ...                                                         | ...                                                                                                                                                                                                                                                        |
| N+9                                                         | <b>Information Identifier N:</b> This field must indicate the supported Information Identifier with the highest value, if there are more than two supported Information Identifiers, where N is the number of Information Identifiers in the list minus 1. |

#### 12.7.3.1.2 Supported Opcodes (Information Identifier 01h) §

This command must return the list of optional and mandatory opcodes (see § Table 12-27) in ascending order of opcode supported by the PCIe-MI Completer on which the Get Information command is received. This command uses the PCIe-MI Command format shown in § Table 12-32 with the Information Identifier field set to 01h (i.e., Supported Opcodes, see § Table 12-31) and with no optional Information Identifier-specific command content.

The PCIe-MI Completion format for this command is shown in § Table 12-34.

*Table 12-34 Get Information – Supported Opcodes Completion* §

| Byte Location                               | Field Description                                                                                                                                                                |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                         | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                                                                                                      |
| <b>Supported Opcodes Completion Content</b> |                                                                                                                                                                                  |
| 8                                           | <b>Number of Opcodes:</b> This field must indicate the number of Opcodes returned (see § Table 12-27).                                                                           |
| 9                                           | <b>Opcode 0:</b> This field must indicate the supported Opcode with the lowest value.                                                                                            |
| 10                                          | <b>Opcode 1:</b> This field must indicate the supported Opcode with the second lowest value, if there is more than one supported Opcode.                                         |
| ...                                         | ...                                                                                                                                                                              |
| N+9                                         | <b>Opcode N:</b> This field must indicate the Opcode with the highest value, if there are more than two supported Opcodes, where N is the number of Opcodes in the list minus 1. |

### 12.7.3.1.3 Supported Parameters (Information Identifier 02h) §

This command must return the list of Parameter Identifiers (see § Table 12-56) in ascending order of Parameter Identifier supported by the PCIe-MI Completer on which the Get Information command is received. If a Parameter Identifier is supported, then the Get Parameters command for that Parameter Identifier must be supported and the Set Parameters command for that Parameter Identifier is optional. This command uses the PCIe-MI Command format shown in § Table 12-32 with the Information Identifier field set to 02h (i.e., Supported Parameters, see § Table 12-31) and with no optional Information Identifier-specific command content.

The PCIe-MI Completion format for this command is shown in § Table 12-35.

*Table 12-35 Get Information – Supported Parameters Completion* §

| Byte Location                                  | Field Description                                                                                                                        |                                                                                                                                                                        |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                            | <b>PCIe-MI Message Header:</b> This field must be set as per <a href="#">§ Table 12-24</a> .                                             |                                                                                                                                                                        |
| <b>Supported Parameters Completion Content</b> |                                                                                                                                          |                                                                                                                                                                        |
| 8                                              | <b>Number of Parameters:</b> This field must indicate the number of Parameter Identifiers returned (see <a href="#">§ Table 12-56</a> ). |                                                                                                                                                                        |
| 10:9                                           | <b>Parameter 0 Information:</b> This field must indicate information about the supported Parameter Identifier with the lowest value.     |                                                                                                                                                                        |
|                                                | Bits                                                                                                                                     | Description                                                                                                                                                            |
|                                                | 15:9                                                                                                                                     | Reserved                                                                                                                                                               |
|                                                | 8                                                                                                                                        | <b>Set Parameters Support:</b> If the Set Parameters command for this Parameter Identifier is supported, then this bit must be Set; otherwise, this bit must be Clear. |
|                                                | 7:0                                                                                                                                      | <b>Parameter 0:</b> This field must indicate the Parameter Identifier with the lowest value that is supported by the Get Parameters command.                           |



| Byte Location    | Field Description                                                                                                                                                                                                                                                     |                                                                                                                                                                        |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11            | <b>Parameter 1 Information:</b> This field must indicate information about the supported Parameter Identifier with the second lowest value, if there is more than one supported Parameter Identifier.                                                                 |                                                                                                                                                                        |
|                  | Bits                                                                                                                                                                                                                                                                  | Description                                                                                                                                                            |
|                  | 15:9                                                                                                                                                                                                                                                                  | Reserved                                                                                                                                                               |
|                  | 8                                                                                                                                                                                                                                                                     | <b>Set Parameters Support:</b> If the Set Parameters command for this Parameter Identifier is supported, then this bit must be Set; otherwise, this bit must be Clear. |
|                  | 7:0                                                                                                                                                                                                                                                                   | <b>Parameter 1:</b> This field must indicate the Parameter Identifier with the second lowest value that is supported by the Get Parameters command.                    |
| ...              | ...                                                                                                                                                                                                                                                                   |                                                                                                                                                                        |
| (N*2)+10:(N*2)+9 | <b>Parameter N Information:</b> This field must indicate information about the supported Parameter Identifier with the highest value, if there are more than two supported Parameter Identifiers, where N is the number of Parameter Identifiers in the list minus 1. |                                                                                                                                                                        |
|                  | Bits                                                                                                                                                                                                                                                                  | Description                                                                                                                                                            |
|                  | 15:9                                                                                                                                                                                                                                                                  | Reserved                                                                                                                                                               |
|                  | 8                                                                                                                                                                                                                                                                     | <b>Set Parameters Support:</b> If the Set Parameters command for this Parameter Identifier is supported, then this bit must be Set; otherwise, this bit must be Clear. |
|                  | 7:0                                                                                                                                                                                                                                                                   | <b>Parameter N:</b> This field must indicate the Parameter Identifier with the highest value that is supported by the Get Parameters command.                          |

#### 12.7.3.1.4 Component Information (Information Identifier 03h) §

This command returns information about the PCIe-MI Component. This command uses the PCIe-MI Command format shown in § Table 12-32 with the Information Identifier field set to 03h (i.e., Component Information, see § Table 12-31) and with no optional Information Identifier-specific command content.

The PCIe-MI Completion format for this command is shown in § Table 12-39.

*Table 12-39 Get Information – Component Information Completion §*

| Byte Location                                   | Field Description                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                             | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                                                                                                                                                                                                                                                                                |
| <b>Component Information Completion Content</b> |                                                                                                                                                                                                                                                                                                                                                            |
| 8                                               | <b>Number of Ports:</b> This field must indicate the number of supported PCIe-MI Ports on the PCIe-MI Component.                                                                                                                                                                                                                                           |
| 9                                               | <b>Maximum Concurrent Commands:</b> This field must indicate the maximum number of PCIe-MI Commands that are able to be processed concurrently by the PCIe-MI Completer that is generating this PCIe-MI Completion. This field must be set to a value in the range of 1 to 32 and all other values are reserved.                                           |
| 11:10                                           | <b>Maximum Command-To-Completion Time:</b> This field must indicate the estimated maximum time in 100 ms units required by the PCIe-MI Completer that is generating this PCIe-MI Completion to process a PCIe-MI Command and begin transmission of the PCIe-MI Completion, assuming the physical transport external to the PCIe-MI Completer is available. |

| Byte Location | Field Description                                                                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | This time is measured by the PCIe-MI Completer from the end of the reception of a <u>PCIe-MI Command</u> to the beginning of the transmission of the corresponding <u>PCIe-MI Completion</u> . |

### 12.7.3.1.5 Port Information (Information Identifier 04h) §

This command returns information about the PCIe-MI Port specified by the PCIe-MI Port Identifier field. This command uses the PCIe-MI Command format shown in § Table 12-40.

*Table 12-40 Get Information – Port Information Command §*

| Byte Location                                    | Field Description                                                                                                                                                                                                                                                     |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                              | <b>PCIe-MI Message Header:</b> This field specifies values as per § Table 12-32.                                                                                                                                                                                      |
| <b>Port Information Command Content</b>          |                                                                                                                                                                                                                                                                       |
| 8                                                | <b>Information Identifier:</b> This field specifies a value of 04h (i.e., Port Information, see § Table 12-31).                                                                                                                                                       |
| <b>Port Information-Specific Command Content</b> |                                                                                                                                                                                                                                                                       |
| 9                                                | <b>PCIe-MI Port Identifier:</b> This field specifies the PCIe-MI Port Identifier. If this field specifies a PCIe-MI Port Identifier that does not exist, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |

The PCIe-MI Completion format for this command is shown in § Table 12-24.

*Table 12-41 Get Information – Port Information Completion §*

| Byte Location                              | Field Description                                                                                                                  |                                                                                                                                                                                        |                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 7:0                                        | <b>PCIe-MI Message Header:</b> This field must be set as per <a href="#">§ Table 12-24</a> .                                       |                                                                                                                                                                                        |                                          |
| <b>Port Information Completion Content</b> |                                                                                                                                    |                                                                                                                                                                                        |                                          |
| 8                                          | <b>Port Description:</b> This field indicates the type of PCIe-MI Port and whether the PCIe-MI Port is exposed to the host system. |                                                                                                                                                                                        |                                          |
|                                            | Bits                                                                                                                               | Description                                                                                                                                                                            |                                          |
|                                            | 7                                                                                                                                  | <b>PCIe-MI Port Exposed:</b> If this PCIe-MI Port is exposed to the host system (see <a href="#">§ Section 12.7.1</a> ), then this bit must be Set; otherwise, this bit must be Clear. |                                          |
|                                            | 6:2                                                                                                                                | Reserved                                                                                                                                                                               |                                          |
|                                            | 1:0                                                                                                                                | <b>PCIe-MI Port Type:</b> This field must indicate the type of this PCIe-MI Port.                                                                                                      |                                          |
|                                            |                                                                                                                                    | Value                                                                                                                                                                                  | Description                              |
| 00b                                        |                                                                                                                                    | PCIe Upstream Port                                                                                                                                                                     | Yes (see <a href="#">§ Table 12-44</a> ) |
| 01b                                        |                                                                                                                                    | PCIe Downstream Port                                                                                                                                                                   | Yes (see <a href="#">§ Table 12-44</a> ) |
|                                            | 10b                                                                                                                                | 2-Wire Interface Port                                                                                                                                                                  | No                                       |

| Byte Location | Field Description |             |             |                                    |
|---------------|-------------------|-------------|-------------|------------------------------------|
|               | Bits              | Description |             |                                    |
|               |                   | Value       | Description | Port-Specific Information Required |
|               |                   | 11b         | USB Port    | No                                 |
|               |                   |             |             |                                    |
| 11:9          | Reserved          |             |             |                                    |

**Port-Specific Information Completion Content**

|      |                                                                                                  |
|------|--------------------------------------------------------------------------------------------------|
| N:12 | <b>Port-Specific Information:</b> This optional field contains port-specific completion content. |
|------|--------------------------------------------------------------------------------------------------|

The PCIe-MI Completion for this command that indicates a PCIe Upstream Port or PCIe Downstream Port is shown in § Table 12-44.

*Table 12-44 Get Information – PCIe Port-Specific Information Completion §*

| Byte Location | Field Description                            |
|---------------|----------------------------------------------|
| 11:0          | This field must be set as per § Table 12-24. |

**PCIe Port-Specific Information Completion Content**

|       |                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 13:12 | <b>Supported Link Speeds Vector:</b> This field indicates the supported Link speeds of this PCIe Port.                                                                                                                                                                                                                                                                                           |                                                                                                                                                   |
|       | Bits                                                                                                                                                                                                                                                                                                                                                                                             | Description                                                                                                                                       |
|       | 15:6                                                                                                                                                                                                                                                                                                                                                                                             | Reserved                                                                                                                                          |
|       | 5                                                                                                                                                                                                                                                                                                                                                                                                | <b>64.0 GT/s Support:</b> If the <u>Link</u> supports 64.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear.                    |
|       | 4                                                                                                                                                                                                                                                                                                                                                                                                | <b>32.0 GT/s Support:</b> If the <u>Link</u> supports 32.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear.                    |
|       | 3                                                                                                                                                                                                                                                                                                                                                                                                | <b>16.0 GT/s Support:</b> If the <u>Link</u> supports 16.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear.                    |
|       | 2                                                                                                                                                                                                                                                                                                                                                                                                | <b>8.0 GT/s Support:</b> If the <u>Link</u> supports 8.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear.                      |
|       | 1                                                                                                                                                                                                                                                                                                                                                                                                | <b>5.0 GT/s Support:</b> If the <u>Link</u> supports 5.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear.                      |
|       | 0                                                                                                                                                                                                                                                                                                                                                                                                | <b>2.5 GT/s Support:</b> If the <u>Link</u> supports 2.5 GT/s, then this bit must be Set; otherwise, this bit must be Clear.                      |
| 15:14 | <b>Current Link Speed:</b> This field indicates information about the current Link speed of this PCIe port per the following table. If there is no Link (i.e., the PCIe-MI Port Exposed bit for the PCIe-MI Port is Clear) or the Link is not active (i.e., the status of the Data <u>Link</u> Control and Management State Machine is not DL_Active), then this field must be cleared to 0000h. |                                                                                                                                                   |
|       | Bits                                                                                                                                                                                                                                                                                                                                                                                             | Description                                                                                                                                       |
|       | 15:6                                                                                                                                                                                                                                                                                                                                                                                             | Reserved                                                                                                                                          |
|       | 5                                                                                                                                                                                                                                                                                                                                                                                                | <b>64.0 GT/s Current Link Speed:</b> If the current <u>Link</u> speed is 64.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear. |

| Byte Location | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Description                                                                                                                                       |
|               | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>32.0 GT/s Current Link Speed:</b> If the current <u>Link</u> speed is 32.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear. |
|               | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>16.0 GT/s Current Link Speed:</b> If the current <u>Link</u> speed is 16.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear. |
|               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>8.0 GT/s Current Link Speed:</b> If the current <u>Link</u> speed is 8.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear.   |
|               | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>5.0 GT/s Current Link Speed:</b> If the current <u>Link</u> speed is 5.0 GT/s, then this bit must be Set; otherwise, this bit must be Clear.   |
|               | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>2.5 GT/s Current Link Speed:</b> If the current <u>Link</u> speed is 2.5 GT/s, then this bit must be Set; otherwise, this bit must be Clear.   |
| 16            | <b>Maximum Link Width:</b> This field must indicate the supported Link width of this PCIe port. Unlike the Maximum Link Width field in the Link Capabilities Register, this value is not permitted to exceed the number of Lanes routed to the slot (Downstream Port), adapter connector (Upstream Port), or in the case of component-to-component connections, the actual wired connection width. This field is not required to account for card carriers (see § Figure 12-12). Card carriers are able to advertise their maximum Link widths in the Connector Subdivision Combinations Descriptor (see § Table 12-20) in the card carrier's FRU Information Device (see § Table 12-12). |                                                                                                                                                   |
| 17            | <b>Negotiated Link Width:</b> If the Link is active (i.e., the status of the Port's Data Link Control and Management State Machine is DL_Active), then this field must indicate the same value as the Negotiated Link Width field in the Port's Link Status Register. If there is no Link (i.e., the PCIe-MI Port Exposed bit for the PCIe-MI Port is Clear), or the Link is not active (i.e., the status of the Port's Data Link Control and Management State Machine is not DL_Active), then this field must be cleared to 0.                                                                                                                                                           |                                                                                                                                                   |
| 18            | <b>PCIe Port Number:</b> This field must indicate the same value as the Port Number field in the Port's Link Capabilities Register.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                   |
| 19            | <b>Starting Lane Number:</b> This field must indicate the lowest-numbered <u>Lane</u> of this PCIe port using the connector's designated lane numbering without lane reversal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                   |
| 20            | <b>Ending Lane Number:</b> This field must indicate the highest-numbered <u>Lane</u> of this PCIe port using the connector's designated lane numbering without lane reversal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                   |

The PCIe-MI Completion for 2-Wire interface ports must be set as per § Table 12-24 with no Port-Specific Information field.

The PCIe-MI Completion for USB ports must be set as per § Table 12-24 with no Port-Specific Information field.

### 12.7.3.1.6 Function List (Information Identifier 05h) §

This command returns a list of non IOV Functions and Physical Functions matching the specified command inputs that are implemented by the PCIe-MI Component. Virtual Functions must not be included in the PCIe-MI Completion. The PCIe-MI Completion for this command must return the following.

- A Function List Data Structure for each non IOV Function and Physical Function implemented by the PCIe-MI Component, up to the number of Functions specified by the Maximum Function List Data Structures field, starting at the value specified by the Starting Function Identifier field.

- A list of Function List Data Structures in ascending order of Function Identifier.
- A list of Function List Data Structures whose Functions match the type specified by one of the bits Set in the Function Type field.

The Function Identifier for each Function in the PCIe-MI Component must be a four-byte value that is unique within the PCIe-MI Component. Function Identifiers in a PCIe-MI Component are permitted to be any four-byte value and are not required to be consecutive.

This command uses the PCIe-MI Command format shown in § Table 12-47.

*Table 12-47 Get Information – Function List Command §*

| Byte Location                                 | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------|------|----------|---|---------------------------------|---|---------------------------------|---|---------------------------------------|---|-------------------------------------|---|---------------------------------------|---|------------------------------|---|-------|---|-----------------------------|---|----------------------|
| 7:0                                           | <b>PCIe-MI Message Header:</b> This field specifies values as per § Table 12-21 with the Opcode field cleared to 00h.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| <b>Function List Command Content</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 8                                             | <b>Information Identifier:</b> This field specifies a value of 05h (i.e., Function List, see § Table 12-31).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| <b>Function List-Specific Command Content</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 12:9                                          | <b>Starting Function Identifier:</b> This field specifies the starting Function Identifier. If this field specifies a Function Identifier that is greater than the maximum Function Identifier supported by the <u>PCIe-MI Component</u> , then an Invalid Input Error Completion with the Error Location field indicating this field must be returned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 14:13                                         | <b>Maximum Function List Data Structures:</b> This field specifies the maximum number of Function List Data Structures that are permitted to be returned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 16:15                                         | <p><b>Function Type:</b> This field specifies a bitmask of Function types (referred to as the Device/Port Type field in the PCI Express Capabilities Register). If a bit in this field is Set and the criteria specified by the Starting Function Identifier field and Maximum Function List Data Structures field are met for a given Function of the type specified by that bit, then that Function must be included in the <u>PCIe-MI Completion</u>; otherwise, that Function must not be included in the <u>PCIe-MI Completion</u>.</p> <p>This field specifying a value of 0 is not an error.</p> <table> <tr> <th>Bits</th><th>Description</th></tr> <tr> <td>15:9</td><td>Reserved</td></tr> <tr> <td>8</td><td>PCI/PCI-X to PCI Express Bridge</td></tr> <tr> <td>7</td><td>PCI Express to PCI/PCI-X Bridge</td></tr> <tr> <td>6</td><td>Downstream Port of PCI Express Switch</td></tr> <tr> <td>5</td><td>Upstream Port of PCI Express Switch</td></tr> <tr> <td>4</td><td>Root Port of PCI Express Root Complex</td></tr> <tr> <td>3</td><td>Root Complex Event Collector</td></tr> <tr> <td>2</td><td>RCiEP</td></tr> <tr> <td>1</td><td>Legacy PCI Express Endpoint</td></tr> <tr> <td>0</td><td>PCI Express Endpoint</td></tr> </table> | Bits | Description | 15:9 | Reserved | 8 | PCI/PCI-X to PCI Express Bridge | 7 | PCI Express to PCI/PCI-X Bridge | 6 | Downstream Port of PCI Express Switch | 5 | Upstream Port of PCI Express Switch | 4 | Root Port of PCI Express Root Complex | 3 | Root Complex Event Collector | 2 | RCiEP | 1 | Legacy PCI Express Endpoint | 0 | PCI Express Endpoint |
| Bits                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 15:9                                          | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 8                                             | PCI/PCI-X to PCI Express Bridge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 7                                             | PCI Express to PCI/PCI-X Bridge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 6                                             | Downstream Port of PCI Express Switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 5                                             | Upstream Port of PCI Express Switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 4                                             | Root Port of PCI Express Root Complex                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 3                                             | Root Complex Event Collector                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 2                                             | RCiEP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 1                                             | Legacy PCI Express Endpoint                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |
| 0                                             | PCI Express Endpoint                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |             |      |          |   |                                 |   |                                 |   |                                       |   |                                     |   |                                       |   |                              |   |       |   |                             |   |                      |

The PCIe-MI Completion format for this command is shown in § Table 12-49.

*Table 12-49 Get Information – Function List Completion* §

| Byte Location                           | Field Description                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                     | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                                                                                                                                                                                                                                                                                           |
| <b>Function List Completion Header</b>  |                                                                                                                                                                                                                                                                                                                                                                       |
| 9:8                                     | <b>Number of Functions:</b> This field must indicate the number of Function List Data Structures returned (see § Table 12-50).                                                                                                                                                                                                                                        |
| 10                                      | <b>Function List Version:</b> This field indicates the version of the Function List Completion Header and Function List Completion Content. This field must be cleared to 0 for this version of the specification.                                                                                                                                                    |
| 11                                      | <b>Function List Completion Header Length:</b> This field indicates the length in bytes of the Function List Completion Header. This field must be set to 5 for this version of the specification.                                                                                                                                                                    |
| 12                                      | <b>Function List Data Structure Length:</b> This field indicates the length in bytes of each Function List Data Structure (see § Table 12-50). This field must be set to 4 for this version of the specification.                                                                                                                                                     |
| <b>Function List Completion Content</b> |                                                                                                                                                                                                                                                                                                                                                                       |
| M+12:13                                 | <b>Function List 0:</b> This field must indicate the Function List Data Structure (see § Table 12-50) matching the specified command inputs of the Function Identifier with the lowest value, if there is at least one supported Function that matches the specified command inputs, where M is the value indicated by the Function List Data Structure Length field. |
| 2*M+12:M+13                             | <b>Function List 1:</b> This field must indicate the Function List Data Structure of the Function Identifier with the second lowest value that matches the specified command inputs, if there are at least two supported Functions matching the specified command inputs.                                                                                             |
| ...                                     | ...                                                                                                                                                                                                                                                                                                                                                                   |
| (N+1)*M+12:N*M+13                       | <b>Function List N:</b> This field must indicate the Function List Data Structure of the Function Identifier with the highest value that matches the specified command inputs, if there are more than two Functions matching the specified command inputs, where N is the number of Function List Data Structures in the list minus 1.                                |

The Function List Data Structure format is shown in § Table 12-50.

*Table 12-50 Function List Data Structure* §

| Byte Location | Field Description                                                                             |
|---------------|-----------------------------------------------------------------------------------------------|
| 3:0           | <b>Function Identifier:</b> This field must indicate the Function Identifier of the Function. |

### 12.7.3.1.7 Function Information (Information Identifier 06h) §

This command returns information about the Function specified by the Function Identifier field. This command uses the PCIe-MI Command format shown in § Table 12-51.

*Table 12-51 Get Information – Function Information Command* §

| Byte Location                               | Field Description                                                                                                     |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 7:0                                         | <b>PCIe-MI Message Header:</b> This field specifies values as per § Table 12-21 with the Opcode field cleared to 00h. |
| <b>Function Information Command Content</b> |                                                                                                                       |

| Byte Location                                        | Field Description                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8                                                    | <b>Information Identifier:</b> This field specifies a value of 06h (i.e., Function Information, see § Table 12-31).                                                                                                                                                                                                                                                                |
| <b>Function Information-Specific Command Content</b> |                                                                                                                                                                                                                                                                                                                                                                                    |
| 12:9                                                 | <b>Function Identifier:</b> This field specifies the Function Identifier. If this field specifies a Function Identifier that exists in the PCIe-MI Component, then the information for the Function corresponding to that Function Identifier must be returned; otherwise, an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |

The PCIe-MI Completion format for this command is shown in § Table 12-52.

*Table 12-52 Get Information – Function Information Completion §*

| Byte Location                                  | Field Description                                                                                       |                                                                                                                                                                                                                                                                                         |                              |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 7:0                                            | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                             |                                                                                                                                                                                                                                                                                         |                              |
| <b>Function Information Completion Content</b> |                                                                                                         |                                                                                                                                                                                                                                                                                         |                              |
| 9:8                                            | <b>Vendor ID:</b> This field must indicate the Vendor ID of the specified Function.                     |                                                                                                                                                                                                                                                                                         |                              |
| 11:10                                          | <b>Device ID:</b> This field must indicate the Device ID of the specified Function.                     |                                                                                                                                                                                                                                                                                         |                              |
| 13:12                                          | <b>Subsystem Vendor ID:</b> This field must indicate the Subsystem Vendor ID of the specified Function. |                                                                                                                                                                                                                                                                                         |                              |
| 15:14                                          | <b>Subsystem ID:</b> This field must indicate the Subsystem ID of the specified Function.               |                                                                                                                                                                                                                                                                                         |                              |
| 18:16                                          | <b>Class Code:</b> This field must indicate the Class Code of the specified Function.                   |                                                                                                                                                                                                                                                                                         |                              |
| 19                                             | <b>Function Description:</b> This field describes the specified Function.                               |                                                                                                                                                                                                                                                                                         |                              |
|                                                | Bits                                                                                                    | Description                                                                                                                                                                                                                                                                             |                              |
|                                                | 7                                                                                                       | <b>PCIe VDM MCTP Endpoint:</b> If the specified Function has a PCIe VDM MCTP endpoint (see the [MCTP PCIe VDM Binding Specification - <a href="https://www.dmtf.org/dsp/DSP0238">https://www.dmtf.org/dsp/DSP0238</a> ]), then this bit must be Set; otherwise, this bit must be Clear. |                              |
|                                                | 6:4                                                                                                     | Reserved                                                                                                                                                                                                                                                                                |                              |
|                                                | 3:0                                                                                                     | <b>Function Type:</b> This field must indicate the specified Function’s type (referred to as the <a href="#">Device/Port Type</a> field in the <a href="#">PCI Express Capabilities Register</a> ).                                                                                     |                              |
|                                                |                                                                                                         | Value                                                                                                                                                                                                                                                                                   | Description                  |
|                                                |                                                                                                         | 0h                                                                                                                                                                                                                                                                                      | PCI Express Endpoint         |
|                                                |                                                                                                         | 1h                                                                                                                                                                                                                                                                                      | Legacy PCI Express Endpoint  |
|                                                |                                                                                                         | 2h                                                                                                                                                                                                                                                                                      | RCiEP                        |
|                                                |                                                                                                         | 3h                                                                                                                                                                                                                                                                                      | Root Complex Event Collector |
| 4h                                             |                                                                                                         | Root Port of PCI Express Root Complex                                                                                                                                                                                                                                                   |                              |
| 5h                                             |                                                                                                         | Upstream Port of PCI Express Switch                                                                                                                                                                                                                                                     |                              |
| 6h                                             | Downstream Port of PCI Express Switch                                                                   |                                                                                                                                                                                                                                                                                         |                              |

| Byte Location | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                |                                 |             |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------|
|               | Bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                    |                                 |             |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                | Value                           | Description |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 7h                                                                                                                             | PCI Express to PCI/PCI-X Bridge |             |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 8h                                                                                                                             | PCI/PCI-X to PCI Express Bridge |             |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Others                                                                                                                         | Reserved                        |             |
|               | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Number of Capabilities:</b> This field must indicate the number of Capability Data Structures in the Capability List field. |                                 |             |
| 21            | <b>Capability Data Structure Length:</b> This field indicates the length in bytes of each Capability Data Structure in the Capability List field. This field must be set to 4 for this version of the specification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                |                                 |             |
| N+21:22       | <b>Capability List:</b> If the Configuration Space Parameter Identifier (i.e., Parameter Identifier 02h) is supported, then this field must indicate a list containing a Capability Data Structure (see § Table 12-55) for each Capability structure supported by the specified Function using the Get Parameters command with the Configuration Space Parameter Identifier, where N is the number of bytes in the list; otherwise, this field must indicate a list containing a Capability Data Structure for each Capability structure supported by the specified Function using a Configuration Read Request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                |                                 |             |
| N+22          | <b>Path Origin:</b> If the specified Function is downstream from an Upstream Port on the <u>PCIe-MI Component</u> , then this field must indicate the PCIe-MI Port Identifier of that Upstream Port.<br><br>If the specified Function is not downstream from an Upstream Port on the <u>PCIe-MI Component</u> (i.e., a Function in a Root Complex) and the Path Length field does not specify a value of 0, then this field must contain a vendor-specific unique identifier that indicates which Segment the specified Function is on. If the specified Function is not downstream from an Upstream Port on the <u>PCIe-MI Component</u> and the Path Length field specifies a value of 0, then no Segment information is indicated by this field and this field must be cleared to 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                |                                 |             |
| N+23          | <b>Path Length:</b> The Path field is optional but strongly recommended for Functions in Root Complexes; otherwise, the Path field is mandatory. If the Path field is supported, then this field must indicate the length of the Path field in bytes; otherwise, this field must be cleared to 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                |                                 |             |
| M+N+23:N+24   | <b>Path:</b> If the value of the Path Length field does not specify a value of 0, then this field must indicate the hierarchical path on the <u>PCIe-MI Component</u> starting with the Function on the <u>PCIe-MI Component</u> that is furthest upstream from the specified Function and ending with the specified Function, where M is the number of bytes in this field (i.e., the value in the Path Length field) and N is the number of bytes in the Capability List field.<br><br>Each byte in this field represents a Function along the path to the specified Function in order from the furthest upstream Function on the <u>PCIe-MI Component</u> in the lowest byte offset of this field to the specified Function in the highest byte offset of this field. Each byte contains the least-significant 8-bits of the Routing ID of the Function (e.g., the Device Number in bits 7:3 and the Function Number in bits 3:0 for non-ARI Functions, or the Function Number in bits 7:0 for ARI Functions).<br><br>If there is only one Function in the path, then the path consists of a single byte representing the specified Function.<br><br>If there is more than one Function in the path, then the byte in this field with the lowest offset is the Function on the <u>PCIe-MI Component</u> that is furthest upstream from the specified Function and every other byte in this field represents the Function on the path to the specified Function that is on the Secondary Bus of (i.e., downstream from) the Function indicated by the next lower byte offset in this field.<br><br>If the Path Length field is 0, then this field must not be included in the <u>PCIe-MI Completion</u> . |                                                                                                                                |                                 |             |



The Capability Structure Data Structure is shown in § Table 12-55.

*Table 12-55 Capability Structure Data Structure §*

| Byte Location | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1:0           | <p><b>Capability ID:</b> If the Capability structure is in PCI-compatible Configuration Space (i.e., the first 256 bytes of Configuration Space), then bits 7:0 of this field must indicate the Capability ID and bits 15:8 of this field must be set to FEh.</p> <p>If the Capability structure is in Extended Configuration Space (i.e., beyond the first 256 bytes of Configuration Space), then this field must indicate the Extended Capability ID.</p> |
| 3:2           | <p><b>Capability Offset:</b> This field must indicate the byte offset of the Capability structure relative to offset 0 of the Function's Configuration Space.</p>                                                                                                                                                                                                                                                                                            |

### 12.7.3.2 Get Parameters (Opcode 01h) and Set Parameters (Opcode 02h) §

The Get Parameters command returns configurable parameters from a PCIe-MI Component. The Set Parameters command sets configurable parameters on a PCIe-MI Component. The Parameter Identifiers supported by the Get Parameters command and Set Parameters command are listed in § Table 12-56. Each PCIe-MI Completer in a PCIe-MI Component may support a different set of Parameter Identifiers.

*Table 12-56 Parameter Identifiers §*

| Parameter Identifier | Description                     | Support <sup>1</sup> | Section              |
|----------------------|---------------------------------|----------------------|----------------------|
| 00h                  | Configuration Space Writability | O <sup>2</sup>       | § Section 12.7.3.2.1 |
| 01h                  | Routing Information             | O <sup>3</sup>       | § Section 12.7.3.2.2 |
| 02h                  | Configuration Space             | O                    | § Section 12.7.3.2.3 |
| 03h                  | Connector Subdivision           | O <sup>4</sup>       | § Section 12.7.3.2.4 |
| 04h to BFh           | Reserved                        |                      |                      |
| C0h to FFh           | Vendor Specific                 |                      |                      |

#### Notes:

1. O/M definition: O = optional; M = mandatory over MCTP for each physical layer binding over which PCIe-MI over MCTP is supported and mandatory over MMPT if PCIe-MI over MMPT is supported.
2. If there are no Configuration Space bits or fields that are writeable using the Set Parameters command with the Configuration Space Parameter Identifier (i.e., Parameter Identifier 02h) to a PCIe-MI Completer, then this Parameter Identifier must not be implemented by that PCIe-MI Completer.
3. If the PCIe-MI Component does not contain any Downstream Ports, then this Parameter Identifier must not be implemented by any PCIe-MI Completer on the PCIe-MI Component.
4. If the PCIe-MI Component does not contain any Upstream Ports, then this Parameter Identifier must not be implemented by any PCIe-MI Completer on the PCIe-MI Component.

The PCIe-MI Command format for the Get Parameters command is shown in § Table 12-57.

Table 12-57 Get Parameters Command §

| Byte Location                                  | Field Description                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                            | <b>PCIe-MI Message Header:</b> This field specifies values as per § Table 12-21 with the Opcode field set to 01h (i.e., Get Parameters, see § Table 12-27).                                                                                                               |
| <b>Get Parameters Command Content</b>          |                                                                                                                                                                                                                                                                           |
| 8                                              | <b>Parameter Identifier:</b> This field specifies the Parameter Identifier (see § Table 12-56). If this field specifies an unsupported Parameter Identifier, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |
| <b>Get Parameters-Specific Command Content</b> |                                                                                                                                                                                                                                                                           |
| N+8:9                                          | <b>Get Parameters-Specific Command Content:</b> This optional field specifies Parameter Identifier-specific command content as defined in the following subsections, where N is the size of this field in bytes. If N is 0, then this field is not present.               |

The PCIe-MI Command format for the Set Parameters command is shown in § Table 12-58.

Table 12-58 Set Parameters Command §

| Byte Location                                  | Field Description                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                            | <b>PCIe-MI Message Header:</b> This field specifies values as per § Table 12-21 with the Opcode field set to 02h (i.e., Set Parameters, see § Table 12-27).                                                                                                               |
| <b>Set Parameters Command Content</b>          |                                                                                                                                                                                                                                                                           |
| 8                                              | <b>Parameter Identifier:</b> This field specifies the Parameter Identifier (see § Table 12-56). If this field specifies an unsupported Parameter Identifier, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |
| <b>Set Parameters-Specific Command Content</b> |                                                                                                                                                                                                                                                                           |
| N+8:9                                          | <b>Set Parameters-Specific Command Content:</b> This optional field specifies Parameter Identifier-specific command content as defined in the following subsections, where N is the size of this field in bytes. If N is 0, then this field is not present.               |

The PCIe-MI Completion format for the Set Parameters command and Get Parameters command for each Parameter Identifier is defined in the following subsections.

### 12.7.3.2.1 Configuration Space Writability (Parameter Identifier 00h) §

This parameter provides commands that perform the following.

- Retrieves a list of Configuration Space bits and fields on the specified Function that are writeable using the Set Parameters command with the Configuration Space Parameter Identifier (i.e., Parameter Identifier 02h) on the PCIe-MI Completer on which the Get Parameters command is received (e.g., writeable by a BMC on an out-of-band management transport).
  - Indicates which of those writeable Configuration Space bits and fields are lockable (i.e., able to be set to read-only) when accessed using a Configuration Write Request (e.g., lockable when accessed by in-band management system software).

- Sets lockable Configuration Space bits and fields to read-only when accessed using a Configuration Write Request.

## IMPLEMENTATION NOTE:

### CONFIGURATION SPACE WRITABILITY EXAMPLE §

The list of Configuration Space bits and fields that are writeable when accessed using the Set Parameters command with the Configuration Space Parameter Identifier are able to be described by an \_OSC Control field bit (see the PCI Firmware Specification) or by a Capability structure or Extended Capability structure (see the Configuration Space Writability Data Structure Type field and the Configuration Space Writability Data Structure Information field in § Table 12-61 and § Table 12-66). If all RW, RW1C, RWS, and RW1CS bits and fields in a Capability structure or Extended Capability structure are writeable when accessed using the Set Parameters command with the Configuration Space Parameter Identifier, then this list is able to indicate that Capability structure or Extended Capability structure.

However, there may be cases where the Configuration Space bits and fields that are required to be writeable when accessed using the Set Parameters command with the Configuration Space Parameter Identifier are not able to be described by a Capability structure or Extended Capability structure. For example, using the \_OSC method, system firmware may retain control of the Completion Timeout Value field and the Completion Timeout Disable bit, while granting control of all other bits and fields in the PCI Express Capability structure (i.e., the PCI Capability structure with a Capability ID of 10h) to the operating system running on the host CPU(s).

A BMC on such a system may require using the Set Parameters command with the Configuration Space Parameter Identifier to write the Completion Timeout Value field and the Completion Timeout Disable bit. A PCIe-MI Completer could support writing all writeable bits and fields in the PCI Express Capability structure using the Set Parameters command with the Configuration Space Parameter Identifier in order to allow the BMC to write to the Completion Timeout Value field and the Completion Timeout Disable bit within the PCI Express Capability structure. However, a PCIe-MI Completer may want to allow the Completion Timeout Value field and the Completion Timeout Disable bit to be writeable using the Set Parameters command with the Configuration Space Parameter Identifier while prohibiting any other bits or fields in the PCI Express Capability structure from being writeable using the Set Parameters command with the Configuration Space Parameter Identifier in order to prevent conflicts with the operating system running on the host CPU(s) that is in control of those other bits and fields.

To satisfy this use case, the PCIe-MI Completer is able to allow a BMC to write the Completion Timeout Value field and the Completion Timeout Disable bit using the Set Parameters command with the Configuration Space Parameter Identifier but prohibit the BMC from writing other bits or fields in the PCI Express Capability structure by advertising support for writability of the bits and fields defined as writeable only by firmware when firmware retains control of the PCI Express Completion Timeout Control bit (i.e., bit 8 of the \_OSC Control field defined by the PCI Firmware Specification). See the \_OSC Control Field Bit Offset Value in § Table 12-61 and § Table 12-66 for additional details.

#### 12.7.3.2.1.1 Get Parameters – Configuration Space Writability §

This command returns a list of Configuration Space bits and fields on the specified Function that are writeable when accessed using the Set Parameters command with the Configuration Space Parameter Identifier on the PCIe-MI Completer on which the Get Parameters command is received (e.g., writeable by a BMC on an out-of-band management transport) and indicates which of the writeable Configuration Space bits and fields are lockable (i.e., able to be set to read-only) when accessed using a Configuration Write Request (e.g., lockable when accessed by in-band management system software).

This command uses the PCIe-MI Command format shown in § Table 12-59.

Table 12-59 Get Parameters – Configuration Space Writability Command §

| Byte Location                                                                    | Field Description                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                                              | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-57.                                                                                                                                                                             |
| <b>Get Parameters - Configuration Space Writability Command Content</b>          |                                                                                                                                                                                                                                                           |
| 8                                                                                | <b>Parameter Identifier:</b> This field is cleared to 00h to specify the Configuration Space Writability Parameter Identifier (see § Table 12-56).                                                                                                        |
| <b>Get Parameters - Configuration Space Writability-Specific Command Content</b> |                                                                                                                                                                                                                                                           |
| 12:9                                                                             | <b>Function Identifier:</b> This field specifies the Function Identifier. If this field specifies a Function Identifier that does not exist, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |

The PCIe-MI Completion format for this command is shown in § Table 12-60.

Table 12-60 Get Parameters – Configuration Space Writability Completion §

| Byte Location                                                              | Field Description                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                                        | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                                                                                                                                                                                                                                                                                             |
| <b>Get Parameters - Configuration Space Writability Completion Header</b>  |                                                                                                                                                                                                                                                                                                                                                                         |
| 8                                                                          | <b>Number of Configuration Space Writability Data Structures:</b> This field must indicate the number of Configuration Space Writability Data Structures returned (see § Table 12-61).                                                                                                                                                                                  |
| 9                                                                          | <b>Configuration Space Writability Completion Version:</b> This field indicates the version of the Configuration Space Writability Completion Header and Configuration Space Writability Completion Content. This field must be cleared to 0 for this version of the specification.                                                                                     |
| 10                                                                         | <b>Configuration Space Writability Completion Header Length:</b> This field indicates the length in bytes of the Configuration Space Writability Completion Header. This field must be set to 4 for this version of the specification.                                                                                                                                  |
| 11                                                                         | <b>Configuration Space Writability Data Structure Length:</b> This field indicates the length in bytes of each Configuration Space Writability Data Structure (see § Table 12-61). This field must be set to 3 for this version of the specification.                                                                                                                   |
| <b>Get Parameters - Configuration Space Writability Completion Content</b> |                                                                                                                                                                                                                                                                                                                                                                         |
| M+11:12                                                                    | <b>Configuration Space Writability Data Structure 0:</b> This field must indicate the first Configuration Space Writability Data Structure (see § Table 12-61), if there is at least one Writeable Configuration Data Structure supported by the specified Function, where M is the value indicated by the Configuration Space Writability Data Structure Length field. |
| 2*M+11:M+12                                                                | <b>Configuration Space Writability Data Structure 1:</b> This field must indicate the second Configuration Space Writability Data Structure supported by the specified Function, if there is more than one Writeable Configuration Data Structures.                                                                                                                     |
| ...                                                                        | ...                                                                                                                                                                                                                                                                                                                                                                     |
| (N+1)*M+11:N*M+12                                                          | <b>Configuration Space Writability Data Structure N:</b> This field must indicate the last Configuration Space Writability Data Structure, if there are more than two Writeable Configuration Data Structures supported by the specified Function, where N is the number of Configuration Space Writability Data Structures in the list minus 1.                        |

The Configuration Space Writability Data Structure format is shown in § Table 12-61.

Table 12-61 Configuration Space Writability Data Structure §

| Byte Location | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | <b>Configuration Space Writability Data Structure Information:</b> This field indicates information about the Configuration Space bits and fields that are writeable when accessed using the Set Parameters command with the Configuration Space Parameter Identifier on the PCIe-MI Completer on which the Get Parameters command is received (e.g., writeable by a BMC on an out-of-band management transport) and information about which Configuration Space bits and fields are able to be locked (i.e., configured as read only) when accessed using a Configuration Write Request (e.g., lockable when accessed by in-band management system software). |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|               | Bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|               | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Lockable In-band:</b> If the Configuration Space Writability Data Structure Type bit is 0b (i.e., a Capability Structure) and all of the writeable Configuration Space bits and fields (i.e., bits and fields that are RW, RW1C, RWS, and RW1CS) in the Capability Structure are able to be locked from modification when accessed using a Configuration Write Request (e.g., by in-band management system software) by setting the Lock bit in a Set Parameters command with the Configuration Space Writability Parameter Identifier (see § Table 12-66), then this bit must be Set; otherwise, this bit must be Clear.<br><br>If the Configuration Space Writability Data Structure Type bit is 1b (i.e., an _OSC Control field bit) and all of the writeable Configuration Space bits and fields (i.e., bits and fields that are RW, RW1C, RWS, and RW1CS) indicated by the _OSC Control field bit (see the PCI Firmware Specification) as writeable only by firmware when firmware retains control of the feature (i.e., the operating system is never permitted to write to the bits or fields) are able to be locked from modification when accessed using a Configuration Write Request by setting the Lock bit in the Set Parameters command with the Configuration Space Writability Parameter Identifier, then this bit must be Set; otherwise, this bit must be Clear. |
|               | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Locked In-band:</b> If the lockable bits and fields described by this data structure are locked from modification (i.e., the writeable bits and fields that are able to be locked are configured as read-only) when accessed using a Configuration Write Request (e.g., by in-band management system software), then this bit must be Set; otherwise, this bit must be Clear.<br><br>The default value of this bit following a Conventional Reset or Function Level Reset of the specified Function is implementation specific (e.g., see the SFI RO In-band Access bit in the SFI Control Register). Any type of reset other than a Conventional Reset or Function Level Reset of the specified Function (e.g., a PCIe-MI Completer Reset caused by the 2-Wire clock being held low for longer than $T_{dcl}$ ) must not cause this bit to change value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|               | 5:1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|               | <b>Configuration Space Writability Data Structure Type:</b> This bit must indicate the type of the Configuration Space Writability Data Structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|               | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|               | 0b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Capability Structure:</b> The type of the Configuration Space Writability Data Structure is a Capability structure or an Extended Capability structure. All writeable Configuration Space bits and fields (i.e., bits and fields that are RW, RW1C, RWS, and RW1CS) in the indicated Capability structure or Extended Capability structure must be writeable when accessed using the Set Parameters command with the Configuration Space Parameter Identifier on the PCIe-MI Completer on which this Get Parameters command is received.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|               | 1b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>_OSC Control Field Bit:</b> The type of the Configuration Space Writability Data Structure is an _OSC Control field. All writeable Configuration Space bits and fields (i.e., bits and fields that are RW, RW1C, RWS, and RW1CS) that are defined as writeable by firmware by the indicated _OSC Control field bit when firmware retains control of the feature must be writeable when accessed using the Set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Byte Location | Field Description                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Bits                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                                                                                                                                                                                                    |
|               |                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                |
|               | Value                                                                                                                                                                                                                                                                                                                                       | Description                                                                                                                                                                                                                                                                                                    |
|               |                                                                                                                                                                                                                                                                                                                                             | Parameters command with the Configuration Space Parameter Identifier on the <u>PCIe-MI Completer</u> on which this Get Parameters command is received.                                                                                                                                                         |
| 2:1           | <b>Configuration Space Writability Data Structure Offset:</b> If the type of the Configuration Space Writability Data Structure is a Capability structure or Extended Capability structure (i.e., the Configuration Space Writability Data Structure Type bit is 0b), then this field must indicate the information in the following table. |                                                                                                                                                                                                                                                                                                                |
|               | Bits                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                                                                                                                                                                                                    |
|               | 15:12                                                                                                                                                                                                                                                                                                                                       | Reserved                                                                                                                                                                                                                                                                                                       |
|               | 11:0                                                                                                                                                                                                                                                                                                                                        | <b>Configuration Space Byte Offset:</b> This field indicates the starting byte offset of the Capability structure or Extended Capability structure.                                                                                                                                                            |
|               | If the type of the Configuration Space Writability Data Structure is an _OSC Control field (i.e., the Configuration Space Writability Data Structure Type bit is 1b), then this field must indicate the information in the following table.                                                                                                 |                                                                                                                                                                                                                                                                                                                |
|               | Bits                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                                                                                                                                                                                                    |
|               | 15:5                                                                                                                                                                                                                                                                                                                                        | Reserved                                                                                                                                                                                                                                                                                                       |
|               | 4:0                                                                                                                                                                                                                                                                                                                                         | <b>_OSC Control Field Bit Offset Value:</b> This field indicates the value of the _OSC Control field bit offset.<br>For example, for the PCI Express Completion Timeout Control bit, which is defined as bit 8 of the _OSC Control field by the PCI Firmware Specification, this field is set to a value of 8. |
|               |                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                |
|               |                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                |

### 12.7.3.2.1.2 Set Parameters – Configuration Space Writability §

This command locks the specified Configuration Space bits and fields of the specified Function from being able to be modified (i.e., configures the bits and fields as read only) or unlocks the bits and fields so they are able to be modified when accessed using a Configuration Write Request (e.g., when accessed by in-band management system software). This command uses the PCIe-MI Command format shown in § Table 12-66.

*Table 12-66 Set Parameters – Configuration Space Writability Command §*

| Byte Location                                                                    | Field Description                                                                                                                                  |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                                              | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-58.                                                                      |
| <b>Set Parameters - Configuration Space Writability Command Content</b>          |                                                                                                                                                    |
| 8                                                                                | <b>Parameter Identifier:</b> This field is cleared to 00h to specify the Configuration Space Writability Parameter Identifier (see § Table 12-56). |
| <b>Set Parameters - Configuration Space Writability-Specific Command Content</b> |                                                                                                                                                    |

| Byte Location | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:9          | <b>Function Identifier:</b> This field specifies the Function Identifier of the Function on which the Configuration Space bit or field writability when accessed using a Configuration Write Request must be set as specified by the Lock bit in the Configuration Space Writability Data Structure Information field. If this field specifies a Function Identifier that does not exist, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                            |
| 13            | <b>Configuration Space Writability Data Structure Information:</b> This field specifies the type of the Configuration Space Writability Data Structure and whether the specified Configuration Space bits and fields are to be locked when accessed using a Configuration Write Request.                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                            |
|               | Bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                            |
|               | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Lock:</b> If this bit is Set and the specified Configuration Space bits and fields contain writeable bits or fields that are lockable, then the specified writeable Configuration Space bits and fields that are lockable must be locked from modification (i.e., set to read-only) when accessed using a Configuration Write Request; otherwise, the specified writeable Configuration Space bits and fields must not be locked from modification when accessed using a Configuration Write Request.<br><br>This command specifying to lock Configuration Space bits and fields that are already locked from modification when accessed using a Configuration Write Request or specifying to unlock Configuration Space bits and fields that are already not locked from modification when accessed using a Configuration Write Request is not an error. |                                                                                                                                                            |
|               | 6:1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                            |
|               | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Configuration Space Writability Data Structure Type:</b> This bit specifies the type of the Configuration Space Writability Data Structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                            |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                                                |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Capability Structure:</b> The type of the Configuration Space Writability Data Structure is a Capability structure or an Extended Capability structure. |
|               | 1h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>_OSC Control Field Bit:</b> The type of the Configuration Space Writability Data Structure is an _OSC Control field bit (see the PCI Firmware Specification).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                            |
|               | 15:14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Configuration Space Writability Data Structure Offset:</b> If the type of the Configuration Space Writability Data Structure is a Capability structure or Extended Capability structure (i.e., the Configuration Space Writability Data Structure Type field is 0h), then this field specifies information about the Capability structure or Extended Capability structure as described in the following table.                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                            |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Description                                                                                                                                                |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 15:12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Reserved                                                                                                                                                   |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 11:0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Configuration Space Byte Offset:</b> This field specifies the starting byte offset of the Capability structure or Extended Capability structure.        |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | If the type of the Configuration Space Writability Data Structure is an _OSC Control field (i.e., the Configuration Space Writability Data Structure Type field is 1h), then this field specifies information about the _OSC Control field as described in the following table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                            |
| Bits          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                            |
| 15:5          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                            |

| Byte Location | Field Description                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Bits                                                                                                                                                                                                                                                                                     | Description                                                                                                                                                                                                                                                                                                                |
|               | 4:0                                                                                                                                                                                                                                                                                      | <p><b>_OSC Control Field Bit Offset Value:</b> This field specifies the value of the _OSC Control field bit offset.</p> <p>For example, for the PCI Express Completion Timeout Control bit, which is defined as bit 8 of the _OSC Control field by the PCI Firmware Specification, this field specifies a value of 8h.</p> |
|               | <p>If the writeable Configuration Space bits and fields specified by this command are not lockable from modification when accessed using a Configuration Write Request, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned.</p> |                                                                                                                                                                                                                                                                                                                            |

The PCIe-MI Completion for this command must be set as per § Table 12-24 with no PCIe-MI Message Content field.

#### 12.7.3.2.2 Routing Information (Parameter Identifier 01h) §

This parameter gets the Segment Number and/or downstream bus number range of the specified Downstream Port or causes the specified Downstream Port to generate a downstream Configuration Write Request in order to set the captured Segment Number, Bus Number, and/or Device Number on the specified downstream Function.

### IMPLEMENTATION NOTE: ROUTING INFORMATION VS. SFI CAM §

If eSFI is supported and a Function has not captured a Segment Number, Bus Number, and/or Device Number (e.g., due to being hidden from the operating system running on a host CPU with the SFI DPF Control field set to a value of 11b to allow MCTP VDMs), then the SFI CAM may be used to generate a Configuration Write Request to a downstream Function in order to cause the Function to capture a Segment Number, Bus Number, and/or Device Number. Programming the Segment Number, Bus Number, and/or Device Number of the Function enables the BMC to configure the Function using the MCTP over PCIe VDM transport which typically has a higher bandwidth than sideband buses.

If the SFI CAM is not supported, then this parameter provides an alternative mechanism for a BMC to set the captured Segment Number, Bus Number, and/or Device Number of a downstream Function.

#### 12.7.3.2.2.1 Get Parameters – Routing Information §

This command returns the routing information (i.e., the Segment Number and/or downstream bus number range) of the specified Downstream Port. This command uses the PCIe-MI Command format shown in § Table 12-71



Table 12-71 Get Parameters – Routing Information Command §

| Byte Location                                                        | Field Description                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                                  | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-57.                                                                                                                                                                                                                                                                                                         |
| <b>Get Parameters - Routing Information Command Content</b>          |                                                                                                                                                                                                                                                                                                                                                                                       |
| 8                                                                    | <b>Parameter Identifier:</b> This field is set to 01h to specify the Routing Information Parameter Identifier (see § Table 12-56).                                                                                                                                                                                                                                                    |
| <b>Get Parameters - Routing Information-Specific Command Content</b> |                                                                                                                                                                                                                                                                                                                                                                                       |
| 12:9                                                                 | <b>Function Identifier:</b> This field specifies the Function Identifier of the Downstream Port whose routing information is requested. If this field specifies a Function Identifier that is not the Function Identifier of a Downstream Port on the PCIe-MI Component, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |

The PCIe-MI Completion format for this command is shown in § Table 12-72.

Table 12-72 Get Parameters – Routing Information Completion §

| Byte Location                                                  | Field Description                                                                                                                                                                   |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                            | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                                                                                                         |
| <b>Get Parameters - Routing Information Completion Content</b> |                                                                                                                                                                                     |
| 8                                                              | <b>Secondary Bus Number:</b> This field must indicate the Downstream Port's Secondary Bus Number.                                                                                   |
| 9                                                              | <b>Subordinate Bus Number:</b> This field must indicate the Downstream Port's Subordinate Bus Number.                                                                               |
| 11:10                                                          | <b>Segment Number Information:</b> This field indicates information about the Downstream Port's Segment Number.                                                                     |
| 15:9                                                           | Reserved                                                                                                                                                                            |
| 8                                                              | <b>Segment Number Valid:</b> If the Downstream Port has a Segment Number, then this bit must be Set; otherwise, this bit must be Clear.                                             |
| 7:0                                                            | <b>Segment Number:</b> If the Downstream Port has a Segment Number, then this field must indicate the Downstream Port's Segment Number; otherwise, this field must be cleared to 0. |

### 12.7.3.2.2 Set Parameters – Routing Information §

This command must cause the specified Downstream Port to generate a Configuration Write Request to offset 0h of the specified downstream Function's Configuration Space in order to set the captured Segment Number, Bus Number, and/or Device Number of the specified Function. The Configuration Write Request must be Type 0 or Type 1 based on the Configuration Request routing and conversion rules defined in § Section 7.3.3 . The size of the data payload for the Configuration Write Request is permitted to be any value permitted for Configuration Write Requests. The content of the data payload for the Configuration Write Request is permitted to be any value. The Routing ID used in the Configuration Write Request is specified in the Routing ID field. If there is a Segment Number for the Downstream Port and the Link is in Flit Mode, then the Segment Number in the Configuration Write Request must be set to the Downstream Port's Segment Number.

If an error external to the PCIe-MI Component is detected with a Configuration Write Request (e.g., a Configuration Write Request to a Function external to the PCIe-MI Component timed out or was completed with a UR or CA Completion

Status), then the PCIe-MI Completer must return a PCIe-MI Completion with the Status field set to 11h (i.e., External Error) and must not signal the error using any other architected mechanisms (e.g., AER).

This command uses the PCIe-MI Command format shown in § Table 12-73.

*Table 12-73 Set Parameters – Routing Information Command §*

| Byte Location                                                        | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                                  | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-58.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Set Parameters - Routing Information Command Content</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8                                                                    | <b>Parameter Identifier:</b> This field is set to 01h to specify the Routing Information Parameter Identifier (see § Table 12-56).                                                                                                                                                                                                                                                                                                                                            |
| <b>Set Parameters - Routing Information-Specific Command Content</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 12:9                                                                 | <b>Function Identifier:</b> This field specifies the Function Identifier of the Downstream Port that is requested to generate the Configuration Write Request. If this field specifies a Function Identifier that is not the Function Identifier of a Downstream Port on the PCIe-MI Component, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned.                                                                  |
| 14:13                                                                | <b>Routing ID:</b> This field specifies the Routing ID that the Downstream Port must use in the Configuration Write Request. If the Bus Number portion of this field (i.e., bits 15:8) specifies a value that is not greater than or equal to the Downstream Port's Secondary Bus Number and less than or equal to the Downstream Port's Subordinate Bus Number, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |

The PCIe-MI Completion for this command must be set as per § Table 12-24 with no PCIe-MI Message Content field.

### 12.7.3.2.3 Configuration Space (Parameter Identifier 02h) §

This parameter provides commands that get and set the contents of the specified Configuration Space.

A Function's Configuration Space layout is recommended to be identical when accessed using Configuration Request TLPs and using this parameter (e.g., each Capability structure, Extended Capability structure, register, bit, and field is accessible at the same offset using either mechanism), unless otherwise specified. If a Capability structure, Extended Capability structure, register, bit, or field is accessible using only one of these mechanisms, then that Configuration Space location when accessed from the other mechanism is recommended to be read-only and return all 0's when read, unless otherwise specified.

If the PCIe-MI Completer supports this parameter on more than one PCIe-MI Completer, then the Configuration Space must be identical when accessed via any of those PCIe-MI Completers.

#### 12.7.3.2.3.1 Get Parameters – Configuration Space §

This command returns the contents of the specified Configuration Space. This command uses the PCIe-MI Command format shown in § Table 12-74. The Configuration Space of a Function that is read using this command may not match the Configuration Space of that Function when read using a Configuration Read Request. For example, if the SFI Extended Capability Hidden bit is Set, then the SFI Extended Capability shows up in the linked list of PCI Express Extended Capabilities when read using this command and does not show up in the linked list of PCI Express Extended Capabilities when read using a Configuration Read Request.

Table 12-74 Get Parameters – Configuration Space Command §

| Byte Location                                                        | Field Description                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                                  | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-57.                                                                                                                                                                                                                                                                       |
| <b>Get Parameters – Configuration Space Command Content</b>          |                                                                                                                                                                                                                                                                                                                                                     |
| 8                                                                    | <b>Parameter Identifier:</b> This field is set to 02h to specify the Configuration Space Parameter Identifier (see § Table 12-56).                                                                                                                                                                                                                  |
| <b>Get Parameters – Configuration Space-Specific Command Content</b> |                                                                                                                                                                                                                                                                                                                                                     |
| 12:9                                                                 | <b>Function Identifier:</b> This field specifies the Function Identifier of the Function from which Configuration Space is read. If this field specifies a Function Identifier that does not exist, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned.                                    |
| 14:13                                                                | <b>Length:</b> This field specifies the number of Configuration Space bytes to read starting at the offset specified by the Offset field. If this value is 0h or greater than 4,096, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned.                                                   |
| 16:15                                                                | <b>Offset:</b> This field specifies the starting byte offset of the Configuration Space to read. If the value of the Length field is less than or equal to 4,096 and the sum of this field and the Length field is greater than 4,096, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |

The PCIe-MI Completion format for this command is shown in § Table 12-75.

Table 12-75 Get Parameters – Configuration Space Completion §

| Byte Location                                                 | Field Description                                                                                                                                                                                                                         |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                           | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24.                                                                                                                                                               |
| <b>Get Parameters – Configuration Space Completion Header</b> |                                                                                                                                                                                                                                           |
| N+7:8                                                         | <b>Configuration Space Data:</b> This field must indicate the contents of the Configuration Space specified by the Get Parameters command, where N is equal to the value of the Length field in the corresponding Get Parameters command. |

### 12.7.3.2.3.2 Set Parameters – Configuration Space §

This command writes to the specified Configuration Space. The PCIe-MI Completer must perform the following steps in the order listed as an atomic operation.

1. Read the DWORD of Configuration Space specified by the Offset field.
2. Perform a logical AND of the data read in step 1 with the value specified by the AND Mask field.
3. Perform a logical OR of the data from the result of step 2 with the value specified by the OR Mask field.
4. Write the DWORD of Configuration Space specified by the Offset field with the data from the result of step 3.

It is not an error if this command specifies a different value than the current value of bits that are not writeable using this command. Those bits are read-only and hence, are not modified by this command.

This command uses the PCIe-MI Command format shown in § Table 12-76.

Table 12-76 Set Parameters – Configuration Space Command §

| Byte Location                                                        | Field Description                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                                  | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-58.                                                                                                                                                                                                                                                       |
| <b>Set Parameters – Configuration Space Command Content</b>          |                                                                                                                                                                                                                                                                                                                                     |
| 8                                                                    | <b>Parameter Identifier:</b> This field is set to 02h to specify the Configuration Space Parameter Identifier (see § Table 12-56).                                                                                                                                                                                                  |
| <b>Set Parameters – Configuration Space-Specific Command Content</b> |                                                                                                                                                                                                                                                                                                                                     |
| 12:9                                                                 | <b>Function Identifier:</b> This field specifies the Function Identifier of the Function whose Configuration Space is written. If this field specifies a Function Identifier that does not exist, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned.                      |
| 14:13                                                                | <b>Offset:</b> This field specifies the byte offset of the DWORD of Configuration Space to write. If the value of this field is not DWORD aligned (i.e., bits 1:0 of this field are not 00b) or is greater than 0FFCh, then an Invalid Input Error Completion with the Error Location field indicating this field must be returned. |
| 18:15                                                                | <b>AND Mask:</b> This field specifies the AND mask to apply to the data.                                                                                                                                                                                                                                                            |
| 22:19                                                                | <b>OR Mask:</b> This field specifies the OR mask to apply to the data.                                                                                                                                                                                                                                                              |

The PCIe-MI Completion for this command must be set as per § Table 12-24 with no PCIe-MI Message Content field.

#### 12.7.3.2.4 Connector Subdivision (Parameter Identifier 03h) §

This parameter gets or sets the connector subdivisions of an Upstream Port.

##### 12.7.3.2.4.1 Get Parameters – Connector Subdivision §

This command returns the supported upstream-facing connector subdivisions of the PCIe-MI Component. This command uses the PCIe-MI Command format shown in § Table 12-77.

Table 12-77 Get Parameters – Connector Subdivision Command §

| Byte Location                                                 | Field Description                                                                                                                    |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                                           | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-57.                                                        |
| <b>Get Parameters - Connector Subdivision Command Content</b> |                                                                                                                                      |
| 8                                                             | <b>Parameter Identifier:</b> This field is set to 03h to specify the Connector Subdivision Parameter Identifier (see § Table 12-56). |

The PCIe-MI Completion format for this command is shown in § Table 12-78.

Table 12-78 Get Parameters – Connector Subdivision Completion §

| Byte Location                                                    | Field Description                                                           |
|------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 7:0                                                              | <b>PCIe-MI Message Header:</b> This field must be set as per § Table 12-24. |
| <b>Get Parameters - Connector Subdivision Completion Content</b> |                                                                             |

| Byte Location | Field Description                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8             | <b>Current Connector Subdivision Index:</b> This field must indicate the index in the Connector Subdivision Combinations Descriptor field of the connector subdivision that is currently set.             |
| N+8:9         | <b>Connector Subdivision Combinations Descriptor:</b> This field must indicate the supported connector subdivisions in the format specified by § Table 12-20, where N is the size of this field in bytes. |

#### 12.7.3.2.4.2 Set Parameters – Connector Subdivision §

This command sets the specified connector subdivision. The PCIe-MI Component reverts to the default Connector Subdivision Descriptor only on a Cold Reset as defined by § Section 12.6.3.1. This command uses the PCIe-MI Command format shown in § Table 12-79.

*Table 12-79 Set Parameters – Connector Subdivision Command §*

| Byte Location | Field Description                                                             |
|---------------|-------------------------------------------------------------------------------|
| 7:0           | <b>PCIe-MI Message Header:</b> This field specifies values per § Table 12-58. |

#### Set Parameters - Connector Subdivision Command Content

|   |                                                                                                                                      |
|---|--------------------------------------------------------------------------------------------------------------------------------------|
| 8 | <b>Parameter Identifier:</b> This field is set to 03h to specify the Connector Subdivision Parameter Identifier (see § Table 12-56). |
|---|--------------------------------------------------------------------------------------------------------------------------------------|

#### Set Parameters - Connector Subdivision-Specific Command Content

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9 | <p><b>Connector Subdivision Index:</b> This field specifies the index in the Connector Subdivision Combinations Descriptor of the connector subdivision to set (e.g., a value of 0 specifies Connector Subdivision 0 and a value of 1 specifies Connector Subdivision 1).</p> <p>If this field specifies a valid index other than the index associated with the current link subdivision and all Links in the current and specified connector subdivision are in DL_Down, then the PCIe-MI Component must set the connector subdivision to the connector subdivision corresponding to the index specified by this field prior to returning the PCIe-MI Completion for this command.</p> <p>If this field specifies a valid index other than the index associated with the current link subdivision and all Links in the current and specified connector subdivision are not in DL_Down, then the following rules apply.</p> <ul style="list-style-type: none"> <li>• The PCIe-MI Component must return a PCIe-MI Completion for this command with the Status field cleared to 00h (i.e., Success) and then set the connector subdivision to the connector subdivision corresponding to the index specified by this field the next time all Links in the current and specified connector subdivision are in DL_Down unless the PCIe-MI Component undergoes a Cold Reset.</li> <li>• If the connector subdivision is set multiple times while all associated Links are not in DL_Down, then the PCIe-MI Component must set the connector subdivision corresponding to the index specified by this field in the most recent occurrence of this command.</li> </ul> <p>If this field specifies the index associated with the current link subdivision, then a PCIe-MI Completion for this command with the Status field cleared to 00h (i.e., Success) must be returned.</p> <p>If this field specifies an invalid index, an Invalid Input Error Completion with the Error Location field indicating this field must be returned.</p> |
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

The PCIe-MI Completion for this command is shown in § Table 12-80

Table 12-80 Set Parameters – Connector Subdivision Completion §

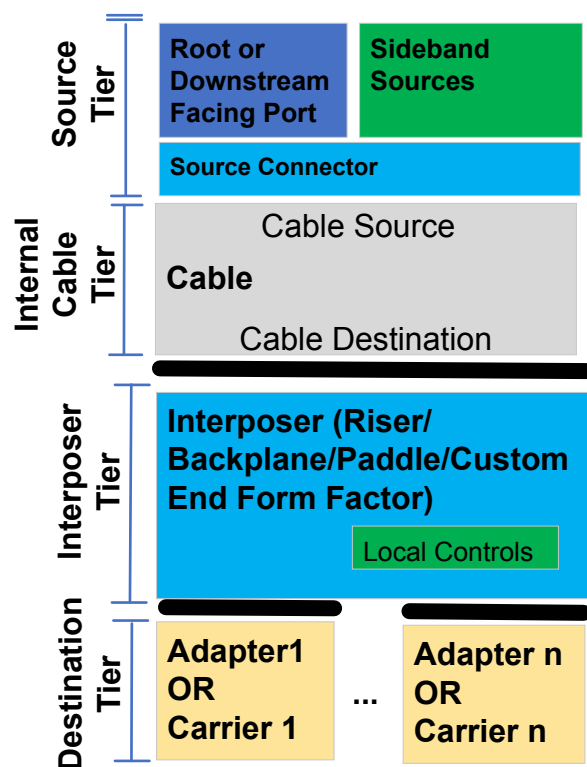
| Byte Location                              | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-----|----------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0                                        | <b>PCIe-MI Message Header:</b> This field must be set as per § <u>Table 12-24</u> .                                                                                                                                                                                                                                                                                                                                                                                               |             |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
| <b>Port Information Completion Content</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
| 8                                          | <b>Status:</b> This field indicates additional status for the command.                                                                                                                                                                                                                                                                                                                                                                                                            |             |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
|                                            | <table><tr><th>Bits</th><th>Description</th></tr><tr><td>7:1</td><td>Reserved</td></tr><tr><td>0</td><td><b>Connector Subdivision Pending:</b> If any of the Links associated with the current or specified connector subdivision are not in DL_Down and must be transitioned to DL_Down due to any condition other than a Cold Reset in order to change the connector subdivision, then this bit must be set to 1b; otherwise, this bit must be cleared to 0b.</td></tr></table> | Bits        | Description | 7:1 | Reserved | 0 | <b>Connector Subdivision Pending:</b> If any of the Links associated with the current or specified connector subdivision are not in DL_Down and must be transitioned to DL_Down due to any condition other than a Cold Reset in order to change the connector subdivision, then this bit must be set to 1b; otherwise, this bit must be cleared to 0b. |
|                                            | Bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Description |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
|                                            | 7:1                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Reserved    |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
| 0                                          | <b>Connector Subdivision Pending:</b> If any of the Links associated with the current or specified connector subdivision are not in DL_Down and must be transitioned to DL_Down due to any condition other than a Cold Reset in order to change the connector subdivision, then this bit must be set to 1b; otherwise, this bit must be cleared to 0b.                                                                                                                            |             |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |
|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |             |     |          |   |                                                                                                                                                                                                                                                                                                                                                        |

## 12.8 Retimer Management §

With increasing interface speeds and varied PCIe topologies, Retimers are expected to be more prevalent. Retimer configuration typically utilizes an SMBus-based dedicated configuration EEPROM with non-standard addressing or required out of band connectivity for in-system updates. Thus, Retimer management is left to the implementer.

## 12.9 Internal Cable Management §

Internal Cable Management guidance is defined for usage with CopprLink.



Example interposer local controls are clocking, PERST, 2-wire and PESTI fanout, discovery, flexible I/O logic, and power delivery.

High speed I/O may bypass an interposer used for sideband management, fanout and power controls.

Figure 12-22 Example Tiers Involving Sidebands §

